STUDIES OF CONIDIAL GERMINATION
IN SOME POWDERY MILDEW FUNGI

by

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A thesis presented in part fulfilment of
the requirements for the degree of Doctor
of Philosophy in the Faculty of Science
of the University of London

October 1972

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ABSTRACT

The germination of several species of powdery mildew fungi was compared on glass and on appropriate host leaf under varying environmental conditions. In no instance, did the germination on the two substrates differ sufficiently to support the hypothesis that the host leaf stimulates germination. On the contrary , experimental evidence suggested that poor germination on glass resulted from unsatisfactory techniques for cleaning the glass. The germination capacity of conidia did , however, vary considerably from collection to collection. In general , conidia collected from young lesions on vigorously - growing plants gave the highest germination rate while those from old lesions on plants grown in artificial light germinated poorly.

In Sphaerotheca pannosa mildew development was retarded when the rose leaves were wetted immediately after inoculation and the degree of retardation was related to the length of the wet period. Leaf wetness prevented the fungus from establishing contact with the host and the most susceptible period was between 6 - 8 h after inoculation when processes leading to penetration of the host were initiated.

The pedicels of rose are colonized more extensively than leaves and this observation is probably due to the faster early growth of the fungus on the pedicels and their longer period of susceptibility which in turn is linked with a slower rate of cuticle formation.

An examination of the fine structure of the haustorium of S. pannosa showed an ellipsoidal body and a tubular neck separated by a perforated septum. Branched lobes investing the body were connected to it mainly at the two poles. The haustorium had a fungal wall which was continuous throughout and the body lay in a sheath which may be fluid. Studies of the host plasmalemma showed that the haustorium was extracellular. The neck was immediately surrounded by host cytoplasm and

outside this was a collar which was probably of host origin and contained callose. Concentric opaque rings occurred on the host and fungus plasmalemmas in the neck region.

Experiments (with Erysiphe cichoracearum on Burdock) indicated that aseptic conidia could be produced but the methods were not reliable enough for further investigation of growth in axenic culture. No growth of this fungus on artificial substrates was obtained beyond the germ tube stage.

ACKNOWLEDGEMENTS

I wish to express my sincere thanks to my supervisor, Dr. B.E.J. Wheeler for his invaluable advice and guidance throughout this work and for his help in the preparation of the manuscript.

I am also grateful to Dr. J.L. Gay for his guidance in the work presented in Section III in this thesis and for his readiness to help at all times.

Special thanks are due to Miss M. Martin for her help with the electron microscopy and in particular for cutting the thin sections; to Mr. C. Sawyer for assistance with photographic work and to Mrs. J. Cheston for typing the tables. I am also grateful to those members of this department who helped me in several ways during the course of my work.

I am indebted to Professor R.K.S. Wood in whose department this work was undertaken, for his continued interest in my welfare.

I am especially grateful to my wife who afforded me constant encouragement throughout the period of work and for typing this thesis.

Finally, I thank the University of Ceylon (Colombo Campus) for granting me study leave.

CONTENTS

INTRODUCTION

Four topics are considered in this thesis, all related to conidial germination in some powdery mildew fungi. The starting point of the investigation was the contradictory evidence on conidial germination on glass and leaf surface and the conflicting hypotheses proposed to explain it. The work then progressed to a detailed examination of germination and early growth of *Sphaerotheca pannosa* on rose. This led to an examination of structural aspects of early infection process especially the fungus haustorium. The possibility of producing aseptic conidia of a powdery mildew and the growth of powdery mildew conidia on artificial substrates was briefly examined.

I. FACTORS INFLUENCING CONIDIAL GERMINATION

INTRODUCTION

Powdery mildew fungi are obligate parasites, depending exclusively on their hosts for nutrition, but the conidia of many of them can germinate *in vitro* and during recent years there have been many studies of such germination and of the factors influencing it (Corner, 1935; Yarwood, 1936; Longree, 1939; Clayton, 1942; Brodie, 1945; Delp, 1954; Nour, 1958; Weinhold, 1961; Peries, 1962; Manners and Hossain, 1963; Griffee, 1967). One possible practical development from this work is the testing *in vitro* of fungicides against powdery mildews (Zaracovitis, 1964 a,b) which otherwise have to be evaluated on whole plants in the field (Sprague, 1949) and the greenhouse (Yarwood, 1951) or on detached leaves (Burchill, 1958) and leaf discs (Dekker, 1962).

Although Zaracovitis (1964 b) obtained over 80% germination with conidia of many powdery mildews on glass slides in a saturated atmosphere and so did Manners and Hossain (1963) with *Erysiphe graminis* in specially-constructed humidity chambers, many other workers have reported poor germination (Corner, 1935; Clayton, 1942; Delp, 1954; Weinhold, 1961; Jhooty and McKeen, 1965 a; Bent, 1967; Griffee, 1967; Jackson, 1968; De Waard, 1971) and have commented, in particular, on the higher germination of the conidia on leaf surfaces of susceptible hosts than on glass (Woodward, 1927; Yarwood, 1936; Nour, 1958; Weinhold, 1961; Peries, 1962; Jhooty and McKeen, 1965; Griffee, 1967; Jackson, 1968). For example, Griffee (1967) recorded only 9% germination of *Sphaerotheca mors-uvae* conidia from black currant on glass coverslips in a saturated atmosphere at 20° compared with 50% on leaf discs under otherwise similar conditions. In experiments with the same fungus, Jackson (1968) obtained higher germination on glass than Griffee but found that the level of germination was extremely variable. On leaf discs, germination was both higher and less variable than on glass. Similar high levels of germination have also been reported for conidia applied to leaves which are resistant to or from plants which are not hosts for, the particular powdery mildew used (Corner, 1935; Longree, 1939; Peries, 1962; Zaracovitis, 1964).

Griffee (1967) found that freezing and wounding leaf discs prior to inoculation increased conidial germination of S.mors-uvae and suggested that this could be attributed to leakage of nutrients which might therefore support a "chemical stimulation" hypothesis but Jackson (1968) was unable to substantiate these results.

Other workers suggest that more conidia germinate on leaves because humidity at the leaf surface is optimal (Longree, 1939; Frampton and Longree, 1941; Delp, 1954; Weinhold, 1961) and that failure to maintain suitable humidities in other situations has often resulted in poor germination.

The type of inoculum used in spore germination tests is also important. Jackson (1968) points out that Zaracovitis grew his mildew-infected plants in the spring and summer when the host grew well. This, he suggests, could account for the production of a highly viable inoculum giving high germination even on glass. In contrast, both Griffee (1967) and Jackson (1968) carried out their investigations during the winter months with spores produced on plants maintained under artificial conditions. This could perhaps explain their low germination on glass; the inoculum itself may have consisted of 'poor' or less viable conidia. Nevertheless, the enhanced germination of such conidia on the leaf still needs to be explained.

The contradictory evidence on conidial germination on glass and leaf surfaces and the conflicting hypotheses put forward by various investigators to explain the differences led to the present studies with conidia of Sphaerotheca mors-uvae (Schw.) Berk., from black currant, Erysiphe cichoracearum DC. from burdock and several other mildews. Certain closely associated problems are also examined.

LITERATURE REVIEW

The literature on powdery mildew fungi contains abundant references to factors influencing conidial germination on leaf surfaces and glass slides and several authors (Yarwood, 1957; Zaracovitis, 1964 a,b; Schnathorst, 1965) have reviewed environmental relationships in the powdery mildews. A brief review of literature connected with factors affecting the process of germination is attempted in this report.

(1) Standardization of techniques

(a) Age of inoculum

Powdery mildew conidia are among the shortest-lived air-disseminated fungal spores (Yarwood *et al.*,1954) and they are reported to diminish in vigour immediately after abscission (Zaracovitis, 1966). Their longevity is markedly affected by the temperature and relative humidity (r.h) of the environment (Prabhu *et al.*, 1962) and delay in harvesting them reduces their chances of germinating successfully (Zaracovitis, 1966). Peries (1962) found non-viable spores of *Sphaerotheca macularis* to be most plentiful on greenhouse-maintained strawberry where spores were not removed by wind in the way they would in the open.

Zaracovitis (1964 a,b) considered the age of the conidia to be the most important factor in ensuring a high percentage germination on glass. He found germinability to be highest when source plants were shaken 24h prior to testing to remove old, mature conidia. Morrison (1966) recorded a two to three fold increase in germination by taking this precaution. Similar claims have also been made by several others (Longree, 1939; Hossain, 1959; Peries, 1962). In contrast, Griffee (1967) found that

48h conidia of <u>Sphaerotheca mors-uvae</u> germinated better than 24h conidia in a series of tests using spores from 24 to 96h old. Jackson (1968) similarly found that the 48h conidia of <u>S.mors-uvae</u> germinated best.

(b) <u>Inoculum production</u>

Mount and Slesinski (1971) consider that one of the greatest sources of experimental variations in germination concerns the production of inoculum. The importance of inoculum quality and uniformity in studying the kinetics of the infection process has been recently analysed by Ellingboe (1968). He considers that rather than deal with the production of large quantities of inoculum increased attention should be focussed on, obtaining a viable homogenous inoculum to enable qualitative analysis of events which occur subsequent to inoculation of the host plant.

Very few investigators have recognized the importance of the age and the general nutritional state of the host plant in ensuring the production of highly viable inocula (Zaracovitis, 1964 a,b). Zaracovitis obtained consistently high percentage germination of conidia from greenhouse plants kept under good conditions for their development.

Powdery mildews are obligate parasites totally dependent on their hosts for their nutrition; any external conditions that affect the growth of the host is expected to affect the growth and reproduction of the parasite (Zaracovitis, 1964 a). Zaracovitis conducted germination tests during spring, summer and autumn because of the "difficulty in growing appropriate host plants for the production of sufficiently highly viable inocula in the winter".

Hammarlund (1925) found that there was a marked reduction in the percentage germination of <u>Erysiphe communis</u> produced in humid air. However, Manners and Hossain (1963) obtained a high percentage germination with *Erysiphe graminis* conidia from different cereal hosts kept in humid atmospheres. Hammarlund (1925) also reported that the conidia of *Sphaerotheca pannosa* on rose produced at near optimum temperature (30°) for the development of the fungus showed a higher germination energy (rate of germination) than those produced near minimum (5°) or maximum (35°) temperatures.

Several investigators have reported the diurnal periodicity of spore release in powdery mildew fungi (Yarwood, 1936 a; Childs, 1940; Cherewick, 1944; Hirst, 1953; Nair and Ellingboe, 1965; Pady *et. al.*, 1969; Benada, 1970; Cole and Fernandes, 1970) but only a few have related this phenomenon to the germinability of the conidia. Yarwood (1936 a) found that the conidia of *Erysiphe polygoni* germinated best when collected in the afternoon. Their germination capacity decreased reaching a minimum in the early hours of the morning, rising again in daylight. Cherewick (1944) reported increased germination of *E. graminis* conidia collected during daylight as opposed to those collected in the night.

(c) Criteria for germination

Very few writers have stated when a conidium was considered germinated (Manners and Hossain, 1963; Zaracovitis, 1964 a; Jackson, 1968). Manners and Hossain (1963) considered a conidium to be germinated when the length of any one germ tube exceeded its breadth. The germ tube length and the rate of germ tube growth as suggested by Tomkins (1932) have also been determined by these workers as additional criteria of germination. Nour (1958) used the state of the conidium (whether turgid or shrunken) as a further criterion of germination.

Percentage germination is the most widely used criterion for the assessment of germination of powdery mildew conidia (Yarwood, 1936; Longree, 1939; Delp, 1954; Manners and Hossain, 1963; Zaracovitis, 1964 a,b;

Jackson, 1968). Percentage germination is based on counts of 200-300 or more conidia including shrivelled ones (Yarwood, 1936; Longree, 1939; Manners and Hossain, 1963; Zaracovitis, 1964 b; Jhooty and McKeen, 1965 a; Griffee, 1967; Jackson, 1968; De Waard, 1971). Conidia in clumps or in chains were found to germinate in lower percentages than single conidia (Brodie and Neufeld, 1942; Brodie, 1945; Delp, 1954; Hossain, 1959; Manners and Hossain, 1963; Zaracovitis, 1964 a) but most investigators have not given details of the types of conidia that were counted.

(d) Incubation period

Almost all investigators recorded the germination of powdery mildew conidia after 24h incubation. Erysiphe polygoni, E. graminis and Uncinula necator germinate on glass and leaf after a very short period of incubation (Brodie and Neufeld, 1942; Delp, 1954; Manners and Hossain, 1963; Zaracovitis, 1966; Stanbridge et al., 1971). Others e.g. Sphaerotheca pannosa, S. fuliginea, S. macularis, S. humuli require much longer periods of germination (Hashioka, 1937; Longree, 1939; Peries, 1962; Zaracovitis, 1966). Zaracovitis (1966) concluded that powdery mildews could be divided into three groups according to their rates of germination and the time required to form appressoria on glass when incubated at 21-23°; viz. over 70% of the conidia germinating and forming appressoria (i) in 5h, (ii) in more than 5h but less than 10h and (iii) in more than 10h (germination less than 30% in 5h).

(2) Effect of temperature

Powdery mildew conidia are known to germinate over a wide range of temperatures. The minimum reported is 0° for E. graminis (Cherewick, 1944) and the maximum for E. graminis and Uncinula necator is a range of 30-35° (Cherewick, 1944; Delp, 1954). The optimum for the various species is within the range 11-28° (Schnathorst, 1965). Yarwood et al. (1954), Yarwood (1957) and Schnathorst (1965) have reviewed the temperature relationships of powdery mildews and agree that the average optimal temperature for germination and growth is about 21°. But there is considerable variations in the optimal temperatures for conidial germination in different powdery mildew fungi. On the basis of independent records the temperature optima for 6 species of powdery mildews are summarized in Table 1. Schnathorst (1965) believes that the optimal temperature for germination, infection and growth tends to approach that of the host. With many powdery mildews however, the optimal temperature is influenced by the r.h. of the environment.

(3) Water relationships

(a) Free water on germination

There is still little agreement in the literature with regard to water relationships in the powdery mildew fungi. Free water is reported to reduce or inhibit germination (Corner, 1935; Yarwood, 1936; Longree, 1939; Brodie and Neufeld, 1942; Brodie, 1945; Delp, 1954; Schnathorst, 1960; Manners and Hossain, 1963; Morrison, 1964) or produce abnormal germination of some spores (Graf-Marin, 1934; Corner, 1935; Longree, 1939; Grainger, 1947; Delp, 1954; Nour, 1958; Schnathorst, 1959). Weinhold (1961) on the other hand showed that free water is an essential requirement for the germination of peach powdery mildew, *S. pannosa* var. *persicae* . In general, many workers agree on the damaging effect of free water on the germination and growth of powdery mildews. There are even several reports of control of powdery mildews by daily spraying of leaves with water (Yarwood, 1939; 1952; McClellan, 1942; Cherewick, 1944; Delp, 1954; Rogers, 1959).

Table 1. Optimum temperatures for germination of powdery mildew conidia

Species of pathogen	Host	Temperature (°C)	Authority
Erysiphe graminis	Barley	20	Manners and Hossain(1963)
Erysiphe cichoracearum	Helianthus	18	Morrison (1964)
Sphaerotheca mors-uvae	Black currant	20	Griffee (1967)
Sphaerotheca pannosa	Rose	21	Longree (1939)
Sphaerotheca fuliginea	Cucumber	28	Hashioka (1937)
Uncinula necator	Grape	25	Delp (1954)

Although immersion of powdery mildew conidia in water is known to inhibit germination there are reports that conidia floating on water are only slightly inhibited or not at all (Corner, 1935; Longree, 1939; Nour, 1958; Peries, 1962; Schnathorst, 1965). Corner (1935) found powdery mildew conidia germinating on the surface of water produced upright germ tubes and he believed this was a result of negative chemotropism to CO_2.

Woodward (1927), Blumer (1933), Graf-Marin (1934), Longree (1939) and Peries (1962) studied germination of conidia on solutions of various nutrients; Longree (1939) observed increased germination of the conidia of S. pannosa when floated on sucrose solutions. Graf-Marin (1934) sometimes obtained similar results when conidia were immersed or floated on 2.5% sucrose solution and extracts of host leaves and concluded that this effect was due to the presence of nutrients.

Corner (1935) maintained that the rapid death of powdery mildew conidia when immersed in water was due to the lack of oxygen. Delp (1954) is of the opinion that this is due to plasmoptysis (extrusion of protoplasm from cells). Conidia have been observed to burst after being immersed in water releasing the protoplasts and vacuoles through pores or ruptures in the conidial wall (Clayton, 1942; Yarwood, 1952). Longree (1939) suggested that the lower germination obtained with conidia floating on the surface of water was a result of excessive intake of water which may have interfered with the normal germination. Zaracovitis (1964 b) who observed changes in the internal vacuolate structure of the cytoplasm of powdery mildew conidia immersed in water maintained that it was due to interference of water with the delicate protoplasmic colloids.

(b) Relative humidity on germination

Conidia of powdery mildews can germinate from 0-100% r.h. Some powdery mildew conidia e.g. Erysiphe polygoni, E.graminis, Uncinula salicis can germinate in very dry atmospheres (Yarwood, 1936; Brodie and Neufeld, 1942; Clayton, 1942; Cherewick, 1944; Brodie, 1945; Delp, 1954; Nour, 1958; Weltzien-Stenzel, 1959; Manners and Hossain, 1963; Zaracovitis, 1964 b) while others e.g. Sphaerotheca mors-uvae, S. pannosa, S. macularis need a high r.h. approaching saturation for germination (Hashioka, 1937; Longree, 1939; Clayton, 1942; Brodie, 1945; Schnathorst, 1960; Peries, 1962; Jackson, 1968). Results of several studies have shown that germination is often high, slightly below saturation, generally in the range of 95 to 99% r.h. (Longree, 1939; Delp, 1954; Schnathorst, 1960; Griffee, 1967; Jackson, 1968). But Zaracovitis (1964 a,b) found that saturated atmospheres provided moisture conditions optimal for germination of powdery mildew conidia. Schnathorst (1965) concluded that powdery mildews could be divided into three groups based on their responses to various levels of moisture stress: (i) those which germinate at low moisture stress-between 75-100% r.h. (ii) those which germinate optimally near 100% (or low moisture stress) but will germinate between 0-100% r.h. (iii) those which will germinate over a wide range of relative humidities.

It has been reported that low humidity has a desiccation action on the conidia and causes shrivelling of both germinated and non-germinated conidia (Yarwood, 1936; Brodie and Neufeld, 1942; Clayton, 1942; Cherewick, 1944; Yarwood, 1952; Nour, 1958; Hossain, 1959; Zaracovitis, 1964 a), especially at high temperatures (Yarwood, 1936; Cherewick, 1944).

Graf-Marin (1934) and Brodie (1945) considered the ability of some powdery mildew fungi to germinate in low relative humidities was a result of water uptake because of the high osmotic pressure in the cell sap. But the results of Schnathorst (1959) and Jhooty and McKeen (1965 a) suggest that the conidia of powdery mildew fungi have low osmotic values indicating a high water activity thus disputing the theory of Graf-Marin and Brodie.

Many investigators consider that powdery mildew as a group do not require an external supply of moisture for germination as they carry sufficient water for germination in their highly vacuolated cytoplasm (Corner, 1935; Horsfall, 1956; Koopmans 1959). Yarwood (1950) demonstrated that the water content of several powdery mildews as a group (Erysiphe polygoni, E. graminis, E. cichoracearum) was 4.5 times as high as the average of other fungi studied (Uromyces, Peronospora, Penicillium, Aspergillus, Botrytis and Monilia). Zaracovitis (1964 b) found that the non-germinated conidia of E. graminis from oats contained 65% of their fresh weight as water. Somers and Horsfall (1966) reported that conidia produced in moist chambers contained about 60-70% of the fresh weight as water. Recently Mitchell and McKeen (1970) reported that more than 50% of the volume of conidia of S. macularis consists of vacuoles in which most of the water in the conidia is stored.

(4) Effect of light

Although some workers have reported that light when applied during germination influences this process (Yarwood, 1936; Weltzien-Stenzel, 1959; Morrison, 1964) others found no such effect (Woodward, 1927; Burchill, 1958; Peries, 1962). It is generally agreed that light has no direct influence on the germination of powdery mildew conidia and can affect this process by acting indirectly through the host during the production of the conidia (Cherewick, 1944; Yarwood, 1957; Zaracovitis, 1964 a; Schnathorst, 1965). However, Mount and Ellingboe (1968) reported that UV light inhibited successful establishment of E. graminis DC.f. sp. tritici with the wheat host by affecting the nucleus 6-12h after inoculation. UV radiation had little effect on mildew development 20h after inoculation when the nuclei in the haustoria were protected by the epidermis.

(5) Effect of substratum

(a) Germination on glass

Many investigators have recorded germination of powdery mildew conidia on glass but relatively few have obtained high percentage germination with some of the species studied (Hammarlund, 1925; Graf-Marin, 1934; Longree, 1939; Nour, 1958; Manners and Hossain, 1963; Zaracovitis, 1964 a,b). A majority of investigators found conidial germination on glass was low and variable. Germination records of conidia for different species of powdery mildew fungi published by various investigators are summarized in Table 2.

Yarwood (1957) stated that germination on glass was abnormal in that germ tubes may grow away from the substratum and appressoria are commonly not formed. Zaracovitis (1964 b) on the other hand obtained over 80% germination and observed appressoria in most of the powdery mildew species he examined. Zaracovitis (1964 b) concluded that the lower germination on glass slides observed by others might have been due to toxic effects of chemicals that are used for cleaning the glass slides.

(b) Germination on agar and on membranes

There are several reports in the literature of germination studies of powdery mildew conidia on agar substrates (Yarwood *et al.*, 1954; Morrison, 1964). Morrison recorded 55% germination of *E. cichoracearum* conidia on 1% water agar. Yarwood *et al.* obtained higher germination (95%)

Table 2. Percentage germination of conidia from different powdery mildews after incubation in a saturated atmosphere

(As reported in the literature)

Pathogen	Host	Percentage germination on glass	Percentage germination on host leaf	Temperature (°C)	Authority
Erysiphe graminis	Barley	24	-	22	Yarwood, 1936
	Barley	5	-	22	Clayton, 1942
	Barley	71	-	15	Nour, 1958
	Barley	71.5	-	15	Graf-Marin, 1934
	Barley	37	-	22	Brodie, 1945
	Barley	94	-	20	Manners & Hossain, 1963
	Oats	39	-	22	Yarwood, 1936
	Oats	30	-	22	Brodie, 1945
	Oats	87	-	20	Manners & Hossain, 1963
	Oats	47	-	22	Grainger, 1947
	Oats	80	-	21	Zaracovitis, 1964
	Wheat	30	-	22	Brodie, 1945
	Wheat	21	-	22	Brodie, 1945
	Wheat	94	-	20	Manners & Hossain, 1963
	Wheat	57.5	-	10	Cherewick, 1944
	Poa pratensis	10	-	22	Brodie, 1945
	Agropyron repens	20	-	22	Brodie, 1945
	Agropyron repens	35	-	22	Brodie, 1945
Erysiphe cichoracearum	Sunflower	9	-	22	Yarwood, 1936
	Hibiscus sp.	33	-	22-31	Nour, 1958
	Helianthus sp.	9	-	22	Brodie, 1945
	Lettuce	27	44	18	Schnathorst, 1960

Table 2 contd....

Erysiphe polygoni	Red clover (Indiana 1930)	56	-	22	Yarwood, 1936
	Red clover (Wisconsin 1933)	76	-	22	Yarwood, 1936
	Red clover (California 1935)	86	-	22	Yarwood, 1936
	Red clover	39	-	22	Clayton, 1942
	Clover	86	-	22	Brodie, 1945
	Clover	92	-	20	Jhooty & McKeen, 1965
	Clover	33	-	25	Yarwood et al., 1954
	Mustard	46	-		Yarwood, 1936
	Polygoma sp	7	-	22	Brodie & Neufeld, 1942
	Delphinium	52	-	22	Brodie & Neufeld, 1942
Erysiphe martii	Red clover	85	-	21	Zaracovitis, 1964
Erysiphe betae	Sugar beet	91	-	21	Zaracovitis, 1964
Erysiphe fisheri	Senecio vulgaris	95	-	21	Zaracovitis, 1964
Erysiphe horridula	Myosotis sp	92	-	21	Zaracovitis, 1964
Erysiphe umbelliferarum	Faba bona	10	-	22-31	Nour, 1958
Leveillula taurica	Euphorbia sp	78	-	31	Nour, 1958
	Abutilon sp	38	-	31	Nour, 1958
	Faba bona	60	-	31	Nour, 1958
Microsphaera alvi	Syringa vulgaris	50	-	22	Brodie, 1945
Oidium begoniae	Begonia	96	-	21	Zaracovitis, 1964

Table 2 contd....

Podosphaera leucotricha	Apple	5	-	22	Berwith, 1936
	Apple	67	-	21	Zaracovitis, 1964
Sphaerotheca fuliginea	Cucurbita pepo	38	-		Nour, 1958
	Cucumber	28	-	28	Haphioka, 1937
Sphaerotheca humuli	Rose	14	-	22	Brodie, 1945
Sphaerotheca pannosa	Rose	97.8	88.0	21	Longree, 1939
	Rose	91.6	-	18	Hammerlund, 1925
	Rose	80	-	21	Zaracovitis, 1964
Sphaerotheca macularis	Strawberry	28	82	20	Peries, 1962
	Strawberry	75	87	20	Jhooty & McKeen, 1965
	Strawberry	91	-	21	Zaracovitis, 1964
Sphaerotheca mors-uvae	Black currant	5.5	61.2	20	Griffee, 1967
	Black currant	23.5	61.0	20	Jackson, 1968
	Black currant	77	-	21	Zaracovitis, 1964
Uncinula salicis	Populus sp	33	-	22	Brodie, 1945
Uncinula necator	Grape	95	-	24	Delp, 1954
	Grape	96	-	21	Zaracovitis, 1964
	Grape	37.8	67.3	21	Weltzien-Stenzel, 1959

of E.graminis conidia on nutrient agar than on plain agar (82%). But Longree (1939) recorded only 8.9% germination of S.pannosa conidia on Potato Dextrose Agar compared to 10% on water agar.

Stoll (1941) and Weltzien-Stenzel (1959) reported higher germination of powdery mildew conidia on collodion membranes than on glass slides. Dickinson (1949) also reported higher germination on collodion membranes but he observed that sometimes the germ tube grew away from the substratum. Jackson (1968) obtained high germination of the conidia of S. mors-uvae on cellophane membranes similar to that on leaf surfaces. More recently, De Waard (1971) reported high percentage germination of the conidia of several powdery mildew fungi on 25µm thick cellulose membranes laid on modified Czapek-Dox agar in Petri dishes, compared to poor germination on glass slides. Penetration of the membranes was not observed.

(c) Germination on susceptible leaves

Several workers have shown that in saturated atmospheres, conidia germinate better on their respective host leaves than on glass (Woodward, 1927; Yarwood, 1936; Longree, 1939; Burchill, 1958; Nour, 1958; Schnathorst, 1960; Peries, 1962; Griffee, 1967; Jackson, 1968; Jhooty and McKeen, 1965 b). Some have stated that the higher germination on leaves is due to stimulation by the hosts (Yarwood and Hazen, 1944; Yarwood et al., 1954; Yarwood, 1957; Rogers, 1959; Peries, 1962; Griffee, 1967).

It has been reported that conidia of some powdery mildew fungi require a high r.h. in order to germinate on glass (Hashioka, 1937; Longree, 1939; Clayton, 1942; Brodie, 1945; Schnathorst, 1960) but will germinate on leaves even when the relative humidity of the environment is very low (Longree, 1939; Burchill, 1958; Weinhold, 1961; Peries, 1962; Jhooty and McKeen, 1965 b). Based on this evidence some authors believe that the increased germination on the leaves is due to the higher r.h. of the latter (Longree, 1939; Frampton and Longree, 1941; Delp, 1954; Weinhold, 1961). Longree (1939) found that conidial germination of S. pannosa on leaves was high at moisture levels unfavourable to germination on glass. Weinhold (1961) made a similar observation with peach powdery mildew S. pannosa . Zaracovitis (1964 b) has shown that a high r.h. is required during the germination of powdery mildew conidia to protect them from losing the water which they contain which supports their germination.

Near the surface of all leaves a humidity gradient exists ranging from that of the leaf surface to that of the environment (Zaracovitis, 1966). But comparatively few attempts have been made to determine this gradient of r.h. above the leaf (Schnathorst, 1960). Thut (1939) showed such a humidity gradient to exist at the surface of Lantana leaves and those of bean, petunia and sunflower. Schnathorst (1960) demonstrated that a humidity gradient which existed above the surface of lettuce leaves was removed when there were air movements. Ramsey et al. (1938) also commented that air movements removed the humidity gradient from leaf surfaces. Although Delp (1954) concluded that there was no accurate method to determine satisfactorily the r.h. of the microclimate at the leaf surface, many workers have recorded relative humidities at various distances from leaf surfaces which are higher than those of the environment (Shaw, 1935; Ramsey et al., 1938; Yarwood, 1939; Yarwood and Hazen, 1944; Rogers, 1959; Schnathorst, 1965).

(d) Germination on resistant and non-related hosts

There are several reports that conidia of some powdery mildews germinate successfully on leaves of resistant hosts and also on leaves of plants totally unrelated to their hosts (Woodward, 1927; Corner, 1935; Smith and Blair, 1950; Peries, 1962; Zaracovitis, 1964 a). These workers agree that the host does not exert any sepcific influence on conidial germination and this has been used by some (Zaracovitis, 1966) as an argument against the theory of host stimulation.

Peries (1962) who recorded low germination on glass found the germination of the conidia of S. macularis was equally high on slides coated with surface waxes obtained from leaves of susceptible and resistant strawberry varieties as well as from apple rootstock. Jhooty and McKeen (1965 b) who repeated this work were unable to substantiate these results. But Jackson (1968) reported a similar stimulation of oonidial germination of S. mors-uvae on glass coated with surface waxes from black currant leaves.

MATERIALS AND METHODS

(1) Inoculum production

Sporulating colonies of the various powdery mildews were maintained on the adaxial surfaces of young, detached leaves of their appropriate hosts. The leaves were kept in a 40.5 x 26.5 x 5 cm polystyrene seed germination tray (Stewarts Plastics Ltd.) with their laminae supported by the standard perspex plate with their petioles protruding through the holes in the plate and reaching the distilled water in the base (Fig 1). Leaves were inoculated in a tray by surrounding this with a cardboard cylinder, 50 cm diam and 65 cm high and then allowing conidia, blown by a teat pipette from an infected leaf held over the cylinder, to settle on them.

The inoculated leaves were then covered with the tray lid and incubated in cabinets. These cabinets were modified refrigerated display units (Frigidaire, R.A.Bott, Ltd.) maintained at either 17 $\pm$ 1^{o}, with a 16h period of 525-760 lumens/sq.ft. provided by a bank of five 65-80W 'warm white' fluorescent tubes (Phillips) outside the unit. Successful inoculations were visible after 5-10 days, depending on the mildew species. Cultures were thus maintained for c. 2-3 weeks. Transfers were then made to fresh young leaves, as described.

(2) Source of host plant material

Young leaves were obtained either from plants grown in the field at Silwood Park (Imperial College Field Station) or from plants raised in the Botany Dept. greenhouse at South Kensington.

Greenhouse Plants

Burdock (Arctium lappa): Plants were raised from seeds collected in early autumn, 1969, from plants in the Walled Garden at Silwood Park. Seeds were first sown on damp blotting paper in germination trays, similar

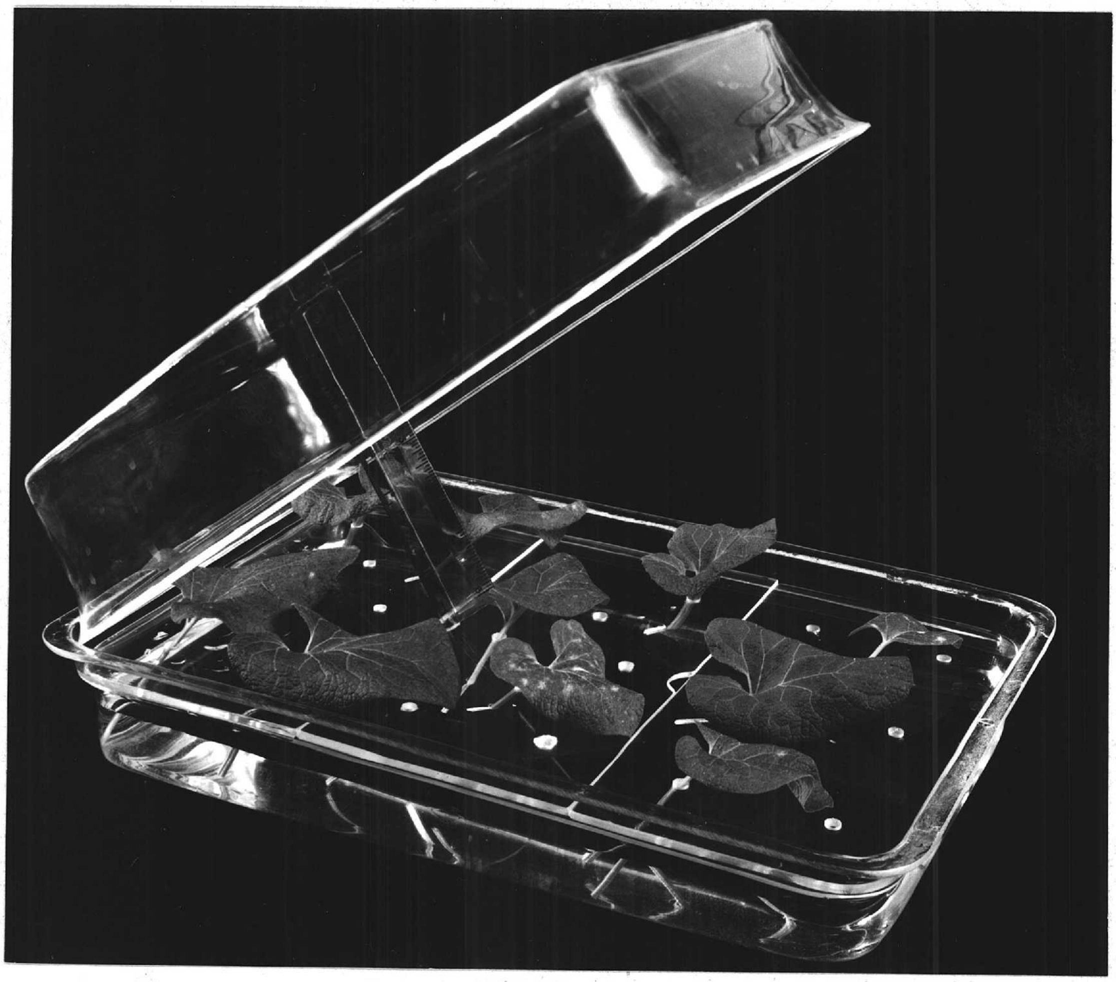


Fig. 1. A germination tray used in maintaining mildew cultures on detached leaves.

to those described above, which were incubated at 25°. After 12-14 days, the young seedlings were potted on in 8 cm diam plastic pots, three seedlings per pot, in John Innes no.1 compost. The pots were then kept in the greenhouse (17 to 20°) for about 6-8 weeks. At this stage the fourth leaf had normally begun to unfold. This and the three successive leaves (5 to 7) were detached about 7 to 10 days after unfolding and used in experiments.

Black currant (Ribes nigrum): Plants (cv. Amos Black) were raised from stem cuttings, 15-20 cm long, taken from bushes at Silwood Park. These were first stood in water in beakers on the laboratory bench for 2-3 weeks until roots formed. They were then potted singly in 20 cm diam pots of John Innes no.1 compost and transferred to the greenhouse. Some 3-4 weeks later many leaves were produced which were suitable for experiments with S. mors-uvae. Subsequently plants were never allowed to grow more than 45 cm high. They were cut back periodically to induce further leaf growth from the axillary buds.

Other hosts: Barley (cv. Proctor) and Cucumber (cv. Improved Telegraph) were also raised from seed and Vine (cv. Riesling) from rootstock after direct planting in potting compost.

The greenhouse was maintained between 17 and 20° through most of the year though during the summer higher temperatures were not infrequent. During the winter, a 16h photoperiod was provided by fluorescent tubes suspended about two feet above soil level. The minimum illumination at soil level was 300 lumens/sq.ft. Occasional infestations of aphids were checked by spraying plants with Malathion 60 (Murphy Chemical Co.) and the greenhouse was also fumigated regularly with Tedion V-18 (Duphar-Midox Ltd.) to control red spider.

(3) Age of conidia

To obtain conidia of approximately uniform age mildew leaves were shaken and air blown over them vigorously several times with a Pasteur pipette fitted with a large rubber bulb to remove old, non-viable conidia 24h before collecting spores for inoculations. Air was also blown over the lower leaf-surfaces and the petioles. The leaves were then left undisturbed in the germination trays in growth cabinets until the conidia were sampled 24h later.

(4) Conidial germination

Uniform deposits of conidia were obtained on coverslips and/or leaf discs by placing these at the base of a 65 cm high settling tower and allowing conidia, blown with a Pasteur pipette from an infected leaf held above and to one side of the tower, to settle on them (Yarwood et al., 1954; Griffee, 1967).

For all in vitro tests, glass coverslips from a newly-opened box (18 x 18 mm, Chance Bros. Ltd.) were washed under running tapwater by rubbing with the fingers. They were dried by blotting with absorbent tissue (Delsey Toilet tissue, Kimberley-Clark Ltd.) and then polished with fresh tissue until this moved smoothly over the surface. The cleaned coverslips were placed on a circle of black cloth at the bottom of the settling tower before conidia were deposited on them. This prevented them sliding and also made possible a rough visual guide of the spore deposit. Unless otherwise stated, in all tests coverslips with conidia were incubated in the special humidity chambers described by Manners and Hossain (1963).

For in vivo tests, discs (1.6 cm diam) were cut from newly - unfolded leaves with a no.6 cork borer. These leaves were previously washed in running tap water, followed by distilled water and then blotted dry or air dried. In some instances, tests were carried out on leaf-pieces or whole leaves. Leaf material was placed randomly on moist filter paper within a large Petri dish lid and the upper leaf surface dusted with conidia in a settling tower as described above. When it was necessary to dust coverslips and leaf discs simultaneously, the latter were placed on discos of moist filter paper and these were placed at random in between the coverslips. Extra care was taken to ensure that the inoculated surface of the leaf disc was kept free of water. Inoculated discs were floated on distilled water in 5.5 x 3.5 x 2 cm transparent polystyrene boxes (Stewarts Plastics Ltd.), covered with lids and incubated under controlled conditions.

In all tests, saturated atmospheres were maintained inside chambers with distilled water. For relative humidities below 100%, saturated salts solutions (as detailed in Appendixlp.225) were used with the solid phase in excess.

Unless otherwise stated, all tests were carried out at 20° ± 1° in a cabinet with a 16h photoperiod.

(5) The assessment of germination

Germination was usually assessed after 24h incubation. A conidium was considered to have germinated when the length of its germ tube exceeded its breadth. Usually germination was assessed on 100–300 conidia selected at random from each coverslip or leaf disc under low power (x 100) microscope. When fewer conidia were used, all were counted by systematically scanning the coverslip or the leaf disc. Germination counts were made only on single spores; spores in chain or in clumps were excluded as often these germinated poorly. A percentage germination between 80–95% was regarded as a satisfactory standard (control) in these tests.

Each test was replicated four or five times and the results are given as an average of the replicates. For analysis, whenever the number of conidia counted varied, the percentage germination was transformed by the angular transformation of Fisher and Yates (1963). The statistical methods used taken from Bailey (1964) were :

(a) Analysis of variance

(b) Significance test

A modified form of 'Student's' t test was used in assessing the significance of differences between two sets of data when the differences were small. 'Student's' t was calculated from the formula

$$t = \frac{\bar{x}_1 - \bar{x}_2}{S\sqrt{\frac{1}{n_1} + \frac{1}{n_2}}}$$

where n_1 and n_2 = the number of observations of the two samples, $\bar{x}_1$ and $\bar{x}_2$ = the mean of the two samples and S = an estimate of the standard deviation based on both samples jointly.

In certain experiments germ tube length was also measured at the end of the incubation period as an additional criterion of germination. For each replicate sample, 20-25 germ tubes were selected at random and measured with the aid of an ocular micrometer under (x 400) magnification.

(a) <u>On glass surface</u>

After the required incubation, the lid of the humidity chamber was removed and the coverslip was examined under the microscope. Very little condensation occurred around conidia on coverslips incubated in the Manners and Hossain chambers. The little condensation that did occur was mostly on the under surface of the coverslip and this was removed with tissue paper. For examination, the coverslip was fixed to a microscope slide with a little vaseline.

(b) <u>On leaf discs</u>

The germination of conidia on leaf discs was determined using the celloidin peel technique of Wenzil (1939) as modified by Delp (1954). A drop of celloidin (5% Gurr's celloidin in 1:3 diethyl ether/ethanol) was placed on the leaf disc with a finely drawn-out glass rod, gently spread over the surface and allowed to air-dry for 1-2 min. Care was taken to prevent excessive drying as this often resulted in the shrivelling up of the film and the loss of conidia. The celloidin was then carefully peeled off the leaf surface with forceps and immediately placed in a drop of Cotton blue in lactophenol on a microscope slide (Appendix1p.225). After 20-30 min the film was mounted in either lactophenol or water and examined under the microscope.

EXPERIMENTAL

(1) Production of conidia

This investigation broadly deals with the germinability of conidia of Sphaerotheca mors-uvae (Schw) Berk., from black currant produced under natural and artificial conditions. It was hoped that if the differences in germination on glass and leaf surfaces were observed as noted by Griffee (1967) and by Jackson (1968), to extend the work to test the validity of the two theories put forward to explain these differences. Conidial germination of Erysiphe cichoracearum DC. from burdock, E. graminis from barley, E. fuliginea from cucumber and Uncinula necator from vine was also examined.

Sphaerotheca mors-uvae

Several preliminary investigations to standardize techniques were conducted with conidia from naturally-infected leaves collected from black currant at Silwood Park. These were transferred in polyethylene bags to the laboratory. Old conidia were removed and the leaves were incubated in germination trays at 20 $\pm$ 1° in an illuminated cabinet. Twenty-four hours later, the fresh conidia produced were dusted on to coverslips and these were incubated at 22° in Manners and Hossain chambers.

Expt 1. Time to assess germination

This preliminary test was conducted with conidia from leaves collected on 28.8.1969. The host leaves supporting the mildew appeared somewhat mature but they were covered completely with sporulating colonies of the fungus. Percentage germination of conidia on coverslips was assessed after 24, 48 and 72h at 22°.

The full results are given in the Appendix p.226 and summarized in Table 3. From these, two facts are apparent. (i) after 24h there was no further increase in conidial germination. In all subsequent tests therefore,

Table 3. Germination of S. mors-uvae conidia after different incubation times

Hours incubation	No. conidia counted	Mean germination %	Mean germination transformed*	S D (transformed) data
24	2085	25.1	29.8	6.2
48	1879	25.7	30.9	4.8
72	980	22.1	27.6	5.1

* Data transformed by the angular transformation of Fisher and Yates (1963)

this was assessed after 24h. (ii) the results agree with those of Griffee (1967) and Jackson (1968) in that germination was poor on glass. The highest germination recorded was 39.6% but the mean remained around 25%.

Expt 2. Effect of spore density on germination

Conidia were obtained as in the previous experiment but from leaves collected on 27.9.1969.

Twelve coverslips were dusted with young conidia to obtain an initial sparse cover. Four coverslips were then gently removed and the spore density on one was quickly estimated by counting the conidia in 20 randomly - selected microscope fields (x 100). These coverslips were then kept aside undisturbed and more conidia were blown on to the remaining coverslips from infected leaves. By repeating these operations three sets of coverslips with increasing spore densities were prepared. The percentage germination on these was assessed after 24h at 22°.

There was no marked difference in the germination of conidia at the three densities tested but, overall, this was again low (Table 4). Bearing in mind that the low germinability of the conidia on glass may have masked the effect of density on germination, it was decided to conduct all further tests using a spore density between 10-25 conidia per microscope field (x 100).

Expt 3. Effect of age of conidia on germination

Young leaves bearing sporulating colonies were collected on 16.6.1970 from the same source plants and were first maintained in germination trays at 17° for 2 days. Old conidia were then removed and 12-15 leaves were distributed in each of three germination trays which were reincubated (at 17°). After 24, 48 and 72h intervals, conidia from one set of leaves in turn were dusted onto coverslips and leaf discs. Germination was assessed on these after 24h at 22°.

Table 4. Effect of spore density on germination of *S. mors-uvae*

Mean No. conidia per field(X 100)	No. conidia counted	No. conidia germinated	Mean germination		S D (transformed) data
			%	transformed	
9.0	1262	330	26.1	30.2	4.4
32.3	1273	262	20.5	26.9	2.8
59.4	1258	289	22.9	28.5	1.6

Contrary to the results obtained in the previous two tests and to those of Griffee (1967) and Jackson (1968), a high percentage germination was obtained on both glass (Fig 2) and leaf and there was no significant difference in germination on these two surfaces (Table 5).

The highest germination was obtained with 24h conidia with a mean of over 85% on both glass and leaf. This also contrasts with the results of Griffee and of Jackson who found that 48h conidia germinated best. The amount of germination progressively decreased with increase in age of the conidia and also became more variable. Nevertheless, even with 3 day-old conidia germination in excess of 60% was recorded. This is above the highest figure quoted by Jackson (1968) for leaf surfaces.

The reasons for the much-improved germination are not clear but this test differed in two small, but possibly important respects, from the previous two. In the latter, old conidia were removed with a Pasteur pipette fitted with a small rubber teat; in this test the pipette was fitted with a large rubber bulb. Possibly the stronger air blasts were more effective in removing old conidia. Additionally and probably more important, the conidia used were from young leaves produced during the early part of the growing season rather from older leaves collected at the end of the host's growth.

Expt 4. <u>Germination of conidia from detached, field-infected leaves</u>

Young, black currant leaves, infected with <u>S. mors-uvae</u> were detached from plants in the field on 9.7.1970 and maintained at 17^{o} in germination trays. Germination of 24h conidia on glass and leaf discs was then assessed at intervals until the leaves began to senesce.

The result (Table 6) are similar to those of Expt 3 in that equally good germination was obtained on both glass and leaf, though as the leaves senesced and the mildew colonies aged germination decreased from an initial 80% to about 50% and, as the standard deviations indicate,

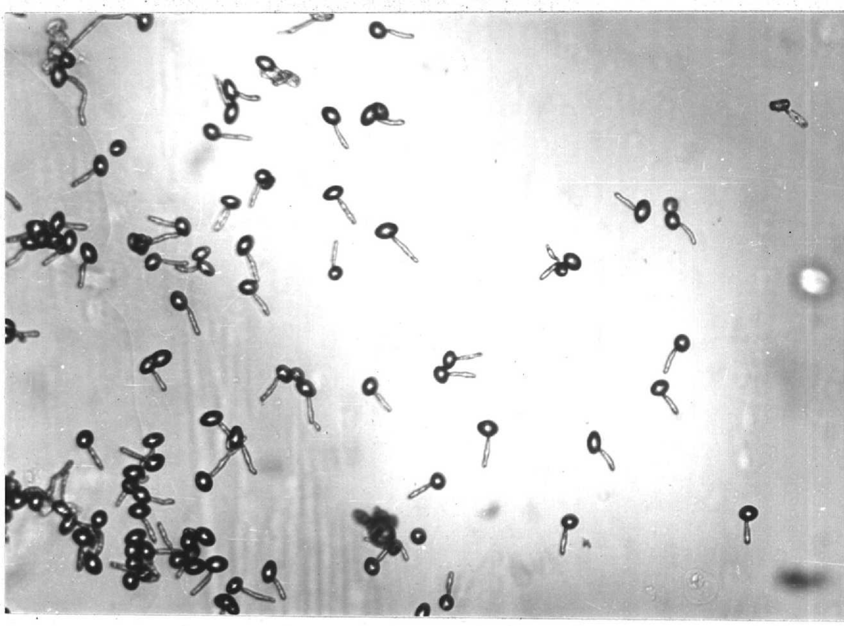


Fig. 2. Germinating conidia of Sphaerotheca mors-uvae on glass x 300.

Table 5. Germination of *S. mors-uvae* conidia of different ages

Age of conidia (h)	Germination site	No. conidia counted	No. conidia germinated	Mean % germination	S D
24	Glass	1500	1312	87.46	1.7
	Leaf	1500	1362	90.8	3.2
48	Glass	1000	718	71.8	3.1
	Leaf	1000	706	70.6	4.2
72	Glass	1000	632	63.2	6.5
	Leaf	1000	616	61.6	9.7

Table 6. Germination of _S. mors-uvae_ conidia from detached, field-infected leaves

Sampling date	Germination site	Mean % germination	S D
10.7.1970	Glass	79.5 (a)	2.2
	Leaf	80.8	4.3
12.7.1970	Glass	73.2	6.8
	Leaf	74.3	10.5
14.7.1970	Glass	81.7 (b)	10.1
	Leaf	79.0	5.1
17.7.1970	Glass	55.6	10.2
	Leaf	58.2	9.5
19.7.1970	Glass	55.3 (c)	1.9
	Leaf	52.1	18.6

(a), t = 0.5 (b), t = 0.4 (c), t = 1.5

generally became more variable. Also towards the end of the sampling period, fewer conidia were produced on the infected leaves.

Expt 5. <u>Germination of conidia from detached leaves, artificially-inoculated with S. mors-uvae</u>

Healthy, young leaves were collected from the field on 10.7.1970 and inoculated as described under materials and Methods. The inooulated leaves were maintained at 17^{o} and when the mildew became visible (after 8-9 days), the germination of 24h conidia was tested. The test was repeated twioe at 3 day intervals using inocula from the same source.

Again equally good germination was obtained on glass and leaf (Table 7). There was only a slight drop in germination-capacity of conidia over the period of the tests.

Expt 6. <u>Effect of increase in age of lesion in the field on conidial germination</u>

The results of Expts 4 and 5 suggest that the conidia of <u>S. mors-uvae</u> germinate equally well on glass and the leaf. These findings are contrary to some previously reported. It is possible that high germinability of conidia is associated with mildew oolonies produced in the summer on rapidly-growing host tissue and that as leaves and colonies age so the germination capacity of the conidia produced falls. It could be that it is these oonidia that germinate well on leaf but not on glass.

An experiment was designed to test this hypothesis. Young leaves (the 2nd leaf not yet fully expanded) from different plants that were just begining to show visible signs of mildew were tagged within a planting of black ourrant at Silwood Park in mid-July 1970. The positions of the mildewed areas on these were sketched on paper for reference. They were mostly on the lower surfaces. The exact age of the mildew colonies was not known but none appeared more than 7-10 days old. In each colony, only a few

Table 7. Germination of S. mors-uvae conidia from detached, artificially-infected leaves

Sampling date	Germination site	Mean % germination	S D
18.7.1970	Glass	85.6	4.9
	Leaf	83.0	5.7
21.7.1970	Glass	78.0	8.1
	Leaf	82.4	4.7
24.7.1970	Glass	78.0	6.2
	Leaf	78.4	9

scattered conidiophores were found. Not all colonies selected were of the same size.

At weekly intervals about 8-12 leaves were brought to the laboratory and their lesions were compared with the original sketches. Any new infections and the non-infected regions were wiped with a piece of wet cotton wool, the old conidia were blown off the remaining lesions and the leaves were incubated at 17° in germination trays. The germination of the young (24h) conidia that were produced was then assessed on glass and on leaf discs. The experiment was continued until finally the conidial production on the leaf samples decreased to such low levels that insufficient conidia could be collected for the tests.

During the 3rd to the 5th week, there was heavy intermittent rain but this did not hamper mildew development on tagged leaves and by the end of the first week, most colonies had spread and other new colonies had formed.

During the middle and the latter part of the experiment it was difficult to obtain mildew-free young leaf tissue for germination tests. Young leaves, apparently uninfected, were washed under running tap water and gently wiped with wet cotton wool to remove any surface mycelium. These were air dried and set up in the germination trays at 17° for a few hours. Sample discs were cut from these, cleared overnight in methanol, stained in cotton blue in lactophenol and examined microscopically for evidence of mildew growth on them. Heavily mildewed leaves were discarded. The leaves that were free of the mildew growth, (which was the case in most instances) or those with little mildew growth were once again washed and dried. Discs were cut from these and inoculated.

Tests could only be carried out for 6 weeks as beyond this period, the mildew did not produce conidia in sufficient quantities for sampling. Also some heavily infected leaves did not expand to their full size but dropped off by this time. The results of this study are given in full in Appendix 3 p 227 and summarized in Table 8. Germination was consistently high irrespective of the surface on which it was tested throughout the growing season.

Maximum germination was recorded in the third week just before the rainy period. Following this period germination declined gradually but again there was no significant difference in germination on the two surfaces though the figures for glass were more variable. Certainly there was no evidence to support any kind of host stimulation even during the sixth week when few spores were produced.

Expt 7. Comparison of germination on glass and leaf with inoculum from greenhouse plants

The above experiments failed to reproduce the results of Griffee (1967) and Jackson (1968). Since these two investigators worked with host material produced under artificial conditions, it was decided to raise black currant plants from stem cuttings (p.29) in the greenhouse and conduct experiments with these. To standardize the experiments the third or fourth leaf was always chosen 7-9 days after it unfolded for inoculation. Also to minimize variation leaves were always selected from as few plants as possible.

(i) Germination of conidia produced on detached leaves:

Leaves were inoculated in the usual manner and the mildew was allowed to develop for 2 weeks before the conidial germination was assessed on glass and leaf discs. The results (Table 9, Test 1) were similar to those obtained earlier with field-produced inoculum; germination in excess of 80% was recorded on both surfaces.

In a further, similar test the mildew was allowed to grow for 3 weeks before conidial germination was assessed. The results were similar (Table 9, Test 2).

Table 8. Percentage germination of *S. mors-uvae* conidia from lesions of different ages

Germination site	Age of lesion Week 1	Week 2	Week 3	Week 4	Week 5	Week 6
Glass	79.5	69.0	89.7	70.0	68.6	64.1
	(3.4)*	(7.4)	(1.7)	(2.8)	(6.3)	(12.0)
Leaf	74.5	72.2	80.0	77.0	75.6	69.1
	(8.0)	(7.9)	(4.6)	(5.6)	(8.2)	(8.1)

* Standard deviation based on percentage figures

Table 9. <u>Germination of *S. mors-uvae* conidia from cultures maintained on greenhouse-produced, detached leaves</u>

	Age of mildew (days)	Germination site	Mean % germination	S D
Test 1	14	Glass	87.4	2.8
		Leaf	84.6	7.6
	17	Glass	84.6	5
		Leaf	81.6	6.3
Test 2	21	Glass	77.6	7.4
		Leaf	73.0	7.3
	24	Glass	68.4	7.4
		Leaf	69.0	5.6

(ii) Germination of conidia produced on attached leaves

(a) In an attempt to produce conidia of low viability, three black currant plants were inoculated and placed under polyethylene bags to encourage maximum germination of the conidia. After 2 days, the bags were removed and the plants allowed to remain in the greenhouse away from the light. Mildew developed on these plants within 8-10 days and the germination of the young (24h) conidia from these plants was tested 2, 4 and 6 weeks after the plants were inoculated. At the two-week stage over 80% germination was recorded on both glass and leaf (Table 10). After 4 weeks however, the germination on glass had decreased to 48.9%, compared to 60.2% (mean values) on the leaf, and was more variable. This increased variability of conidia germinated on glass was particularly varied at the 6-week stage. During the sixth week although conidia were produced in sufficient numbers to test, cleistothecia were abundant on the infected leaves and conidial production ceased soon afterwards.

(b) In a similar experiment inoculated plants were kept in the greenhouse for 10 days only and then were transferred to the laboratory bench, near the window but not in direct sunlight. They were watered every second day. A thermograph alongside the plants recorded temperatures fluctuating between 14.5 and 21.5^{o}; the daily mean was about 18^{o}. Conidial germination was estimated just before and 5 and 12 days after the plants were transferred to the laboratory.

Although the plants appeared relatively healthy during this period they did not produce any new growth nor did the mildew colonies on them increase substantially.This may have been owing to the low moisture level in the centrally-heated laboratory. Conidial production was also affected; noticeably fewer conidia were produced even after 5 days in the laboratory and none were produced after 18 days.

There was a rapid decline in the germinability of the conidia produced under these conditions (Table 11). Before the plants were introduced

Table 10. Percentage germination of S. mors-uvae conidia produced on greenhouse plants

Age of mildew (weeks)	Mean percentage germination Glass	Leaf
2	82.0 (5)	85.1 (3.1)
4	48.9 (6.6)	60.2 (4)
6	57.9 (6.6)	62.2 (13.5)

S D in brackets

Table 11. Germination of *S. mors-uvae* conidia produced on plants at room temperature (Mean 18°)

Date	Germination site	Mean percentage germination	S D
29.10.1970	Glass	90.4	3.2
	Leaf	90.6	3.7
3.11.1970	Glass	72.3	15.5
	Leaf	65.2	12.2
10.11.1970	Glass	55.4 (a)	2.7
	Leaf	60.4	8.9

(a), t = 1.1

Table 12. Germination of *S. mors-uvae* conidia produced on plants at 29 ± 3^{o}

Date	Germination site	Mean percentage germination	S D
30.10.1970	Glass	88.8	2.3
	Leaf	89.5	2.9
1.11.1970	Glass	79.8	5.9
	Leaf	84.8	3.8
6.11.1970	Glass	61.8	9.7
	Leaf	66.2	3.3

to the test situation the mean germination was over 90% on both glass and leaf. After 12 days in the laboratory (10.11.70) germination decreased to 55.4% on glass and to 60.4% on the leaf. Differences in germination on these two surfaces were not statistically significant.

(c) A further experiment was conducted on similar lines. Four plants were inoculated with conidia as described in (a) above and 10 days later they were transferred to an illuminated growth cabinet at $29^{o} \pm 3^{o}$ and conidial germination was assessed, 2 and 5 days after this transfer. The plants were watered daily.

The results (Table 12) were similar to those obtained in (b).

Expt 8. Production of conidia on potted plants maintained at Silwood Park and at South Kensington

The preceding studies show conclusively that the conidia of S. mors-uvae are capable of germinating in high proportions equally well on glass and leaf surface irrespective of their method of production. Yet, the work of Griffee (1967) and Jackson (1968) has demonstrated differences in the germinability of conidia on these two surfaces. In a final attempt to reproduce this effect the work of these two investigators was repeated exactly under similar conditions.

Twenty, 2-month-old, vigorously growing, potted black currant plants were subjected to a dense conidial inoculum. Within a 2 week period in the greenhouse mildew developed rapidly on all the young tissue and sporulated profusely. The plants were then divided arbitrarily into two groups. One group was transferred to the illuminated constant temperature (20^{o}) room at Silwood Park in which Griffee and Jackson kept their plants. The other group was left in the greenhouse at South Kensington and was regarded as a control. The experiment was set up in mid-November 1970. A comparison of germination on leaf discs and glass was made of the conidia produced on plants just before introduction to the test situation and thereafter at regular intervals over a 6-week period.

Although the conidial production appeared similar, the plants that were maintained in the greenhouse in South Kensington appeared greener and more vigorous than those maintained at Silwood Park. The results of this study are summarized in Table 13. They do not differ substantially from those from similar experiments in this investigation. In both situations the germination was appreciable initially (over 80%). In the control plants at South Kensington, this changed little throughout the 6-week period. A slight reduction in germination was observed in conidia from the plants at Silwood Park after only 10 days. But subsequently this level was maintained over the test period. In the final germination test 48h-old conidia were used as in Griffee's and Jackson's experiments. These, too, germinated satisfactorily (although germination of conidia from control plants was slightly lower than with 24h conidia) and there were no significant differences in germination on glass and leaf surfaces.

Expt 9. <u>A comparison of spore germination methods</u>

Following the results of the last experiment it was decided to compare spore germination methods of other workers with those employed in the present study to test their reliability. The methods examined were:

1) Germination of conidia on new glass slides hung vertically over water in sealed Kilner jars as described by Zaracovitis (1964 a).
2) Germination on coverslips cleaned by different methods (see below) and fixed by Vasalene to slides which were then incubated, over water, in Petri dishes sealed with Vasalene. Coverslips were cleaned:
   (a) with 'Chemico' (Griffee, 1967), see Appendix 4 p.229
   (b) with detergent (Stergene), followed by soaking in 5% acetic acid (Purkayastha and Deverall, 1965)
   (c) as described on p.30 (new coverslips only).

Table 13. Germination of *S. mors-uvae* conidia produced in two situations

Date	Germination site	Mean percentage germination	
		Silwood Park	South Kensington
14.11.1970	Glass	87.0 (4.0) (a)	87.0 (3.7) (c)
	Leaf	87.6 (4.0)	83.8 (6.0)
24.11.1970	Glass	63.2 (5.8)	91.2 (3.7)
	Leaf	67.6 (4.1)	86.4 (1.5)
2.12.1970	Glass	74.0 (5.5)	81.6 (1.6)
	Leaf	69.8 (3.1)	73.4 (4.2)
5.12.1970	Glass	- (-)	85.0 (3.0)
	Leaf	- (-)	81.0 (7.4)
18.12.1970*	Glass	68.0 (4.4) (b)	73.0 (5.1) (d)
	Leaf	67.2 (7.6)	78.0 (4.4)

S D in brackets

* 48 h conidia

(a), t = 0.2 (b), t = 0.2 (c), t = 1.0 (d), t = 1.6

3) Germination in Manners and Hossain chambers as described on p. 30.

4) Germination on leaf discs as described on p. 30.

Appropriately labelled coverslips and slides were randomly distributed among leaf discs and inoculated simultaneously with 24h conidia from infected plants maintained in the greenhouse at South Kensington. The conidia were detached by slight shaking of the plant and a simultaneous blast of air produced with the mouth as suggested by Zaracovitis (1964 a).

The experiment was later repeated separately using 48h conidia from plants maintained in the constant temperature room at Silwood Park and in the greenhouse at South Kensington.

The results of this study are summarized in Table 14. From these it is clear that high levels of germination can be obtained on leaf discs and on glass that was cleaned without the use of any chemicals irrespective of the method of incubation. Germination in some instances was low and variable with large standard deviations on glass that was cleaned either with 'Chemico' or with detergent/acetic acid. The pattern of conidial germination on glass cleaned by the different methods was generally similar with all inocula, but overall conidia from plants grown at Silwood Park germinated less well than those from the greenhouse-grown plants at South Kensington.

Cleaning with Chemico gave the poorest result although conidia from South Kensington germinated on glass cleaned by this method in appreciable numbers when incubated in Manners and Hossain humidity chambers. Condensation was always a problem on slides incubated in the Petri dish moist chambers, whereas there was little condensation in the Manners and Hossain chambers. This could account for the different amounts of germination. Variable results were obtained on coverslips cleaned with detergent/acetic acid method. Although this was a less satisfactory method the results obtained by this method were certainly better than those with Chemico method.

Table 14. Comparison of spore germination methods

Method	Mean percentage germination					
	Artificially-produced conidia			Field-produced conidia		
	24 h conidia	48 h conidia				
	Sth.Ken 5.12.70.	Sth.Ken 18.12.70.	Silwood Pk 18.12.70.	10.8.71.	12.8.71.	14.8.71.
Zaracovitis (1964)	72.8(8.5)	74.8(6.3)	68.2(6.8)	72.6(9.2)	73.0(13.6)	77.2(3.5)
Petri dish moist chambers						
Chemico-cleaned	53.4(8.6)	39.8(10.1)	39.6(11.6)	56.8(24.2)	64.8(13.6)	65.0(10.9)
Detergent/Acetic acid	79.6(7.0)	50.6(12.0)	61.8(10.1)	-	-	-
New coverslips	79.0(8.6)	77.7(7.2)	72.2(4.1)	-	-	-
Manners & Hossain chambers						
New coverslips	85.0(3)	73.0(5.1)	68.0(4.4)	80.0(2.0)	86.8(4.7)	87.2(3.5)
Chemico-cleaned	-	63.2(14.0)	35.2(11.5)	-	-	-
Leaf disks	81.0(7.4)	78.0(4.4)	67.2(7.0)	80.2(3.2)	87.2(3.1)	89.0(2.1)

S D in brackets

Contrary to Jackson's findings Zaracovitis' method gave good germination results, almost as good as on new coverslips in Manners and Hossain humidity chambers. Condensation vapour was also found on these coverslips but obviously they did not interfere with the germination of the conidia.

Conidia incubated in the Manners and Hossain chambers on new coverslips produced the highest germination percentages with the lowest standard deviations.

The above results suggest that the factors which lead to poor and variable germination on glass are the methods of cleaning the glass and condensation of water vapour on the glass in certain types of incubation chambers. Conidial germination may therefore be inhibited either by traces of chemicals or by water itself, which is known to have a deleterious effect on powdery mildew conidia (Zaracovitis, 1964 a,b; Corner, 1935; Yarwood, 1936; Manners and Hossain, 1963). The results do not provide any evidence of increased germination on leaf surface which might be considered a stimulation of germination; on the contrary, they suggest that in some instances, on glass germination is inhibited.

This experiment was repeated in August 1971 using conidia produced in the field at Silwood Park. It was repeated three times at two day intervals using inocula from the same source maintained on detached leaves at 17°.

The results (Table 14) agree with what has already been established, in that consistently high germination can be obtained on glass cleaned without the use of any chemicals.

<u>Erysiphe cichoracearum</u>

Studies on burdock mildew were undertaken to find out if the conditions under which conidia were produced would influence their ability to germinate. <u>E. cichoracearum</u> on Burdock is useful for such work because both host and fungus can be easily raised under artificial conditions in a relatively short time. The burdock plants were grown in the greenhouse as described on p.29. The mildew was maintained on detached leaves at 17°. Two preliminary tests were made using 24h conidia to standardize the techniques. These are briefly described below.

Expt 10. Time to assess germination

Coverslips dusted with conidia were incubated at 20° and conidial germination assessed at 3,6,12,24 and 48h intervals.

The results (Table 15) show that maximum germination is reached after 24h.

Expt 11. Spore density on germination

The effect of conidial density on germination was studied as described on p.36 . As the results in Table 16 demonstrate, at the spore densities that were tested there were no marked differences in spore germination.

In all subsequent studies with E. cichoracearum about 10-20 conidia per microscope field (x 100) were used and their germination was assessed after 24h.

Expt 12. Effect of different temperature regimes on conidia germination

Two-week old burdock leaves were detached from greenhouse - maintained plants and were inoculated with conidia produced in the field. The inoculated leaves were re-distributed and placed in three germination trays with approximately 15 leaves to a tray. The trays were then covered with their lids and maintained at the following temperature regimes.

1. 17° refrigerated growth cabinet with a 16h photoperiod.
2. room temperature with alternating light and dark donditions; the tray was kept on the laboratory bench and a thermograph was placed alongside.

Table 15. Germination of *E. cichoracearum* conidia after different incubation times

Hours incubation	No. conidia counted	Mean percentage germination
3	988	4.9
6	961	15.6
12	929	31.6
24	935	87.4
48	1613	81.1

Table 16. Effect of spore density on germination of *E. cichoracearum*

Mean No. conidia per field (x 100)	19.2	26.8	41.8	62.4
Mean % germination	87.7	88.5	87.0	89.0

3. 25° (constant temperature room) with continuous 'poor' (170 lumens/sq.ft.) light provided by two 65-80W flourscent tubes.

At 17° mildew developed rapidly, 8-9 days after inooulation, but the leaves remained turgid and in otherwise reasonable condition. At room temperature (19-25.5°), mildew also developed well but the leaves fairly soon began to yellow. No mildew developed at 25° and the leaves themselves died in about 7-10 days. Tests on germination were therefore confined to conidia produced on leaves at 17° and at room temperature. The germination of these conidia on glass coverslips in Manners and Hossain chambers was examined daily for five consecutive days. Usually at each sampling time 4-5 leaves picked at random provided sufficient inoculum. Afterwards leaves were returned to the regimes but conidia on all the leaves were blown-off to ensure the production of conidia of the same age for subsequent tests. This experiment was repeated three times using different leaf material. The results are summarized in Tables 17-19 for each replicate experiment.

The two temperature regimes under whioh the conidia were produced had no effect on their viability. In all tests germination was consistently high, over 90% in most, and there was no noticeable decrease in germination over the 5-day period. However, it was observed that with the increase in age of the mildew culture, fewer conidia were produced though this was not estimated quantitatively.

Next, the germinability was examined of conidia produced on leaves at $13.5 \pm 1^{\circ}$, 20° and 23.5° in growth cabinets each with a 16h photoperiod.

At 13.5° the development of the mildew was delayed and visible sporulation was noted about 10-12 days after inoculation. Hence the conidial samples were taken 14 days after inoculation when sufficient spores were produced. Spores from the other two temperature regimes were

Table 17. Germination of *E. cichoracearum* conidia produced under two different temperature regimes

Sampling date	Temperature regime	No.conidia counted+	No.conidia germinated	Mean % germination
26.9.1969	RT*	761	632	83.0
	17°	1417	1372	96.8
27.9.1969	RT	744	690	92.7
	17°	2487	2316	93.1
28.9.1969	RT	739	699	94.5
	17°	2458	2358	95.9
29.9.1969	RT	1229	1121	91.2
	17°	2674	2492	93.1
30.9.1969	RT	1762	1694	96.1
	17°	2237	2066	92.3

+ Total of five replicates

* Room temperature; range 19 - 25.5°, mean 23°

Table 18. Germination of *E. cichoracearum* conidia produced under two different temperature regimes

Sampling date	Temperature regime	No. conidia counted+	No. conidia germinated	Mean % germination
1.10.69	RT*	1741	1603	92.0
	17°	1361	1292	94.7
2.10.69	RT	1032	990	95.9
	17°	861	800	92.9
3.10.69	RT	1202	1131	94.1
	17°	2371	2150	90.6
4.10.69	RT	204	188	92.1
	17°	2505	2360	94.2
5.10.69	RT	318	284	89.3
	17°	3000	2689	89.6

\+ Total of five replicates

* Room temperature; range 17° - 24°, mean 21°

Table 19. Germination of *E. cichoracearum* conidia produced under two different temperature regimes

Sampling date	Temperature regime	No. conidia counted[+]	No. conidia germinated	Mean % germination
13.10.69	RT[*]	421	390	92.6
	17°	974	904	92.8
14.10.69	RT	186	161	86.5
	17°	1555	1507	96.9
15.10.69	RT	647	568	87.7
	17°	1711	1621	94.7
16.10.69	RT	229	215	93.8
	17°	1780	1587	89.1
17.10.69	RT	971	836	86.1
	17°	1532	1431	93.4

\+ Total of five replicates

* Room temperature; range 19° - 24.5°, mean 21.5°

also sampled at the same time and the tests were repeated on three consecutive days. Although the sporulation was delayed at 13.5^{o} the conidia produced at this temperature regime had marginally the highest germination rate (Table 20), but there were no marked differences in germination of conidia produced in the different regimes.

Expt 13. <u>Assessment of spore production on detached leaves</u>

Detached leaves (4th leaf 10 days after unfolding) were inoculated with conidia and divided into two groups; one was incubated at 17^{o}, the other at room temperature as described on p58 . The mildew was allowed to develop for 12 days and then old conidia were removed by blowing with a Pasteur pipette over the leaves. Twenty four hours later, three leaves per tray were carefully picked up at random and 1.6 cm diam discs were cut from well-mildewed areas using a cork borer. About 5-7 discs were removed from the three leaves at each sampling. This was done in a draught-free room, extra care being taken to ensure that the leaves bearing sporulating colonies of the mildew were disturbed as little as possible during the operation. These leaves were then discarded but conidia on the remaining leaves were blown-off to ensure the production of conidia of uniform age for the following day.

The mildewed leaf discs were carefully placed in M^{c}Cartney bottles using forceps and shaken in 10ml 0.01% Tween 80 (in distilled water) in a mechanical shaker for 10 min at half speed. The leaf discs were then removed, the conidial suspension was concentrated by centrifugation at 3000 r.p.m. for 10 min and then the conidia were finally resuspended in 0.5ml distilled water. The number of conidia was assessed in each sample from five replicate counts in a haemocytometer. Conidial production at these two temperature regimes was assessed over a 5-day period.

Although the main aim of this experiment was to study the effect of temperature on conidial production and the results (Table 21) do

Table 20. <u>Germination of *E. cichoracearum* conidia produced under three different temperature regimes</u>

	Mean percentage germination		
Sampling date	13.5^{o}	20^{o}	23.5^{o}
23.4.70	96.7	92.3	91.2
24.4.70	92.1	88.1	84.8
25.4.70	92.4	87.9	89.3

Table 21. Production of *E. cichoracearum* conidia under two different temperature regimes

	No. conidia (thousands) / cm^2 infected tissue				
Day No.	1	2	3	4	5
17^{o}	38.8	41.2	33.0	32.7	35.8
Room temp.($17-24^{o}$)	15.5	16.3	17.1	7.8	-

indicate substantial differences in this respect between the two regimes, it is unlikely that temperature alone is responsible for these. Not only was lesion development sparse on leaves at room temperature (17-24°), whereas leaves at 17° became fully covered with mildew, but also within colonies the production of conidial chains was less dense than on leaves at 17° as is shown by the relative amounts of conidia in Table 21. Additionally, leaves at room temperature appeared to senesce more rapidly. Particularly towards the latter part of the experiment conidial production at room temperature fell rapidly and ceased altogether whereas at 17° it continued unaltered. Probably in this situation light was also a factor limiting the production of conidia.

Other mildews

Expt 14. The germination of conidia produced under different temp/light regimes was also studied with the following mildew:

Erysiphe graminis on barley

Sphaerotheca fuliginea on cucumber

Uncinula necator on vine

Barley seedlings (about 15-25 per pot) were inoculated 6-8 days after sowing the seed, cucumber (1 plant per pot) after 5-6 weeks and vine (1 plant per pot) 2-3 months after planting a rootstock.

The plants were inoculated with conidia of the appropriate mildew and kept under polyethylene covers in the greenhouse for 24h to ensure maximum spore germination. The polyethylene covers were then removed and the plants left for 7 days when mildew growth became visible. Plants were then transferred to the following regimes.

1. 14° illuminated growth cabinet
2. A non-refrigerated growth cabinet with 16h photoperiod
   A thermograph placed inside the cabinet before the experiment began recorded above 27° when the lights

were turned on. The daily temperature inside the cabinet ranged from a little above room temperature (19 ± 2^{o}) to 33^{o}. This cabinet is referred to as the high temperature cabinet.

3. Greenhouse where the daily average temperature was 16^{o}.

In each regime there were: Barley, 1 pot with 20 seedlings; Cucumber, 3 plants; Vine, 2 plants. The pots were placed in trays containing water. After 5 days, old conidia were removed and the germination of the young (24h) conidia was then tested on glass and on leaf surface in the usual manner. The tests were repeated two days later.

The mildews developed best on plants in the greenhouse; the young tissue of all plants were covered with profusely-sporulating colonies before the sampling began. Despite this, the host plants remained in reasonable condition and produced new leaf tissue during this period. At 14^{o} there were no signs of sporulation of U. necator on vine but at high temperature, this sporulated profusely. E. graminis also grew and sporulated well in the high temperature cabinet but cucumber leaves became brittle after a few days.

The results of the germination tests are given in Table 22. Conidia of all species produced at 14^{o} and in the greenhouse germinated well. So did those of U. necator produced in the high temperature cabinet but not conidia of E. graminis and S. fuliginea. The percentage germination of these was less than that of conidia produced at the lower temperatures. Additionally, with E. graminis germination of conidia on glass was markedly lower than on the leaf.

This last result with E.graminis is the only piece of evidence in this present study to support increased germination of powdery mildew conidia on leaf surface. The experiment was therefore repeated with slight modifications using only E. graminis.

Two batches of 6-day old barley seedlings in two separate pots were inoculated with conidia kept under polyethylene bags for 24h. The bags

Table 22. Germination of powdery mildew conidia produced in different environments

Mildew species	Germination site	Mean percentage germination					
		14°		Greenhouse		High temperature	
		12.10.70	14.10.70	12.10.70	14.10.70	12.10.70	14.10.70
E. graminis	Glass	90.8	91.6	91.6	82.6	42.6	43.4
	Leaf	90.8	89.6	91.2	89.0	63.4	74.8
S. fuliginea	Glass	73.4	89.0	77.6	-	50.6	48.6
	Leaf	85.2	87.8	86.6	-	47.0	53.6
U. necator	Glass	-	-	89.6	-	92.2	95.2
	Leaf	-	-	93.0	-	92.4	90.6

Table 23. Germination of E. graminis conidia produced under two different environments

	Germination site	Mean percentage germination				
		28.10.70	31.10.70	2.11.70	4.11.70	6.11.70
High temperature cabinet	Glass	73.8	55.0	45.2	42.0	57.4
	Leaf	75.8	71.0	65.0	62.4	59.0
Greenhouse	Glass	76.0	73.6	77.0	74.2	73.0
	Leaf	75.2	76.0	76.6	77.0	78.4

were then removed and the plants left in the greenhouse for 7 days until sporulating colonies appeared. The germination of conidia produced on the two batches of plants was estimated . Subsequently, one batch of plants was transferred to the high temperature cabinet, the other was left in the greenhouse. Germination of the conidia was then assessed after 3 days and at 2-day intervals until conidial production reached low levels.

The results are summarized in Table 23. At the beginning of this experiment (28.10.70) conidial germination from both batches of plants was high-over 75% on glass and on leaf. This remained relatively unchanged in the greenhouse-maintained plants. But the germination of the conidia produced in the high temperature regime was reduced and this reduction in germination was more pronounced on glass than on the leaf surface. This same result was consistently obtained until the sporulation ceased.

A high percentage of conidia produced under high temperature was shrivelled but these were excluded from the germination counts. Also, it was observed that some of the non-germinated conidia on glass were surrounded by a halo of exudates, possibly due to some bacterial growth. These were ofcourse not observed in the celloidin peels removed from the leaves but thought to shrivel up when celloidin was placed on the leaves. Since shrivelled conidia were excluded from counts this could perhaps explain the higher percentage figures obtained on the leaves.

(2) The leaf surface

The results of the preceding, sections demonstrate that conidia of S. mors-uvae and E. cichoracearum can germinate in high percentages, often 80-90% on glass surfaces. The increased germination on leaf surface as reported by other workers was not found. The results further suggest that with the increase in age of the mildew colonies and their hosts, the amount of germination of their conidia decreased only slightly. This work was extended to cover a wide range of powdery mildews collected from

Silwood Park during the months of July, August and September 1970 and their conidial germination on glass was compared with that on their leaf surfaces. A few greenhouse-maintained mildews such as <u>E. graminis</u> on barley and <u>U. necator</u> on vine were also included in this study.

Method: The mildewed leaves were maintained in germination trays at 17°. The old conidia were removed and the leaves were incubated for 24h to produce a new crop of conidia. The percentage germination was assessed on glass and on leaf surfaces. Whenever possible the tests were carried out for several days using the same source of inoculum until either the conidial production ceased or the leaves began to senesce. Often, tests were repeated several times throughout the growing season to see if there were changes in the amount of germination with increase in maturity of the host plants in the field.

The following powdery mildew fungi and their hosts were used. The identifications were checked in the host plant indexes published by Blumer (1933, 1967).

1) <u>Erysiphe graminis</u> on Barley
2) <u>Erysiphe cichoracearum</u> on Burdock (<u>Arctium lappa</u>)
3) <u>Erysiphe cichoracearum</u> on Michelmas Daisy (<u>Aster</u> sp)
4) <u>Erysiphe horridula</u> on Forget-me-not (<u>Myosotis</u> sp)
5) <u>Erysiphe fisheri</u> on Groundsel (<u>Senecio vulgaris</u>)
6) <u>Erysiphe martii</u> on Red Clover
7) <u>Erysiphe polygoni</u> on Knotgrass (<u>Polygonum aviculare</u>)
8) <u>Erysiphe polygoni</u> on <u>Delphinium</u>
9) <u>Erysiphe polygoni</u> on Pea (<u>Pisum sativum</u>)
10) <u>Erysiphe lamprocarpa</u> on Plantain (<u>Plantago major</u>)
11) <u>Erysiphe galeopsidis</u> on Hedge woundwort <u>Stachys sylvaticus</u>)
12) <u>Erysiphe galeopsidis</u> on Deadnettle (<u>Lamium purpureum</u>)

13) *Erysiphe artemesiae* on *Artemesia* sp

14) *Erysiphe communis* on Turnip

15) *Erysiphe communis* on Swede

16) *Erysiphe heraclei* on Parsnip

17) *Oidium* sp on Marrow

18) *Oidium begoniae* on Begonia

19) *Microsphaera* sp on Oak (*Quercus* sp)

20) *Podosphaera leucotricha* on Apple

21) *Sphaerotheca mors-uvae* on Black currant

22) *Sphaerotheca mors-uvae* on Gooseberry

23) *Sphaerotheca fuliginea* on Cucumber

24) *Sphaerotheca pannosa* on Rose

25) *Uncinula necator* on Vine

26) *Uncinula bicornis* on Sycamore (*Acer pseudoplatanus*)

Results: Except for a few species, a consistently high percentage germination was obtained with the conidia of the powdery mildews that were investigated. The full results are given in Appendix 5 p.230. Table 24 presents the maximum germination obtained. From these it can be seen that there were no marked differences in conidial germination between glass and leaf surfaces.

In contrast to other reports, except that of Zaracovitis (1964 a,b), mean conidial germination over 80%, sometimes as high as 95%, was recorded for most species studied. With the conidia of *E. lamprocarpa* from Plantain and *U. bicornis* from Sycamore a mean germination of *c.* 65% was consistently obtained but there was no evidence of increased germination on the leaf. Germination was poor on both substrates with conidia of *E. cichoracearum* from *Aster* sp, *E. artemesiae* from *Artemesia* sp and *Podosphaera leucotricha* from apple. In these instances a mean germination between 20-35% was obtained (see Table 24). Difficulties in

Table 24. Percentage germination of conidia of different powdery mildew fungi incubated at 20° in a saturated atmosphere for 24 h

Fungus	Source of conidia	Date	Germination site	Germination * Mean %	S D
1) Erysiphe graminis	Barley	1.7.70	Glass	89.4	6.0
			Leaf	85.1	2.4
2) E. cichoracearum	Burdock	25.8.70	Glass	96.2	2.4
	Arctium lappa		Leaf	96.7	1.5
3) E. cichoracearum	Michelmas Daisy	29.7.70	Glass	36.7	1.9
	Aster sp.		Leaf	39.2	2.2
4) E. horridula	Forget-me-not	29.7.70	Glass	96.0	2.8
	Myosotis sp.		Leaf	96.0	2.8
5) E. fisheri	Groundsel	11.8.70	Glass	86.4	4.1
	Senecio vulgaris		Leaf	89.8	1.9
6) E. martii	Red clover	12.8.70	Glass	92.8	1.9
			Leaf	91.6	3.0
7) E. polygoni	Knotgrass	11.8.70	Glass	85.8	6.4
	Polygonum aviculare		Leaf	97.2	1.9

Table 24 contd....

8) E. polygoni	Delphinium sp.	14.7.70	Glass Leaf	92.0 87.7	4.5 7.4
9) E. polygoni	Pea Pisum sativum	2.9.70	Glass Leaf	82.5 85.5	5.1 5.5
10) E. lamprocarpa	Plantain Plantago major	2.9.70	Glass Leaf	64.0 64.5	3.6 3.0
11) E. galeopsidis	Hedge woundwort Stachys sylvaticus	2.9.70	Glass Leaf	86.0 85.0	2.8 3.1
12) E. galeopsidis	Deadnettle Lamium purpureum	9.9.70	Glass Leaf	78.6 81.4	4.2 3.3
13) E. artemesiae	Artemesia sp.	9.9.70	Glass Leaf	20.2 25.2	2.8 5.4
14) E. communis	Turnip	2.9.70	Glass Leaf	94.7 95.5	2.3 2.3
15) E. communis	Swede	2.9.70	Glass Leaf	95.0 96.7	2.9 1.8
16) E. heraclei	Parsnip	9.9.70	Glass Leaf	97.0 97.8	2.0 1.4
17) Oidium sp.	Marrow	5.9.70	Glass Leaf	95.0 92.6	2.2 2.7
18) O. begoniae	Begonia	2.9.70	Glass Leaf	92.7 95.5	2.7 3.1

Table 24 contd....

19) *Microsphaera* sp.	Oak *Quercus* sp.	19.7.70	Glass Leaf	97.0 98.5	3.8 1.0
20) *Podosphaera leucotricha*	Apple	29.9.70	Glass Leaf	23.5 30.0	3.1 2.0
21) *Sphaerotheca mors-uvae*	Black currant	19.7.70	Glass Leaf	88.2 83.7	6.8 6.0
22) *S. mors-uvae*	Gooseberry	26.7.70	Glass Leaf	87.0 87.5	5.2 1.2
23) *S. fuliginea*	Cucumber	25.7.70	Glass Leaf	87.2 90.5	6.3 4.2
24) *S. pannosa*	Rose	9.9.70	Glass Leaf	87.2 89.6	1.3 3.1
25) *Uncinula necator*	Vine	25.8.70	Glass Leaf	94.7 97.0	0.9 1.8
26) *U. bicornis*	Sycamore *Acer pseudoplatanus*	5.9.70	Glass Leaf	66.6 67.0	5.2 4.8

* Average of 4 - 5 replicates

obtaining a good germination with P. leucotricha have also been reported by Zaracovitis (1964 b) although he obtained a value between 50 and 70% for this fungus.

(3) Physical factors and conidial germination

The effects of various environmental factors on the behaviour of powdery mildew conidia are known to vary with the different species (Yarwood, 1957; Schnathorst, 1965). Temperature and relative humidity are among the more important environmental factors which influence conidial germination. Hence the influence of these factors and their interactions on the behaviour of conidia of E. cichoracearum, S. mors-uvae and S. pannosa were investigated. The mildews for this study were collected from naturally-infected plants grown at Silwood Park during the summer months of 1969 and 1970. The infected leaves were maintained in germination trays at 17° and the conidia were sampled within 1-2 days after they were brought to the laboratory. The old conidia were removed 24h before the tests were conducted.

The germination studies were carried out mainly on glass coverslips but comparative tests were also carried out in vivo, particularly at different relative humidities. In all studies percentage germination was estimated after 24h. In some experiments, the germ-tube length was also measured.

Erysiphe cichoracearum (from Arctium lappa)

Expt 15. Effect of temperature

Coverslips dusted with conidia were incubated at temperatures ranging from 0-35°. The results are given in Table 25. No germination occured at 5° or 35°. At 5° although the conidia did not germinate they appeared turgid whereas at 35°, they had shrivelled. A sharp increase in germination was observed at 12° and at 15°, over 50% of the conidia had

Table 25. Mean percentage germination of *E. cichoracearum* conidia on glass at different temperatures

Temperature (^{o}C)	5	9	12	15	17	20	22	24	27	30	33	35
Percentage germination	0	7.7	18.6	57.2	81.7	91.8	94.2	94.6	92.6	89.6	5.6	0

germinated. The optimum temperature for germination was between 22° and 27° where a mean of over 90% was recorded. Above 30°, germination decreased rapidly. At 30°, although nearly 90% of the conidia had germinated, nearly all were shrivelled and appeared dead. A conidium was considered to be shrivelled when it had lost its turgidity. In the germinated conidia, shrivelling always took place from the main body of the conidium and not from the germ tube. A little shrivelling, about 8-9%, was also evident at 24°, well within the optimum temperature range. At 22° however, no shrivelling was observed.

Expt 16. Effect of light

Coverslips dusted with conidia were incubated in growth cabinets at 20°, 22° and 24° with a 16h photoperiod. Replicate batches were incubated in the dark (placed inside empty biscuit tins and closed with lids) in the same cabinets.

The results (Table 26) indicate that the light supplied in this experiment did not effect conidial germination.

Expt 17. Effect of relative humidity

The conidia were incubated at 20° under different relative humidities and the percentage germination was estimated after 24h.

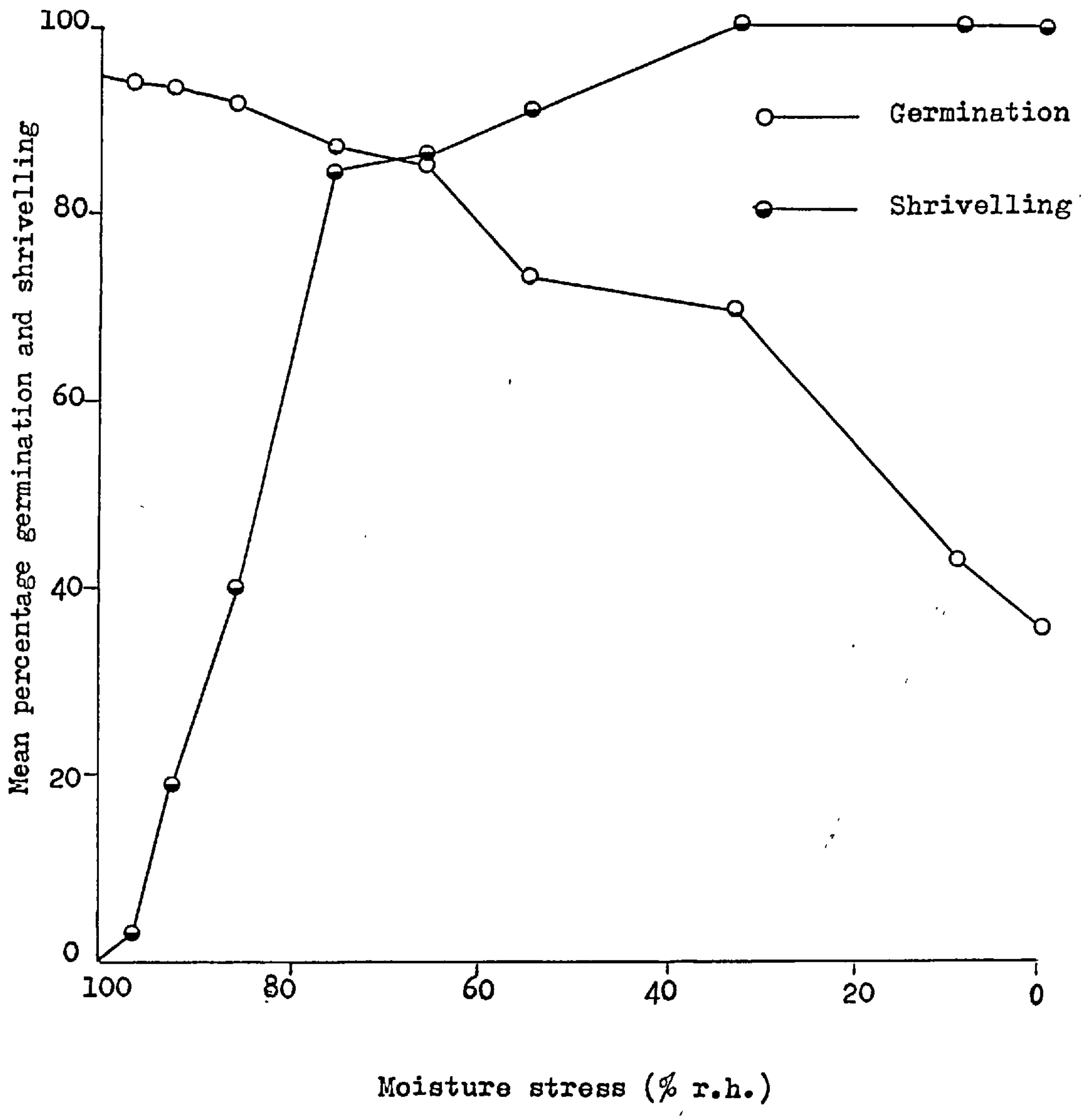
E. cichoracearum conidia germinated in appreciable quantities at low relative humidities. Even at 0% r.h. a mean germination of 35.8% was obtained and at 55%r.h. over 70% of the conidia germinated (Fig.3). Germination was optimal between 97% and 100% r.h.

At these moisture levels, there was little or no shrivelling of the conidia. At moisture levels below these however, some conidia shrivelled either without germinating or during germination. Thus at 93% r.h. a mean of nearly 20% of the germinated conidia had shrivelled. At moisture levels below 93% shrivelling increased rapidly (see Fig. 3) and

Table 26. Effect of light on the germination of *E. cichoracearum* conidia

Temperature (°C)	Mean percentage germination	
	Light	Dark
20	92.5	89.2
22	95.7	90.4
24	97.3	94.4

Fig.3.Effect of moisture stress on germination and shrivelling of
E. cichoracearum conidia at 20°.



at 55% r.h. although a mean germination over 70% was obtained more than 90% of the germinated conidia were shrivelled. At 33% r.h., there was 100% shrivelling of both germinated and non-germinated conidia.

In this experiment the germ tube length was not measured but shorter germ tubes were observed in germinated conidia at low moisture levels. This suggest that the conidia of E. cichoracearum were capable of germinating at low moisture levels but their further growth was arrested due to shrivelling of the germ tubes.

Expt 18. Interaction of temperature and relative humidity

Coverslips dusted with conidia were incubated at 0,34,76,86,97 and 100% r.h. and at temperatures ranging from 5° to 30°. After 24h percentage germination and shrivelling of the conidia were assessed. The germ tube length was also measured.

From results summarized in Table 27 a number of conclusions can be made. First, the results show that the conidia can germinate over a wide range of temperature and relative humidities. However, appreciable germination took place between 15° and 25° at moisture levels above 34% r.h. There was no germination at 5° even at high moisture levels and at 30°, although germination was satisfactory at 100% r.h., there was little or no germination at moisture levels below 97% r.h.

Second, the results show that the shrivelling was more prominent at low moisture levels particularly at higher temperatures. This was observed on both germinated and non-germinated conidia. Above 25°, all the conidia shrivelled irrespective of moisture stress but at lower temperatures, shrivelling was evident only at low moisture levels.

The results further suggest that the germ tube growth was influenced to a greater extent by the temperature. The germ tube growth was limited at low temperatures presumably due to its slow growth but at 30° germ tube growth was arrested due to the shrivelling of the conidia. At

Table 27. Interaction of temperature and relative humidity on germination of *E. cichoracearum* conidia

Temperature (°C)		% RH 0	34	76	86	97	100
5	A	0	0	0	0	0	0
	B	100	84.0	31.0	0	0	0
	C	0	0	0	0	0	0
10	A	17.7	18.9	16.9	17.0	17.6	23.0
	B	100	87.1	35.9	13.0	0	0
	C	3.9	7.2	7.1	8.7	8.3	10.3
15	A	22.9	70.9	77.7	83.3	84.4	75.5
	B	100	90.6	32.9	31.5	0	0
	C	4.1	9.7	14.4	19.7	19.6	21.0
20	A	23.6	67.3	90.1	93.0	92.2	93.9
	B	100	100	85.3	43.6	0	0
	C	6.3	9.2	12.9	16.8	19.5	24.5
25	A	38.5	63.3	85.3	92.4	88.8	93.3
	B	100	100	96.6	93.4	21.7	10
	C	4.8	11.9	15.2	16.9	22.9	22.7
30	A	0	0	4.0	8.8	16.2	67.4
	B	100	100	100	100	100	100
	C	0	0	3.7	4.6	6.8	8.6

A = % germination; B = % shrivelling; C = germ tube length, µm (Mean of 20 germinated conidia) temperatures between 15° and 25° this was similar at all relative humidities. At low moisture levels within this temperature range germ tube growth was checked due to desiccation and the germ tubes appeared shrivelled. Although at low relative humidities the germ tubes were killed soon after their initiation, the results suggest that the rates of germination (the time required for the germ tubes to appear) were similar at these moisture levels. The germ tubes length was at a maximum at 100% r.h. at 20°.

Sphaerotheca mors-uvae (from Black currant)

Expt 19. Effect of temperature

Conidia were incubated on glass coverslips at 17°, 20°, 22° and 25° and the percentage germination was assessed after 24h. The optimum temperature for germination was around 22° (Table 28) where a mean of over 80% was obtained without any signs of shrivelling. At 25°, about 8-12% of the germinated conidia were shrivelled. Replicate batches of conidia incubated in the dark at the same temperature showed that there were no significant differences in germination with those incubated in the light.

Expt 20. Effect of relative humidity

The conidia were incubated at 20° at 97%, 98% and 100% relative humidities. The results (Table 29) show that below 98% r.h., little or no germination occured but at 100% r.h., a mean of over, 80% was obtained. Obviously, the relative humidity is a critical factor for the germination of conidia of S. mors-uvae.

Sphaerotheca pannosa (from Rose)

Expt 21. Effect of temperature

As for the other two species, the conidia of S. pannosa were dusted on to coverslips and incubated at 18°, 20° and 25° and the percentage germination assessed after 24h. The results (Table 30) suggest that these

Table 28. Effect of temperature on conidial germination of <u>S. mors-uvae</u> (100% r.h.)

Temperature (^{o}C)	Mean percentage germination	
	Light	Dark
17	68.6	69.4
20	78.0	78.6
22	84.4	80.0
25	78.4	78.0

Table 29. Effect of relative humidity on germination of *S. mors-uvae* conidia (20°C)

	Conidial germination	
% R H	% Mean	S D
97	1.2	0.8
98	4.8	5.4
100	81.6	7.6

Table 30. Effect of temperature on the germination of <u>S. pannosa</u> conidia (100% r.h)

Temperature (^{o}C)	conidial germination % Mean	S D
18	81.4	3.9
20	83.2	5.1
25	80.0	4.4

three temperatures included the optimum range where over 80% of the conidia germinated. At 25° however, nearly 70% of the germinated conidia were shrivelled. No shrivelling occurred at the other two temperatures.

Expt 22. Effect of relative humidity

Of the three relative humidities at which tests were carried out, maximum germination occurred at 100% (Table 31). The results show that germination decreased rapidly from 84% at 100% r.h. to 7.2% at 97% r.h. At 93% r.h., there was no germination and nearly 70% of the conidia (ungerminated) were shrivelled.

Expt 23. Comparison of conidial germination on glass surface with that on leaf discs

Many workers have reported on the enhanced germination of powdery mildew conidia on leaf surfaces as compared to glass and have suggested that this occurrence was due to the more favourable moisture conditions maintained at the leaf surface (Frampton and Longree, 1941; Delp, 1954; Weinhold, 1961). But the results of this study have shown that the conidia of E. cichoracearum, S. mors-uvae and S. pannosa can germinate well on glass between 20° and 22° at humidities approaching saturation. At low moisture levels however, germination was either reduced or ceased altogether and this was particularly evident at higher temperatures. Conidia that germinated at very low relative humidities eg. E. cichoracearum, often shrivelled and died. Their tolerance to low humidity decreased with the increase in temperature (Table 27). Since these studies were done on glass, it was decided to investigate the effect of the host leaf surface on the germination of the conidia of the three species of powdery mildews.

Leaf discs were used for this purpose and the conidial germination on these was compared with that on glass. Tests were carried out only at a selected range of relative humidities at 20°.

Table 31. Effect of relative humidity on the germination of *S. pannosa* conidia (20°)

R H (%)	conidial germination % Mean	S D
93	0	0
97	7.2	3.7
100	84.0	3.6

The leaf discs were not floated on the solutions controlling the relative humidity as it was feared that the salt solutions might bring about changes in the leaf tissue that would mask any effects the surrounding air might have on the conidia. They were therefore placed on coverslips attached to the lids of the Manners and Hossain humidity chambers and incubated in these. A control was set up for the 100% r.h. where the discs were floated on water. No significant difference in spore germination was observed on discs incubated 'dry' at 100% r.h. and those that were floated on water.

All discs at the end of the 24h incubation appeared 'healthy' except for those maintained at 0% r.h. which had withered, presumably due to the lack of water in the surrounding air.

Erysiphe cichoracearum

The results indicate that the conidia of E. cichoracearum germinated under very low moisture levels (Table 32). Although the amount of shrivelling was not estimated, it was observed that both non-germinated and germinated conidia were shrivelled after 24h exposure to low moisture levels. At 76% r.h. conidia germinated satisfactorily on glass but their germination on the leaf was significantly higher. At 100% r.h. on the other hand, equally good germination was obtained on both glass and leaf. The results therefore suggest that in atmospheres of high relative humidity, the conidia of E. cichoraoearum germinated in high percentages, often over 90%, and that their germination was not influenced by the substratum. The increased germination on the leaf at lower moisture levels, suggest that this was due to the existence of a more favourable microclimate on the leaves which was different from the surrounding atmosphere. Comparatively low germination on the leaf under very dry conditions eg. 0% r.h. was probably due to the leaf itself being killed due to desiocation.

Table 32. <u>Effect of relative humidity at 20° on the germination of</u> <u>*E. cichoracearum* conidia</u>

Relative humidity (%)	Mean percentage germination	
	Leaf	Glass
0	33	33
33	54	42
76	86	65
100	94	89

Sphaerotheca mors-uvae

The moisture requirements for conidial germination of S. mors-uvae were very much different from those of E. cichoracearum. The results presented in Table 33 show that the conidia of S. mors-uvae germinated satisfactorily only above 98% r.h. Higher germination on the leaf at this moisture level suggests the existence of an improved moisture condition at the leaf surface. At 100% r.h. maximum germination was obtained on both glass and leaf.

Sphaerotheca pannosa

The maximum percentage germination of the conidia of rose powdery mildew was also obtained at 100% r.h. (Table 34). At this moisture level, over 90% of the conidia germinated. On glass, germination decreased sharply to 5.8% at 97% r.h., and below this, germination ceased altogether. On leaf on the other hand, the decrease in percentage germination was gradual with the increase in moisture stress. Even at 9% r.h. 12.5% of the conidia had germinated. Although the conidia germinated on leaf at very low relative humidities there was a marked increase in percentage shrivelling of the germinated conidia. But the fact that conidia germinated on leaf at low moisture levels suggests that the microolimate at the leaf surface was different from the ambient atmosphere even for a short period during which time germination was initiated. These results therefore provide further evidence in support of the hypothesis that better germination of conidia on the leaf is due to favourable moisture conditions maintained at the leaf surface.

Table 33. Effect of relative humidity at 20° on the germination of *S. mors-uvae* conidia

Relative humidity (%)	Mean percentage germination	
	Leaf	Glass
97	0.8	0.4
98	22.8	2.8
100	84.6	87.2

Table 34. **Effect of relative humidity at 20° on the germination of *S. pannosa* conidia**

Germination site		Approximate relative humidity (%)								
		9	33	55	66	76	86	93	97	100
Glass	A	0	0	0	0	0	0	0	5.8	94.0
	B	100	100	100	100	100	100	75	0	0
Leaf	A	12.5	27.5	35.0	54.8	60.1	63.1	64.5	69.5	91.4
	B	100	95	50	40	25	18	10	0	0

A = Mean % germination

B = Approximate % shrivelling

DISCUSSION

Contrary to many previous reports of poor germination of powdery mildew conidia on glass and their enhanced germination on host leaves, the results of this investigation indicate that powdery mildew conidia germinate well on both surfaces. The results also indicate that there are no significant differences in germination on these two surfaces. In this respect, the results agree with those of Zaracovitis (1964 a,b).

Griffee (1967) obtained only 9% germination of S. mors-uvae conidia from black currant on glass compared with 50% on leaf discs under similar conditions. Jackson (1968) also recorded poor and variable levels of germination of conidia of the same fungus on glass compared with consistently high (about 60%) germination on leaves. In this study however, over 80% germination of the conidia of this fungus and several other mildew fungi was obtained on glass. In the light of this study, the differences in germination noted by previous investigators seem to be mainly due to the lack of standardization of some factors of great importance in ensuring high germination on glass.

Although most workers have realized the importance of basic factors that influence conidial germination such as the use of youngest conidia, avoiding the use of instruments in detaching the conidia, incubating conidia in saturated atmospheres, ensuring they do not come in contact with free water, incubating at optimum temperatures at least for 24h etc., certain other factors (discussed below) are still overlooked by many.

Powdery mildew fungi being obligate parasites totally dependent on their hosts for their existence are affected by the general state of their host's growth. Any changes which the environment may induce in the host is likely to influence the growth and reproduction of the parasite and the subsequent behaviour of the parasitic units (inoculum). The results have shown that one of the most important factors that should not be overlooked in studies concerning germination of powdery mildew conidia is the production of conidia of high germinability. In this connexion it is important to maintain the host plants supporting the mildews under good growing conditions. Also it is important to use inooula from newly infected plants, as plants that have supported the mildew for prolonged periods may produce conidia of low germinability. Conidia produced on plants maintained in the 20° constant temperature room provided with artificial light in which Griffee (1967) and Jackson (1968) kept their plants showed a slight but nevertheless significant reduction in germination compared to those produced on greenhouse plants where the latter were greener and appeared to grow more vigorously. Although mildew cultures were maintained on newly developed leaves after stripping - off the old ones, the plants used by Jackson (1968) in his studies could not have been in the most healthy state desired. These plants had borne the mildew continuously for over 12 months in addition to being attacked by red spider. This could perhaps partly explain why he obtained a poor inoculum. Jackson however obtained higher germination on glass with conidia produced on young plants that had been struck as cuttings and freshly inoculated with the mildew.

The results show that there are no differences in the germinability of the conidia produced naturally in the field or artificially produced on detatched leaves or on whole plants in the greenhouse. However, marked reductions in germination, as much as 30% were obtained with all conidia with increase in age of lesions and the host. Germination also became variable as the lesions and the hosts aged. Most workers seem to have undervalued the importance of using conidia from young lesions and have assumed that inocula taken from sporulating colonies, irrespective of their age, would have the same germination potential.

It is interesting to note that even with batches of conidia of poor germinability obtained from older lesions, no significant differences could be found between germination on glass and that on the leaf. The differences in germination on glass and leaf observed by others were possibly due to one or more of the following.

(i) Failure to remove old conidia from the source of inoculum. The results in this study suggest that stronger blasts of air produced with a Pasteur pipette with a larger rubber bulb were probably more effective in removing the old conidia than with a Pasteur pipette fitted with a smaller rubber teat. Also extra care was taken to remove old and shrivelled conidia from petioles and underside of leaves, a precaution not taken by other workers. Griffee (1967) and Jackson (1968) both removed old conidia from infected plants by "gently shaking the plants", a method in the light of present results, considered to be less satisfactory, especially when whole plants were used as a source of inoculum. These workers used 48h conidia which they found germinated best and this was contrary to all published work (Longree, 1939; Peries, 1962; Hossain, 1959). In this investigation, highest germination (nearly 90%) was obtained on both glass and leaf with 24h conidia of S. mors-uvae. The 48h conidia showed c. 20% reduction in germination.

(ii) The use of chemically cleaned glassware as reported by some workers, for example, Griffee (1967). In this study glass slides cleaned chemically produced poor and variable results. Relatively few workers have realized the importance of the use of clean glass surfaces for spore germination tests (Hashioka, 1937; Zaracovitis, 1964 a,b; Jackson, 1968). Indeed most workers have not given any details on the methods which they used to clean slides and have assumed that slides used in their studies were free of toxic chemicals which could have interfered with normal germination. Zaracovitis (1964 b) emphasized that cleaning of glass slides with chemicals such as H_2SO_4/dichromate mixture, commonly used in cleaning laboratory glassware should be strictly avoided as the presence of chemicals, even in traces after several washings would interfere with conidial germination.

(iii) The use of certain unsatisfactory incubation chambers which permit free water to condense on glass slides/coverslips. The results indicated that in saturated atmospheres condensation drops of water occurred on slides which inhibited conidial germination, especially in Petri-dish moist chambers. In this respect, Manners and Hossain chambers proved to be ideal for spore germination tests as very little condensation occurred on coverslips around the conidia and often this too was on the undersurface of the coverslips.

From results of this investigation there is no reason to suppose that the leaf "stimulates" conidial germination. Jackson (1968) found leaf waxes of black currant when applied to glass slides stimulated germination of S. mors-uvae conidia (68.3% and 67.5% germination) at 30 and $120 \mu g/cm^2$ respectively, compared to the control - plain glass with no wax coating (39.0% germination). A closer examination of his results showed that these two treatments gave moderately low standard deviations (2.6 and 3.2 respectively) while the SD of the control was very much higher (9.8). This suggests a greater variability of germination on plain glass surface as opposed to glass coated with waxes. Hence what appeared to be a stimulatory effect of the waxes could well have been a masking effect of unknown factors on glass cleaned with Haemo - sol.

Peries (1962) reported enhanced germination of conidia of S. macularis from strawberry on host leaves compared to glass and supported the theory of host stimulation. He also reported that S. macularis conidia germinated equally well on slides coated with surface waxes from susceptible and resistant strawberry leaves as well as from plants totally unrelated to the host. Jhooty and M^{c}Keen (1965 b) who repeated Peries' work found that surface waxes from strawberry leaves were neither stimulatory nor inhibitory. Presumably the leaf surface waxes are therefore inert substances that do not interfere with conidial germination. Hence coating glass surfaces containing toxic substances with the inert surface waxes of leaves, will make them appear as stimulatory which was what probably Jackson (1968) and Peries (1962) observed in their studies.

Studies show that under optimum conditions germination on glass and leaf is equally good. But r.h. at the leaf surface probably favours germination in many instances – especially for those powdery mildews which germinate well over a wide range of r.h. e.g. E. cichoracearum. The leaf surface probably protects such conidia from injury caused by desiccation . However, the humidity at the leaf surface can be limiting for species such as S. mors-uvae and S. pannosa which have such exacting moisture requirements for germination.

II. GERMINATION AND EARLY GROWTH OF ROSE POWDERY MILDEW

INTRODUCTION

Rose powdery mildew caused by the fungus Sphaerotheca pannosa (Wallr.) Lev. var rosae Wor. is one of the most widespread and troublesome diseases of rose, occuring annually on outdoor roses in the summer and throughout all seasons on greenhouse roses. No cultivated rose is completely immune but cultivars differ considerably in susceptibility (Croyier, 1961). Croyier and Hildebrandt (1962) have demonstrated the biological specialization of the fungus.

The youngest leaves and pedicels are the most susceptible; the leaves becoming resistant to infection a few days after unfolding (Rogers, 1959; Bartlett, 1964; Weinhold and English, 1964). The fungus also attacks young stems, buds and even petals of the flower.

Although the causal organism was described by Wallroth in 1819 and the disease itself was known even earlier (Croyier, 1961), field control of the disease is frequently erratic. One reason for this is the relative lack of information on the biology of the fungus and its build - up and spread in the field. These aspects were, however, recently studied by Price (1969).

Early reports related the high incidence of the disease to soil factors such as dryness (Knight, 1818; Yarwood,1949) and nutrient inbalance, particularly excess of nitrogen and deficiency of phosphorus (Schaffnit and Volk, 1930; Schaffnit, 1932). Others (see Butler and Jones, 1955) suggest that mildew outbreaks in the summer are associated with warm periods of high relative humidity and this is supported by several papers describing the effects of temperature and r.h. on the development of the fungus (Hammarlund, 1925; Longree, 1939; Rogers, 1956). Two aspects of the field development of the disease have so far received little attention. These are the effect of leaf wetness on the early growth and development of the fungus and the (apparent) extreme susceptibility of the pedicels of many cultivars to fungal attack. These were the subjects of the present investigation.

MATERIALS AND METHODS

(1) Source of host plant material

Miniature rose cv Perle d' Alcanada and cv Little Buckeroo were grown in 15.5 cm diam plastic pots, 1 plant per pot in John Innes No 1 compost. The pots were kept in the open garden for 2 - 3 weeks in November 1970 and then transferred to a greenhouse at the Chelsea Physic Garden. The greenhouse was maintained between 17 and 19^{o} and provided with a 16h photoperiod by several mercury vapour lamps during the winter. About 5 - 8 plants were transferred to the greenhouse every 10 days to ensure a continuous supply of fresh host tissue. Once the flowers were produced and used in experiments the plants were cut back and left in the greenhouse until they produced another crop of flowers, which was usually in about 6 - 8 weeks.

A similar batch of plants was maintained in the Botany Department greenhouse at South Kensington to supply young leaves for maintaining the mildew cultures. The same host varieties were used throughout; mildew cultures were maintained on leaves of Little Buckeroo and the experiments were conducted on leaves and flowers of Perle d' Alcanada

(2) Mildew cultures

S.pannosa was maintained on the adaxial surfaces of young leaves which had unfolded 3 - 5 days previously. The leaves were floated on distilled water in 5.5 x 3.5 x 2.0 cm polystyrene boxes and inoculated with 24h conidia from an infected leaf in the manner described on page 27. The inoculated leaves were then covered with the box lids and incubated in a cabinet at 17 $\pm$ 1^{o} provided with a 16h photoperiod. At this temperature, the mildew sporulated effectively for 15 - 17 days but usually subcultures were made every 10 - 12 days.

(3) Standardization of host tissue

For all studies, 3 - day old, fully expanded leaves with five well developed leaflets were chosen from a shoot nearest the end of a rose branch. The leaves were tagged as their terminal leaflets unfolded and this day was referred to as day 1. The flowers were tagged as soon as they were noticed after they emerged from the enclosing bracts; this day was also arbitrarily referred to as day 1. Unless otherwise stated, in all studies 4 - 6 day old flower buds were used.

The leaves and flower buds (with pedicels) were detached just before they were used. They were washed three times in distilled water and dried between paper towelling. They were then randomly placed on a 15cm diam Petri dish lid and dusted with 24h conidia in a 18 cm diam 72 cm high settling tower. The level of inoculum was determined on coverslips interspersed with the leaves and pedicels. Three to six conidia per high power field (x 400) was taken as a standard.

The inoculated material was transferred to a 17.8 x 10.2 x 3.8 cm polystyrene box lined with moist tissue paper, closed with the lid and incubated in an illuminated cabinet at 20 $\pm$ 1°. Extra care was taken to ensure that the inoculated surfaces were kept free of moisture. The cut ends of the petioles and pedicels were inserted into the wet tissue.

(4) Measurement of sporulation

The number of conidiophores per unit area was counted as a measure of sporulation using an ocular grid micrometer under a binocular microscope. The conidiophores were counted from ten randomly selected fields (x 30).

(5) Measurement of cuticle thickness

Measurements were made from 15 μm thick freezing microtome sections of pedicels and leaves under the oil immersion objective (x 1500) of a microscope and an ocular micrometer. Ten to twenty measurements were made for each surface. The cuticle and epidermal cell wall were measured together.

(6) <u>Assessment of fungal growth</u>

The growth of the fungus on leaves and flower buds was examined by the celloidin peel technique described on p.32. With flower buds, the celloidin peels were removed from the entire inoculated surface which included the swollen receptacle and the pedicel. These were stained in lactophenol cotton blue and examined under the microscope (x 100).

Stages in the development of the fungus over a 72h period at 20^{o} as illustrated by Price (1970) were divided into 4 grades as shown below:

Incubation period (h)	Growth stage	Grade
0	Ungerminated conidium	0
5 - 8	Single germ tube and appressorium	1
8 - 14	Appearance of a second germ tube	2
14 - 24	Appearance of a third germ tube; extension of the first and the second germ tubes	3
24 - 36	Branching of the germ tubes	
36 - 48	Production of conidiophore initials	4
72	Sporulation	

Occasionally a fourth germ tube was observed but this was included in grade 4. The production of conidia which was observed at 72h was not graded. These grades were used to assess fungal growth as a Growth Index.

$$\text{Growth Index} = \frac{\text{sum of grades}}{\text{number of conidia counted}} \times \frac{100}{\text{highest grade}}$$

One hundred conidia were counted at random for each replicate celloidin peel of a sample. Usually 3 - 4 replicates were prepared for each sample and the results given as an average of the total.

EXPERIMENTAL

(1) Effect of leaf wetness

In addition to the many reports on the injurious action of water on powdery mildew conidia, some of which are described on p.15 , there are others which indicate that when water is applied to plants mildew development is severely restricted. Yarwood (1936), for example, found that mildew on peas was destroyed by subjecting infected leaves to a fine water spray and later (Yarwood, 1939) he reported on the use of water sprays as an effective control measure for rose, bean, cucumber and barley powdery mildew. Inhibition of mildew development on rose by water sprays has been confirmed by other workers. These do not appear to be commercially satisfactory for disease control (McClellan, 1942) but may be useful in raising disease - free rose plants for experimental use (Rogers, 1959). Mist applications appear especially effective (Kohl, 1955; Langhans, 1955), particularly when applied immediately after leaves are inoculated with conidia (Rogers, 1959).

The stage at which germinating conidia are most vulnerable to the damaging effect of water were investigated in the experiments described below.

(A) Stages in the development of the fungus

(i) Light microscopy

The development of the fungus was studied at 20 $\pm$ 1^{o} by inoculating detached leaves with 24h conidia and examining them at intervals. The methods used were:-

(1) Celloidin peel technique as described on p.32

(2) Leaves cleared overnight in methanol and stained for 30 min in either aqueous aniline blue or lactophenol cotton blue (Griffee, 1967).

(3) Leaves boiled in alcoholic cotton blue, left in the

stain for 2 days and cleared in chloral hydrate (Shipton and Brown, 1962; see Appendix 6 p.235).

(4) Leaves treated with periodic acid followed by staining with basic fuchsin (the Periodic - Schiff method of Preece, 1965; see Appendix 6 p.235).

Sections (15 - 25 μm) cut with a freezing microtome were also examined after staining them in acetic aniline blue or lactophenol cotton blue.

Results

Celloidin peels were useful in examining the initiation of germ tubes and their subsequent growth. Leaves cleared in methanol and stained in aniline blue or cotton blue also provided information about the sequence of germ tube development and their rate of growth. Although the host cells were cleared with the Periodic - Schiff method the fungus failed to take up the red stain of basic fuchsin. The technique of Shipton and Brown was not entirely satisfactory as the chloral hydrate often cleared the leaf as well as the fungus, of the stain. However, better results were obtained when leaves were examined immediately after boiling in alcoholic cotton blue when the leaves were cleared but the fungus was stained blue. Still better results were obtained when these were restained in lactophenol cotton blue for 10 - 15 min , washed in distilled water before examining them mounted in 50% glycerol. This last method was useful in examining haustoria on whole leaves. Only fully developed haustoria were seen in the freezing microtome sections. Penetration and early stages of haustorial formation were not seen.

From the observations made using the above methods, the growth of the fungus can be divided into several phases;

(a) Formation of appressorial initials

At 20 $\pm$ 1° and near 100% r.h. the conidia started to germinate between 2 - 4h after inoculation. At 3h, a short germ tube — the primary germ tube was produced from one side of the conidium (Fig. 4a). This elongated and enlarged and by 6h a club shaped appressorial initial was formed (Fig. 4b). By this time approximately 75 - 80% of the conidia formed these structure.

(b) Maturation of the appressorium

At the same time a septum developed at the proximal end of the appressorial initial (Fig. 4c). The septum was first detected in 6h and by 8h nearly 90% of the germinated conidia formed mature appressoria. A slight swelling on one or both sides about the middle of the appressorium was often seen at this stage (Fig. 4d).

(c) Penetration of the host cell

A central, darkly stained (with cotton blue) region surrounded by a clear halo (Fig. 4e) was also seen on the appressorium. Although these were interpreted as points of origin of infection pegs conclusive evidence is lacking.

(d) Formation of the secondary germ tube

Between 8 - 10h, a second germ tube was produced — the secondary germ tube, usually on the opposite side (Fig. 4f) but occasionally on the same side as the primary germ tube (Fig. 4g). At the same time the primary germ tube elongated from the distal end of the appressorium. Evidence suggests that this took place after penetration of the host cell.

(e) Formation of the haustorium

Between 16 - 20h after inoculation, in cleared tissue (Shipton and Brown), small spherical bodies — haustorial initials were seen in the epidermal cells directly underneath the appressoria (Fig. 4h). A third germ tube — the tertiary germ tube usually developed at this stage from a polar position giving the conidium a tripartite appearance (Fig i).

(f) Maturation of haustorium

Between 20 - 24h, mature haustoria were seen arising from nearly 70 - 75% of the appressoria that developed. In cleared cells these appeared

Fig. 4. Stages in the development of Sphaerotheca pannosa on rose loavos.

(a) , (b). Formation of appressorial initials (x 1800 approx.)

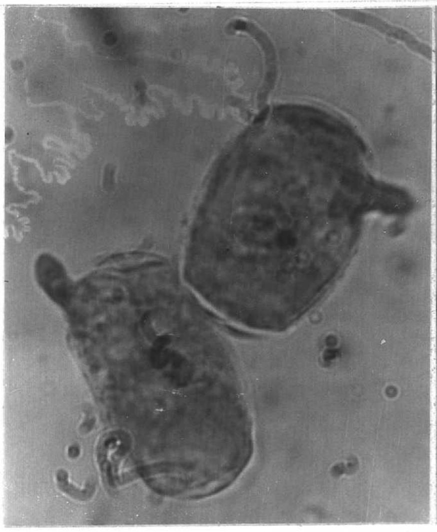
(c) , (d). Maturation of the appressoria (x 1800 approx.)

(e) . Penetration of the host cell (x 1600 approx.)

(f) , (g). Formation of the secondary germ tubes(x 450 approx.)

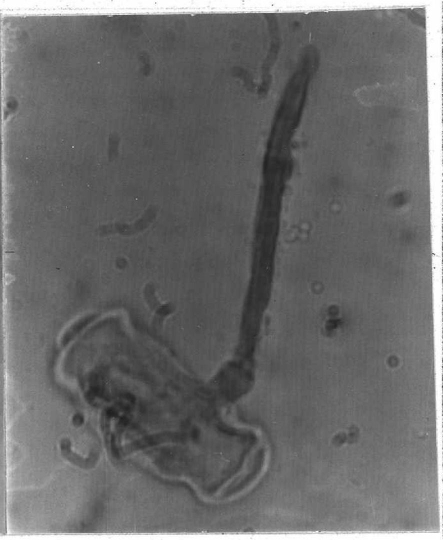
(h) . Development of the haustorial initial (x 3400 approx.)

(i) . Formation of the tertiary germ tube (x 450 approx.)



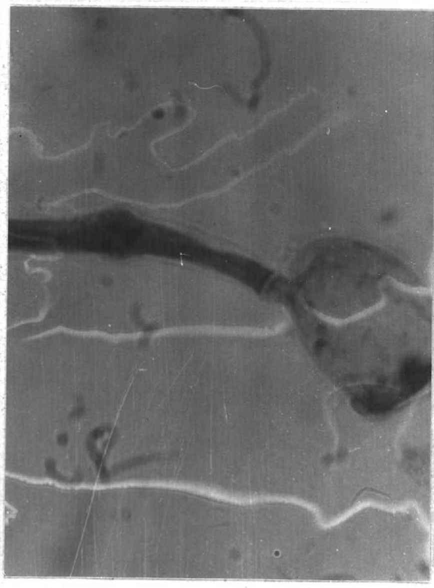
(a)

(b)



(c)

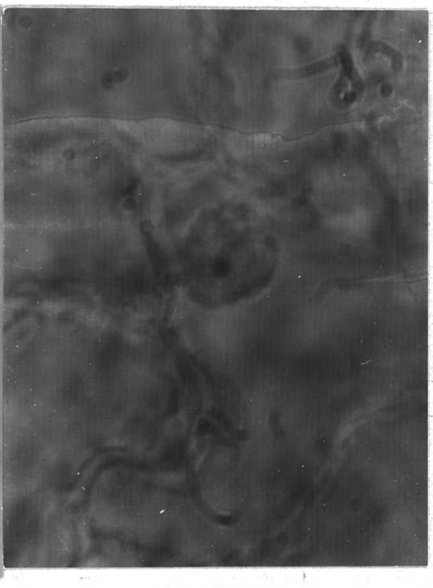
(d)



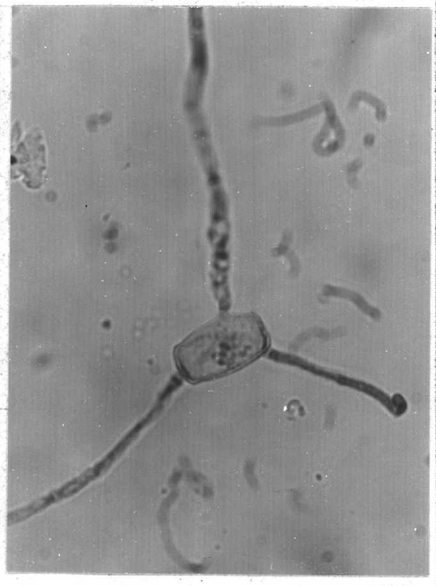
(e)

(f)

(g)



(h)



(i)

as glistening spherical bodies. In sectioned leaves, they appeared somewhat oblong surrounded by a granular sheath, connected to the appressorium outside by a narrow neck (Fig. 5).

Germ tube growth was rapid after the initiation of the third germ tube. Sometimes a fourth germ tube was produced. Between 22 - 36h the germ tubes branched and by 48h the conidiophore initials were formed. In 72h sporulation had begun.

(ii) Scanning electron microscopy

Early stages in the development of the fungus were also examined by scanning electron microscopy. The methods of specimen preparations are given below.

(1) Leaf discs bearing germinating conidia were mounted on stubs with Durofix. These were dried in a desiccator for 24h. Some were fixed in formalin vapour for 40 min (see Appendix 6 p.236) before drying in the desiccator.

(2) Celloidin peels (unstained) were mounted on stubs with Durofix, Araldite or gum and dried overnight in a desiccator.

The specimens were finally coated with gold/palladium alloy in a vacuum coating unit and viewed in a Cambridge Stereoscan at 60 keV.

Results

Scanning electron microscopy did not provide any more information than from material examined under the light microscope. Although there are obvious advantages of using the scanning electron microscope (SEM) in the study of powdery mildews, such as following the effect of application of chemotherapeutants on the development of the mycelium and of sporulation etc. this microscope has little value in the study of mildew growth on leaf surface other than providing a greater depth of focus and a magnification range from 20 upto 50,000 times.

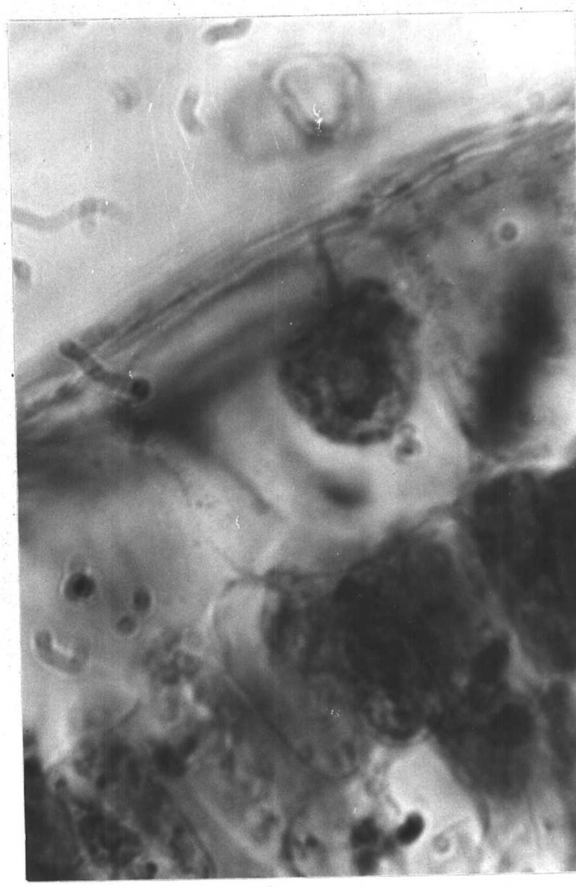


Fig. 5. Haustorium of S. pannosa in epidermal cell of rose (cv. Perle d' Alcanada) 28h after inoculation of leaf with conidia(x 5280 approx.)

Fixing conidia with formalin vapour did not have any noticeable effect. Araldite and Durofix reacted with celloidin during mounting causing considerable shrivelling and shrinkage of the celloidin peels. However, gum was satisfaotory for this purpose.

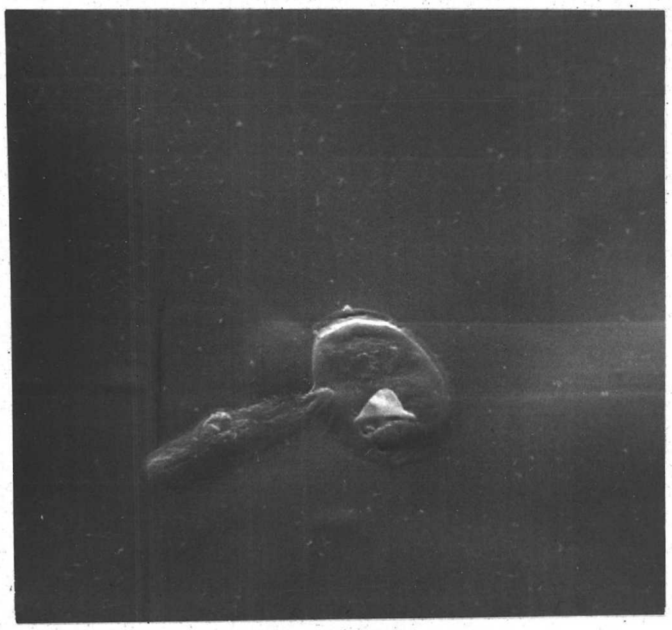
Celloidin peels viewed under the SEM failed to reveal the origins of infection pegs on the appressoria. This was possibly due to collapse of the fungal material either during mounting or when drying.

Swellings on the appressoria that were observed at 6 - 8h under the light microscope were sometimes seen on the celloidin preparations as crater - like papillae (Fig. 6a,b). These probably were lateral outgrowths of the germ tubes of no particular significance. They were not observed in the later stages.

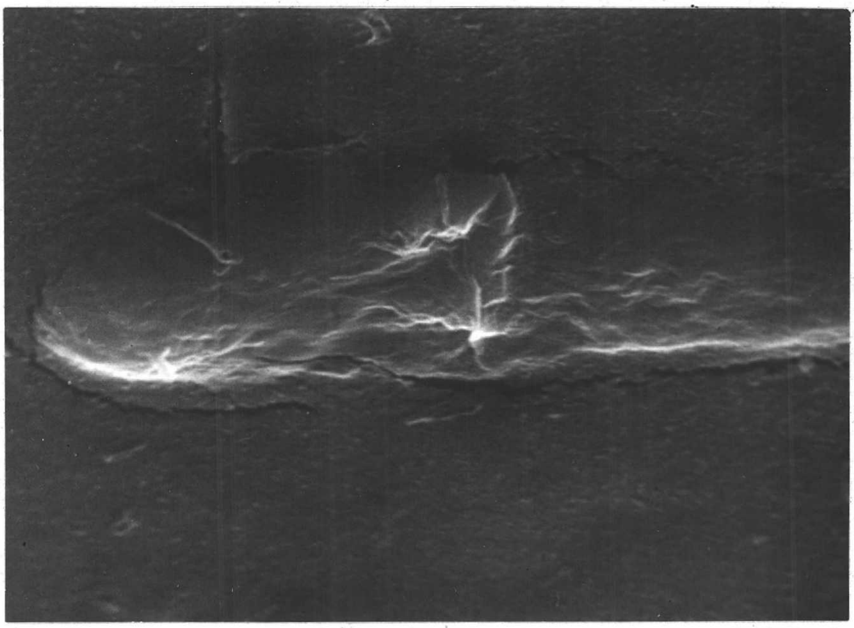
(B) Leaf wetness experiments

This section deals with the effect of water in the form of disorete droplets on the early growth and development of S. pannosa on rose leaves. Leaves dusted with conidia were subjected to different periods of wetness immediately after inoculation and several hours after incubation.

The inoculated leaves were placed in a polystyrene box (17.8 x 10.2 x 3.8 cm) lined with wet tissue and sprayed for 2 - 3 min with a fine mist of distilled water from a hand spray (Shandon Laboratory spray gun) held about 60 - 90 cm away until they were covered with minute but discrete droplets. The lid of the box was sealed with 'Sellotape' and the box incubated at 20 $\pm$ 1^{o} in a cabinet with a 16h photoperiod. After the required incubation, the lid was removed and the leaves were dried at room temperature under a fan placed 120 - 150 cm away. This operation took about 30 min. The leaves were then re-incubated at 20 $\pm$ 1^{o} in the same box making sure that the tissue lining was sufficiently moist and the leaves were free from any traces of moisture. The cut ends of the petioles were embeded in the wet tissue.



a



b

Fig. 6. Germinating conidium of S. pannosa viewed in the scanning electron microscope.

(a) Germinated conidium showing a single germ tube x 2150 approx.

(b) A close - up of the germ tube showing a papilla x 5750 approx.

Each treatment had 7 - 10 samples with 3 - 5 replicates for each sample. The leaves were examined usually at 6, 12, 18, 24, 36, 48, and 72h intervals and the Growth Index (GI) of the fungus assessed.

Expt (1) Effect of 2, 4, 6 and 8h wetness immediately after inoculation and 6h later

In a preliminary investigation the following tests were carried out.

A Control - no wetting.

B 2, 4, 6 and 8h wet periods immediately after inoculation.

C 2, 4, 6 and 8h wet periods, starting after a 6h incubation of inoculated leaves.

The results are summarized in Table 35 and Fig. 7. In general, periods of leaf wetness delayed mildew development. This was more marked when the wet period immediately followed inoculation with conidia than when leaves were wetted 6h later. There were also some indications that the reduction in fungal growth was directly related to the length of the wet period though the results, especially for the 2h wet - period, are somewhat variable.

Expt (2) Effect of 8h wet period immediately after inoculation and 6h later

Following the results of the preliminary studies in Expt (1) a further test involving an 8h wet period was conducted on similar lines.

The results of this study are given in full in Appendix 7 p.237 and summarized in Table 36. The results were similar to those obtained in Expt (1). The growth patterns of the treatments were similar to that of the control but at reduced levels. A plot of growth index against time (Fig. 8) showed that for each treatment, there was a short lag phase followed by an exponential period of growth from 10 - 36h. After c. 40h the rate of growth slowed and at 48h appeared constant.

Table 35. Effect of leaf wetness on conidial germination of S. pannosa

Wet Period	Hours incubation	Growth Index* A	B	C
2 Hrs	6	17.8	10.6	18.0
	18	24.5	22.9	22.6
	24	25.5	24.3	24.7
	48	53.4	29.9	31.5
	72	47.0	44.8	31.2
4 Hrs	6	18.3	10.5	16.9
	18	34.2	26.5	22.0
	24	38.0	32.4	22.3
	36	37.0	32.3	24.6
	48	44.4	45.0	45.3
	72			
6 Hrs	6	18.1	8.2	17.8
	18	23.5	22.6	23.5
	24	32.3	24.6	30.0
	36	35.3	28.2	30.4
	48	48.1	33.5	44.6
	72			
8 Hrs	6	18.4		17.5
	18	26.4	19.4	20.4
	24	29.0	19.8	23.2
	36	29.8	23.0	37.0
	48	56.5	21.5	30.6
	72	52.9	26.6	39.1

A - control - no wetting
B - wetted immediately after inoculation
C - wetted 6 hrs after inoculation
* - Average of 3 replicates

Fig. 7 Changes in the growth pattern of S. pannosa caused by leaf wetness.

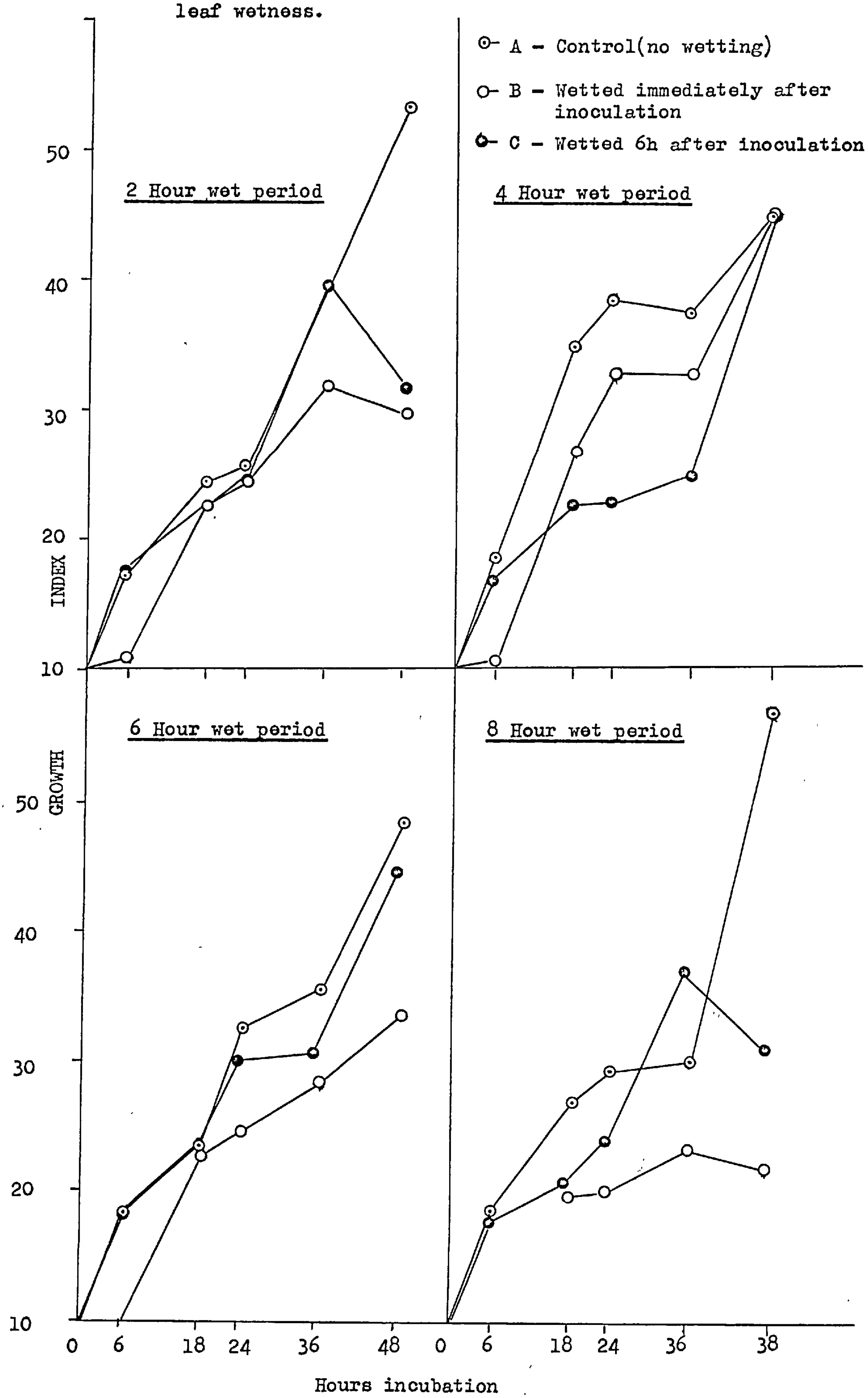
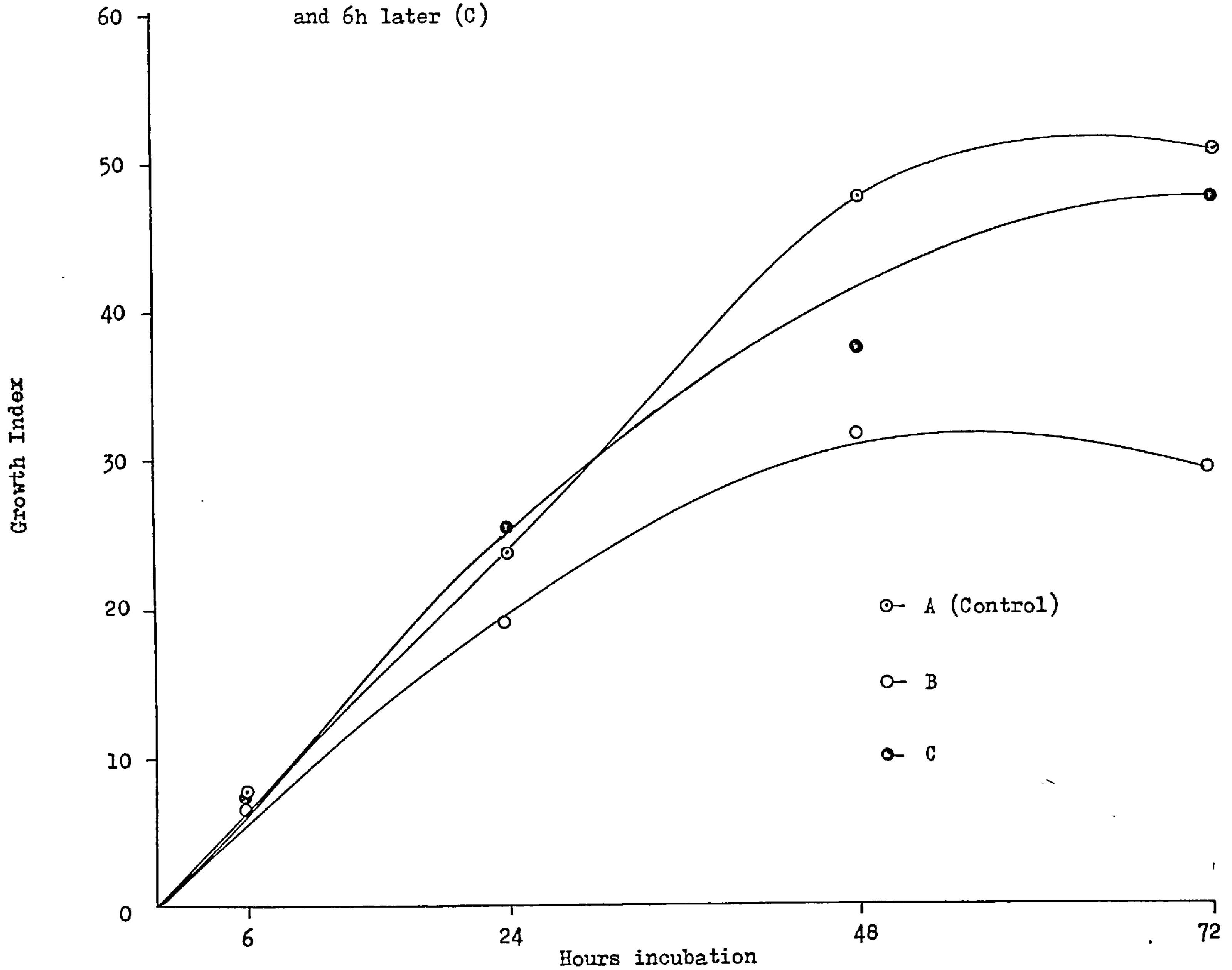


Table 36. Effect of 8 h wet period on conidial germination of S. pannosa

Hours incubation	Growth Index *		
	A	B	C
6	7.6	(6.5) +	7.1
24	23.8	19.0	25.5
48	47.5	31.5	37.3
72	50.7	29.4	47.8

A - control - no wetting
B - Wetted immediately after inoculation
C - Wetted 6 hrs after inoculation
+ - Leaf surface dried and examined immediately
* - Average of 3 replicates

Fig. 8. Effect of 8h wetness on the growth of S. pannosa immediately after inoculation (B) and 6h later (C)



The results also show that not all conidia in the population had equal potentials for germination (see Appendix 7). Some 25% of the conidia never germinated and about the same number formed one germ tube only (grade 1). Approximately half the conidia produced a second germ tube (grade 2). Of these, nearly 90% developed further but only about 35% of the total population went on to produce conidiophore initials. This was determined at 72h but at this stage GI was difficult to measure due to ramification of hyphae even when low levels of inocula were used. Also, it was observed that beyond 72h, non - germinated conidia were either pushed aside or dislodged by the growing hyphae often giving an incorrect picture of the GI.

A wet period applied 6h after inoculation with conidia did not greatly influence the growth of the fungus and the results were similar to those of the control. On the other hand a wet period immediately after inoculation reduced fungal growth although some conidia germinated during the wet period (Table 36). However, nearly 55% of conidia of the total population that germinated remained at the 1 - germ tube stage and did not develop further. This was more than double the number relative to the control.

Leaves from treatment A (control) and B (wet period after inoculation) were further incubated for a total of 6 days and the mildew growth was examined visually. A considerable difference in the amount of mildew coverage between these two was observed. Mildew growth was dense and evenly spread over the entire leaf surface on control leaves A whereas it was sparse and patchy on leaves in treatment B.

Expt (3) <u>Effect of 8h wetness 12 and 24h after incubation</u>

The following batches of leaves were prepared:-

A. control - no wetting.

D. 8h wetness, 12h after incubation.

E. 8h wetness, 24h after incubation.

Wetness after the initial establishment of the mildew on the leaves had little effect on its development (Table 37). There were no adverse effects of wetness on germinating conidia after 12h incubation by which time the conidia had developed the second germ tube. Studies on the development of the fungus (p. 108) suggested that the second germ tube is initiated after the penetration of the host by an infection peg of the appressorium. This, therefore suggests that once the fungus has penetrated the host surface, although the haustorium is not fully formed, leaf wetness has little effect on its development.The results further suggest that during the wet periods, conidial development was merely suspended, and that as soon as the leaves were dried, they recovered slowly and resumed their growth, lagging behind the control by about 8 - 10h (Fig. 9). After 72h, their growth was equal to that of the control.

In another experiment the effect of a 12h wet period immediately after inoculation (F) and 12h after incubation (G) was studied. The results of this study (Table 38) were similar to those of Expt (1) in that wetness immediately after inoculation reduced mildew growth more than wetness 12h after incubation. The results also suggest that prolonged wet periods caused greater reductions in mildew growth than shorter wet periods (Figs. 9,10).

Expt (4) <u>A comparison of methods for assessing fungal growth</u>

Although the growth index of the fungus based on graded conidia was satisfactory in bringing out the overall differences between wet treatments, there were marked visual differences in mildew growth that were not shown with this method. This was particularly noticed when the mildew was allowed to develop for 5 - 7 days (Fig. 11).

An experiment was undertaken to compare the estimated growth index with the actual mycelial growth. This was essentially a repeat of part of Expt (1) in which the effects of 2, 4, 6 and 8h wetness immediately after inoculation were studied. A control was also set up where leaves were not

Table 37. Effect of 8 h wet period on conidial germination of <u>S. pannosa</u> 12 and 24 hours after inoculation

Hours incubation	Growth Index*		
	A	D	E
12	19.0	18.5	20.6
24	25.6	21.7	29.5
36	49.0	41.7	40.0
48	53.4	43.4	44.8
72	58.0	51.5	50.0
96	56.5	55.0	58.7

A - control - no wetting
D - Wetted 12 hours after inoculation
E - Wetted 24 hours after inoculation
* - Average of 3 replicates

Fig. 9. Growth pattern of S. pannosa after a 8h wet period 12 and 24h after inoculation

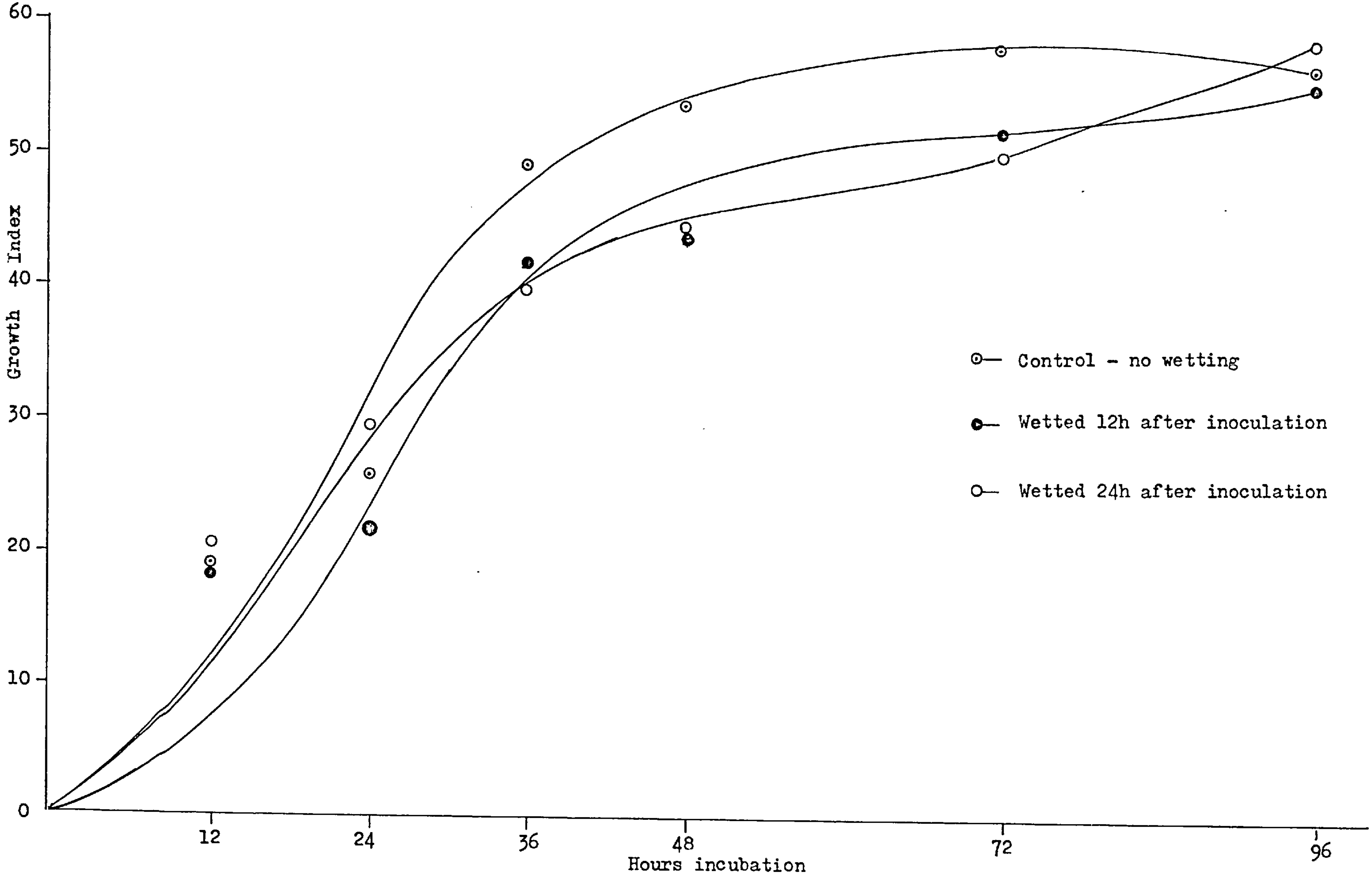
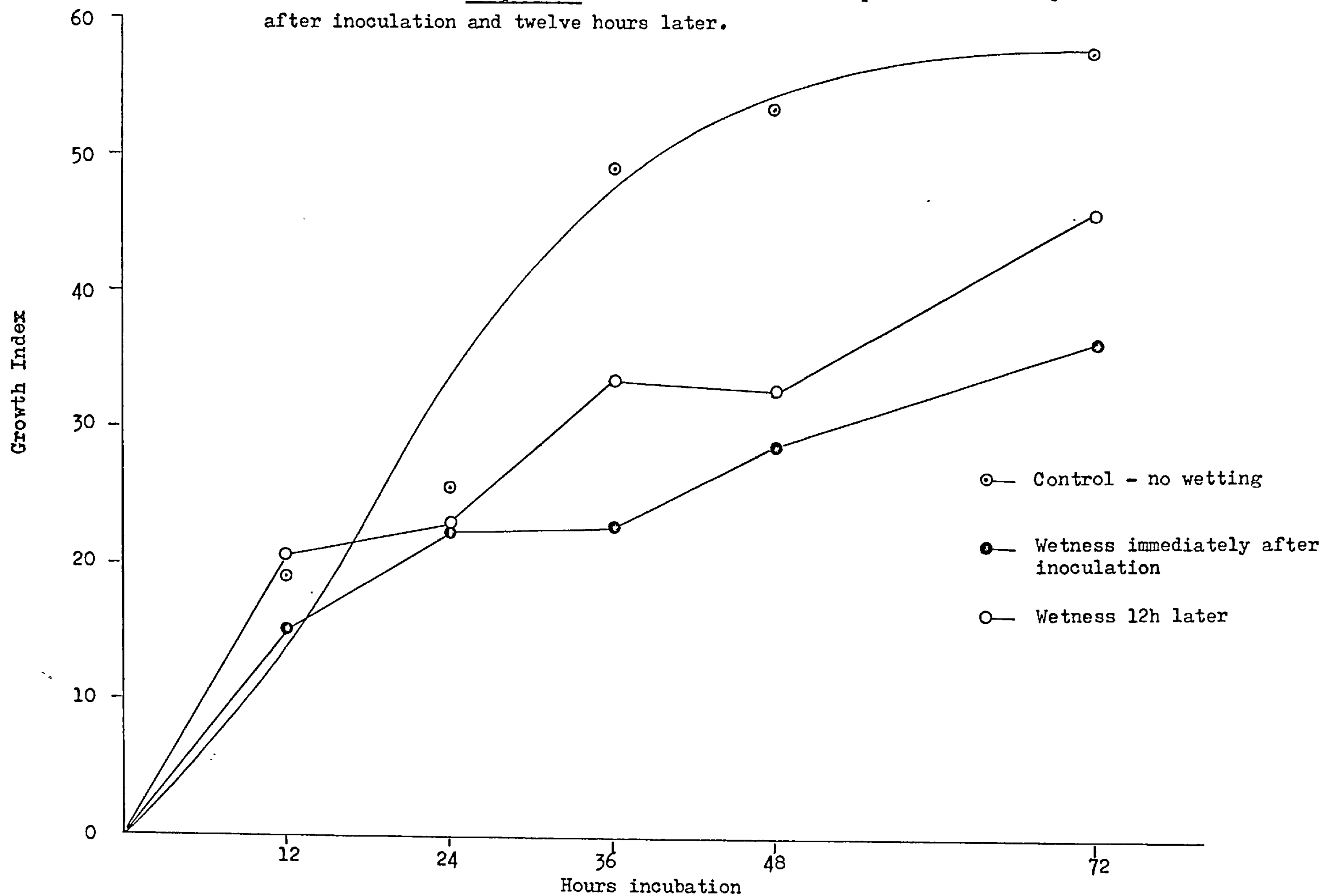


Table 38. Effect of 12 h wetness on germination of conidia of *S. pannosa* immediately after inoculation and 12 h after incubation

Hours incubation	Growth Index *		
	A	F	G
12	19.0	15.0	20.9
24	25.6	22.3	23.0
36	49.0	22.8	33.7
48	53.4	28.9	32.9
72	58.0	36.8	46.0

A - control
F - Wetness immediately after inoculation
G - Wetness 12 h after incubation
* - Average of 3 replicates

Fig. 10. Growth pattern of <u>S. pannosa</u> after a twelve hour wet period immediately after inoculation and twelve hours later.



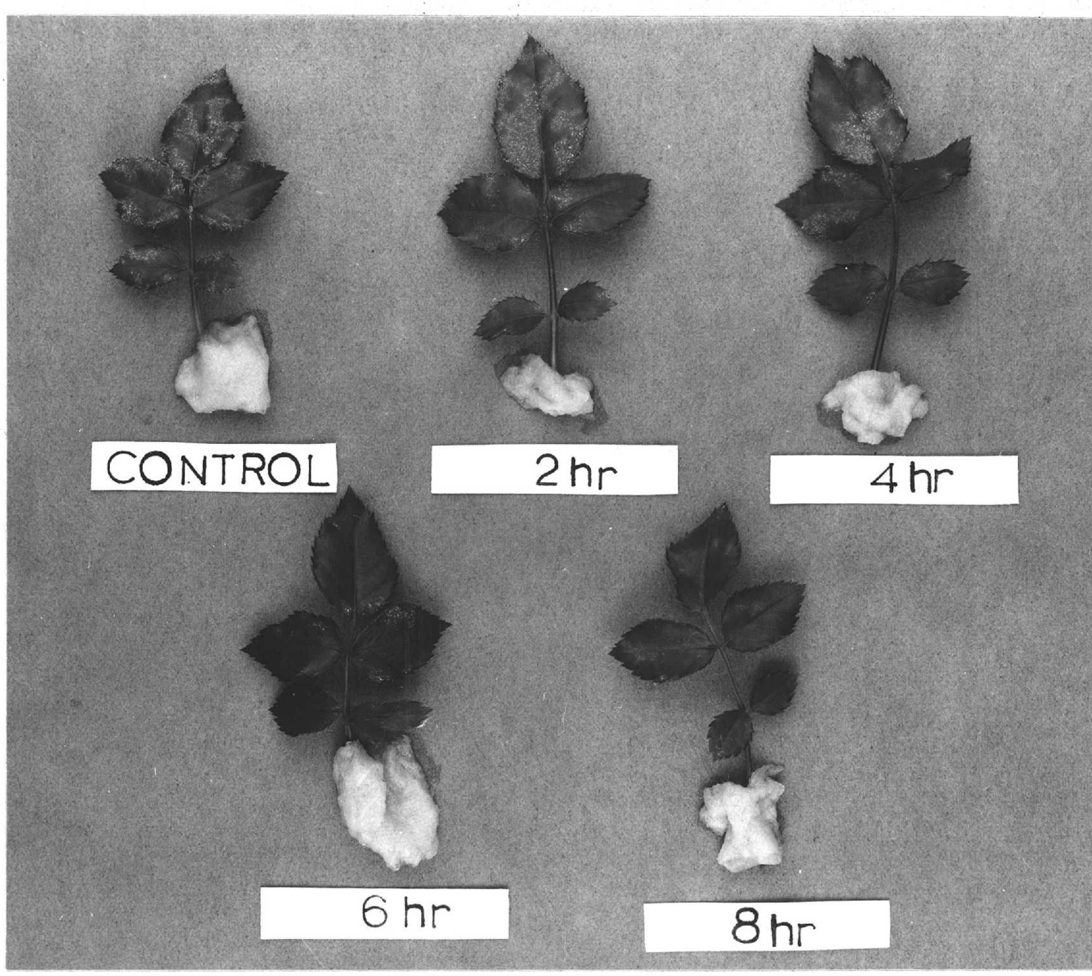


Fig. 11. Effect of different periods of wetness on mildew development on rose leaves.

subjected to wetting. The leaves were sampled at 6, 24, 48 and 72h. The stained celloidin peels were mounted in 50% glycerine and the GI of the fungus assessed in the usual manner. Using the same celloidin peels germ tube growth of 100 randomly - selected conidia for each replicate peel was traced onto paper using a camera lucida (mag 613 x 10). This was then measured with a map measurer (Curvimeter Universal map measurer) and the total mycelial growth given in mm.

Table 39 summarizes the results. Both methods were equally good in illustrating the influence of leaf wetness on mildew growth which was directly related to the length of wet period. But as shown in Figs. 12 and 13 the differences in fungal growth, especially in later stages, was better expressed when the actual mycelial growth was measured as opposed to the estimated growth index.

These results therefore suggest that for an accurate estimation of fungal growth the Growth Index measurement is satisfactory upto grade 3 ie. the first 24h. Beyond this, mycelial growth measurement provides a more reliable estimation of the overall growth of the fungus.

Expt (5) Effect of leaf wetness on the development of mildew colonies

This experiment was undertaken to pin point the origin of mildew colonies on leaves subjected to wetness. The leaves of cv Little Buckeroo were chosen because of their larger size.

Six 2 mm diam rings were marked with Indian ink on the terminal leaflets of each of 6 leaves. When the ink dried, the leaves were inoculated with 24h conidia. Drops of distilled water were placed within three of the marked rings on each leaflet using an aglar syringe. The remaining three rings on each leaflet were left dry and were referred to as the control.

Five of the leaves placed in a polystyrene box and lined with wet tissue were incubated at $20 \pm 1^{\circ}$. After 8h, they were dried under a fan at room temperature and re-incubated at $20 \pm 1^{\circ}$. After 72h celloidin

Table 39. Effect of different periods of leaf wetness immediately after inoculation on conidial germination of *S. pannosa* - A comparison of growth index and mycelial growth

Hours incubation	Measurement of growth	Growth Index* / mycelial growth**(mm)				
		control no wetting	Hours wetness			
			2	4	6	8
6	G. I.	15.8	10.5	11.8	13.8	-
	Mycelial growth	1.2	0.72	0.85	1.0	-
24	G. I.	37.4	27.7	28.5	25.0	21.8
	Mycelial growth	12.6	6.4	5.8	3.6	3.2
48	G. I.	45.5	42.4	38.0	28.7	31.8
	Mycelial growth	78.0	42.9	27.2	12.7	17.0
72	G. I.	47.8	44.8	37.5	34.4	33.8
	Mycelial growth	121.5	102.4	50.6	39.4	39.5

*, ** Average of 3 replicates

Fig. 12 Effect of different periods of leaf wetness immediately after inoculation on the Growth Index of S. pannosa.

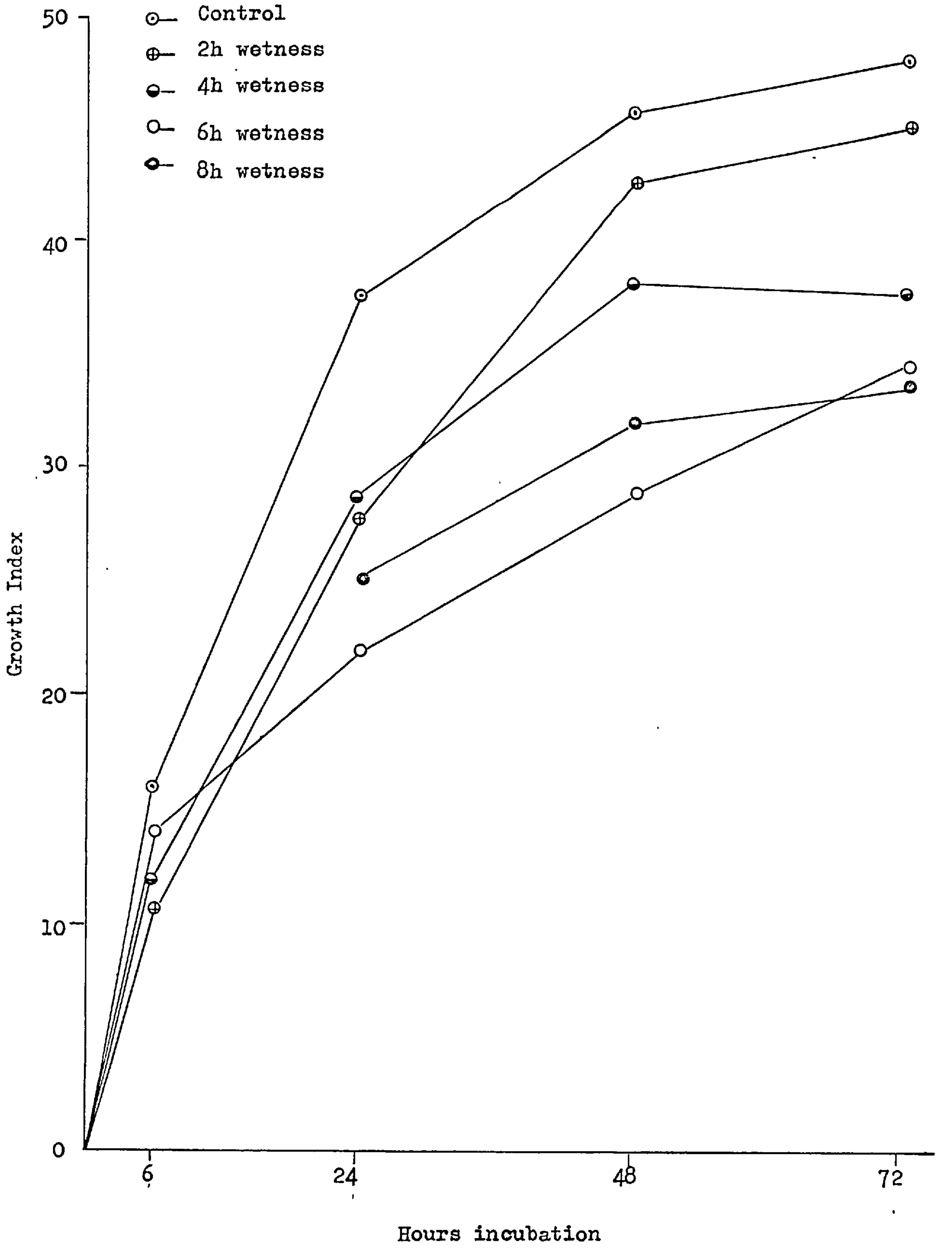
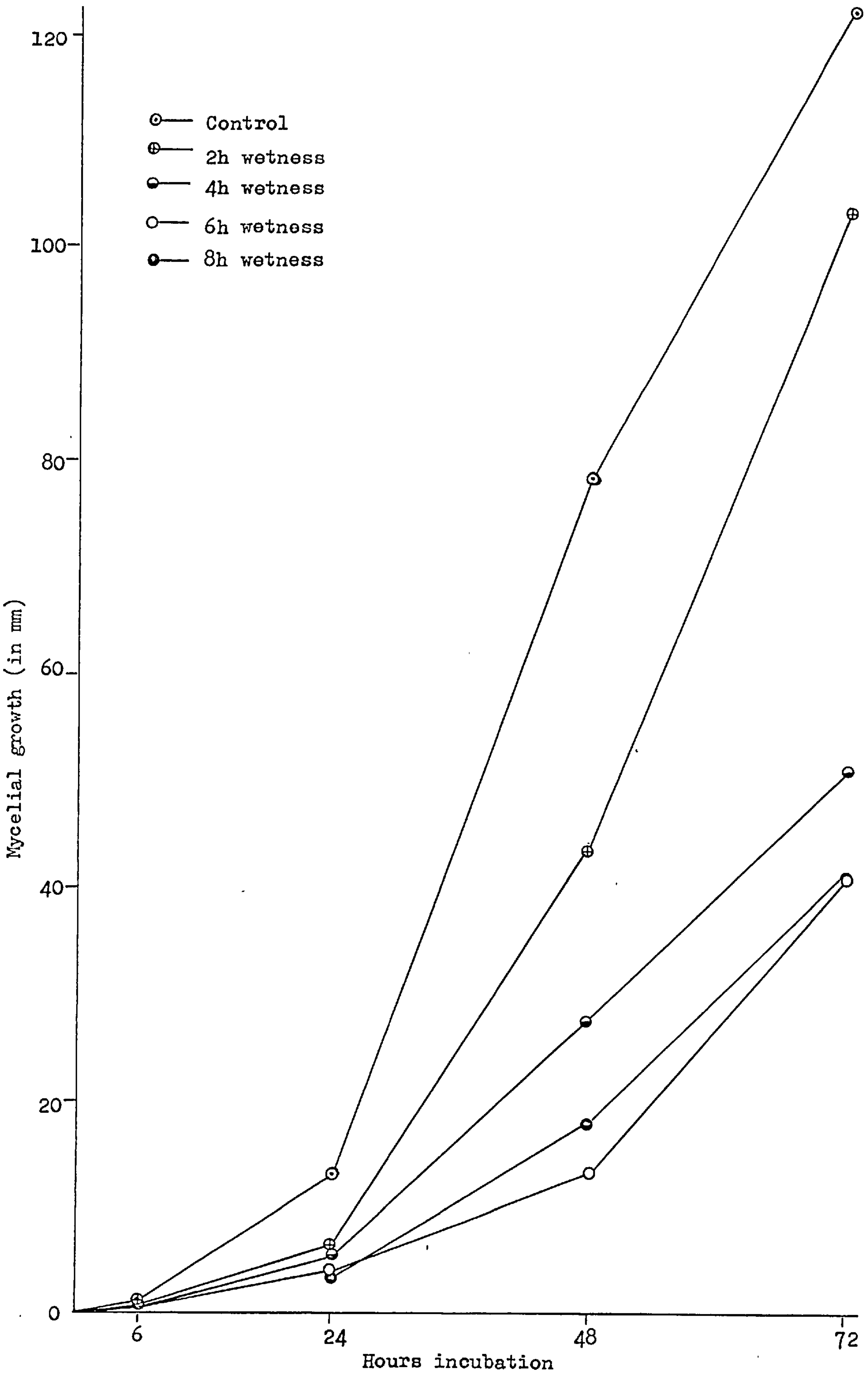


Fig. 13 Changes in the growth rate of S. pannosa caused by different periods of leaf wetness immediately after inoculation.



peels were prepared from these which also removed with them the Indian ink rings from the leaf surfaces. These were stained with lactophenol cotton blue and the GI of the fungus within each ring was assessed. At 72h there were very few conidia in grade 2 stage. For convenience all grades above this stage were grouped together but were placed under the category grade >2.

The 6th leaf was incubated separately with the drops of water in a 5.5 x 3.5 x 2 cm polystyrene box closed with the lid and sealed with "Sellotape" at 20 $\pm$ 1^{o}. The leaf was examined microscopically with the rest at 72h before and after drying under a fan.

Microscopic examination of the leaf that was incubated continually wet revealed that no mildew colonies developed under the drops of water. But colonies were found around these, in between the marked rings and within the control rings. Conidia were found floating on the surface of water, Mostly in clusters around the edges of the droplets but most of them were ungerminated and had collapsed. A few however, had produced long germ tubes that either grew away from the water or vertically upwards without branching.

There were few or no mildew colonies originating from within the test rings. However, they were invaded by hyphal extensions from colonies arising from outside the rings. The results of this study (see Appendix 8 p. 239) show that the mean GI of the controls was nearly twice that of the test rings.

The absence of mildew colonies from within the test rings does not necessarily suggest inhibition of germination. It is possible, that when the drops of water were placed on the leaves the conidia were either pushed to the edges or floated away and germinated in these positions. The non - germinated oonidia inside the rings could have been the ones that were submerged and the ones that would never have germinated in any case.

Within the control rings, mildew colonies developed equally distributed.

(2) <u>A comparision of germination and growth on leaf and flower pedicel</u>

In the field, the flower stalks (pedicels) of the rose including the enlarged receptacle region often seem especially susceptible to powdery mildew, becoming densely covered with closely packed conidiophores. Severely infected pedicels sometimes become swollen (Fig. 14). Price (1970) reported that infected pedicels were an important source of inoculum for the build-up of the fungus in the field but he could not explain why pedicels were so susceptible. A search in the literature on rose powdery mildew revealed that no other work has been done on the infection of the flower stalk. This work deals with the investigations into conditions necessary for infection of pedicels and a comparison of germination and early growth of the fungus on leaves and pedicels. The effect of the fungus on the pedicel was also studied.

Unless otherwise stated, the experiments were conducted on detached host tissue of cv. Perle d' Alcanada as described on p.104. Whenever it was necessary to inoculate attached pedicels and leaves a 25 cm high settling tower was used. This was slightly cone - shaped, being 8 cm diam at one end and 4.5 cm at the other. A 3 mm slit cut along the length of the cylinder at the narrow - end enabled a leaf or a pedicel to be inserted over which the settling tower was held in position with the aid of a clamp attached to a retort stand. The leaves and pedicels were inoculated individually in the usual manner (p.104). A glass slide was held underneath to determine the level of inoculum and for each test, the same level of inoculum was used. After inoculation, the plants were placed under polyethylene bags for 24h at the required temperature to encourage maximum germination of conidia. After the required incubation, the GI of the fungus was assessed.

(A) <u>Mildew growth experiments</u>

Expt (6) <u>Rate of growth of leaves and pedicels</u>

A preliminary investigation was made on the growth patterns of leaves and flowers for a better understanding of the susceptibility of these

Fig. 14. Infected rose plant of cv. Perle d' Alcanada. Note swollen pedicel (ringed).

organs to mildew infection. Increase in length of the pedicel was regarded as a measure of growth of the flower (Leopold 1964) and increase in leaf area as that of the leaf. Leaf area was derived from measurements of the width of the apical leaflet and the length of the leaf (Price, 1969).

Growth was measured every 2 - 3 days on attached, tagged leaves and flowers from 15 potted rose plants maintained in the greenhouse. During the first two assessements however, pedicels were detached for ease of measurement as they were still covered by the surrounding bracts. The cuticle thickness of the terminal leaflets and pedicels of comparable age was also measured.

The patterns of development of the leaf and the flower were similar in that they both conformed to the characteristic Sigmoid pattern of growth (Fig. 15). The leaf reached its maximum size around the 14th day whereas the maximum length of the pedicel was reached 3 - 4 days later. Initially, the cuticle thickened at the same rate in both leaf and pedicel but subsequently it increased much more rapidly in the leaf. By the 10th day, the leaf cuticle had reached its maximum thickness and at this stage was nearly twice that of the pedicel. The cuticle of the pedicel reached its maximum thickness about 4 days later but it was still much less than that of the leaf (see Appendix 9 p.241).

Expt (7) <u>A comparison of mildew growth on leaves and pedicels</u>

Young leaves and flower pedicels were inoculated and incubated at 20 $\pm$ 1°. The GI of the fungus was assessed after 6, 24, 48 and 72h.

The growth of the fungus was similar on both organs upto 24h (Table 40). During the next 48h however, mildew growth was better on leaves while that on the pedicels declined. The inooulated material was incubated for a further 3 days but sporulation was observed only on leaves. All pedicels at the end of the incubation period (6 days) had begun to senesce. In this experiment however, the exact age of the flower was not known.

Fig. 15.   Rate of growth of the attached leaves and flowers of rose

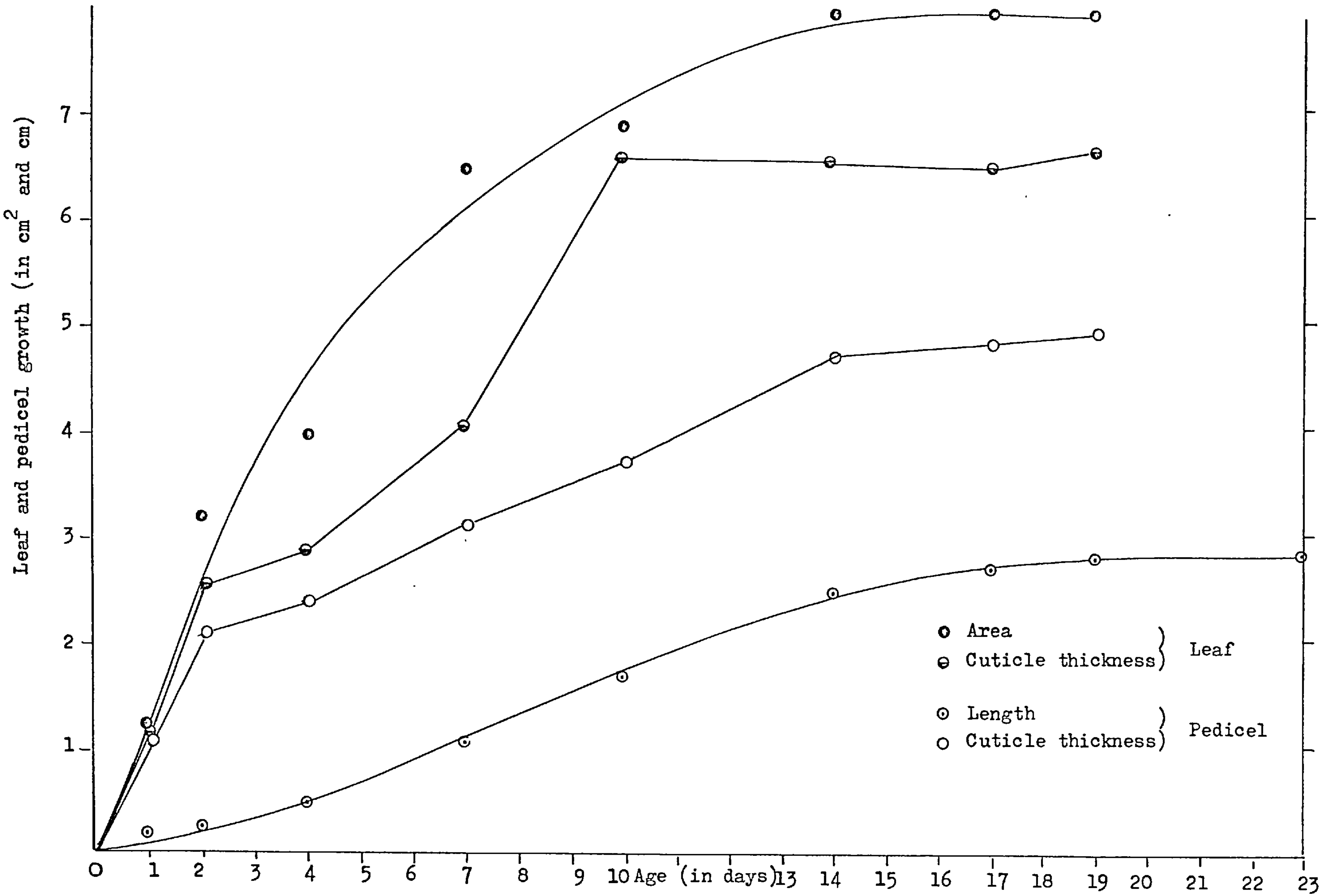


Table 40. Mildew growth on detached leaves and pedicels

Hours incubation	Growth Index *	
	Leaves	Pedicels
6	11.1 (5.0)	11.0 (4.0)
24	16.6 (2.5)	19.3 (14.0)
48	32.5 (14.4)	20.6 (12.2)
72	40.1 (23.7)	29.1 (21.6)

* Average of 3 replicates

S D in brackets

Expt (8) Relationship between increase in age of leaves and pedicels to susceptibility to mildew attack

One day old leaves and pedicels were tagged. At 2 - day intervals samples of these were detached, inoculated and incubated at $20 \pm 1^{\circ}$. The GI was assessed at 24, 48 and 72h. Samples were incubated for a further 5 days to obtain sporulation.

The following facts emerge from results shown in Table 41. (i) Youngest leaves and pedicels were most susceptible to the mildew and susceptibility decreased with increase in age of the tissue. (ii) The flower pedicels remained susceptible for a longer period, upto about the 10th day, whereas the susceptibility of the leaves diminished rapidly after the 6th day. (iii) The results also suggest that initially mildew growth was more rapid on pedicels (see 24h estimates) than on leaves.

There was no sporulation on pedicels of all ages;these began to senesce after 5 - 7 days. There was abundant sporulation on leaves upto 5 - 6 days old but little or none on leaves older than 7 days.

Expt (9) Early growth of the fungus on leaves and pedicels

The results of Expt 8 suggested that early growth of the mildew was faster on the pedicel. This was tested by assessing percentage germinatio of conidia at 3, 6 and 9h on 4 - day - old leaves and pedicels incubated at $20 \pm 1^{\circ}$. The germ tube length was also measured.

The results (Table 42) indicated only a marginal increase in conidial germination and germ tube growth on pedicels.

Expt (10) Mildew growth on leaves and pedicels at 12°

Detached 4 - day - old leaves and pedicels were inoculated and maintained at 12° in a cabinet with a 16h photoperiod. The GI was assessed at 24, 48 and 96h.

The results (Table 43) are similar to those of Expt (7). Mildew

Table 41. Effect of increase in age of pedicels and leaves on the mildew development on rose

Hours incubation	Surface tested	Growth Index*							
		Day 1	Day 2	Day 4	Day 6	Day 8	Day 10	Day 12	Day 14
24	Leaf	38.5	24.1	27.9	23.3	21.8	22.9	20.2	20.1
	Pedicel	-	33.1	30.3	28.6	23.5	23.9	21.6	22.9
48	Leaf	41.7	43.0	36.2	32.1	33.6	24.8	22.0	23.4
	Pedicel	-	42.6	48.1	37.1	43.0	31.3	29.0	29.1
72	Leaf	50.6	48.3	40.5	41.4	34.0	29.0	22.0	23.5
	Pedicel	-	47.0	48.6	42.1	44.0	31.6	31.2	30.6

* Average of 3 replicates

Table 42. Early growth of *S. pannosa* on detached pedicels and leaves

Hours incubation	Pedicel		Leaf	
	Percentage(a) germination	Germ tube (b) length (μm)	Percentage germination	Germ tube length (μm)
3	24.0 (1.1)	9.75 (0.2)	15.5 (1.9)	8.25 (0.2)
6	62.7 (5.3)	26.5 (0.6)	52.0 (14.1)	24.2 (0.8)
9	70.0 (2.2)	31.7 (0.6)	63.7 (7.8)	28.5 (0.7)

(a) Average of 4 replicates

(b) Average of 4 replicates, 20 germ tubes measured for each replicate

S D in brackets

Table 43. <u>Mildew growth on detached leaves and pedicels at 12°</u>

Hours incubation	Growth Index *	
	Pedicel	Leaf
24	24.3	19.4
48	29.5	35.3
96	59.0	54.5

* Average of 4 replicates

growth was slightly higher on pedicels at 24h but with further incubation, growth was similar to that on the leaf. At 96h although the amount of mildew growth was similar on both leaf and pedicel there was no sporulation on the latter. The fungus and pedicel died within 6 – 9 days while leaves remained alive and supported abundant sporulating colonies even after 12 days.

Expt (11) <u>Effect of the sucrose and kinetin on the logevity of detached pedicels and mildew development</u>

The results of Expts 8 – 10 indicated that although the initial growth of the fungus was rapid on detached pedicels, no sporulation occurred. It was therefore decided to feed detached pedicels with sucrose and kinetin solutions and to observe their effects on the logevity of the pedicels and the growth of the fungus.

Five day old flower buds were dusted with 24h conidia and placed upright in 0.5 cm diam, 2 cm high glass tubes containing the following; (i) 1%, 2%, 3% and 4% sucrose solutions and (ii) 10 ppm, 20 ppm, 40 ppm and 80 ppm kinetin solutions. The cut ends of the pedicels were dipped in the solutions. Three replicate samples of each solution were prepared with three flowers to a tube. The tubes were stood in a rack, placed inside a 18.0 x 9.0 x 7.5 cm polystyrene box lined with moist tissue paper, closed with the lid and incubated at 20 $\pm$ 1^{o} in a cabinet with a 16h photoperiod. The specimens were examined for mildew growth over a 2 week period and the glass tubes were topped up with the appropriate solutions when necessary. Pedicels in distilled water were used as controls.

Although mycelial growth was visible on all pedicels, sporulation was sparse and was observed after 6 days only on 3 flowers fed with 40 ppm kinetin solution. Pedicels in 20 ppm and 40 ppm kinetin solution appeared healthier and greener than pedicels fed with sucrose solution and distilled water which had begun to yellow after 5 – 7 days. Pedicels fed with sucrose solutions were contaminated. All pedicels started to decay by 10 – 12 days.

Expt (12) Mildew growth on attached leaves and pedicels in relation to their age

The above experiments failed to produce sporulation of rose mildew on detached flower pedicels possibly due to their rapid senescence. Hence an experiment was conducted using attached material.

One day old leaves and pedicels were tagged on vigorously growing, greenhouse - maintained rose plants. At 4 day intervals, for 24 days, each plant was transferred to the laboratory and the tagged leaves and pedicels (7 - 9 replicates of each type) were inoculated individually using a settling tower. They were then stood in a trough of water and incubated at 20 $\pm$ 1° in a cabinet with a 16h photoperiod. The plants were covered with polyethylene bags for 24h to ensure maximum germination of the conidia.

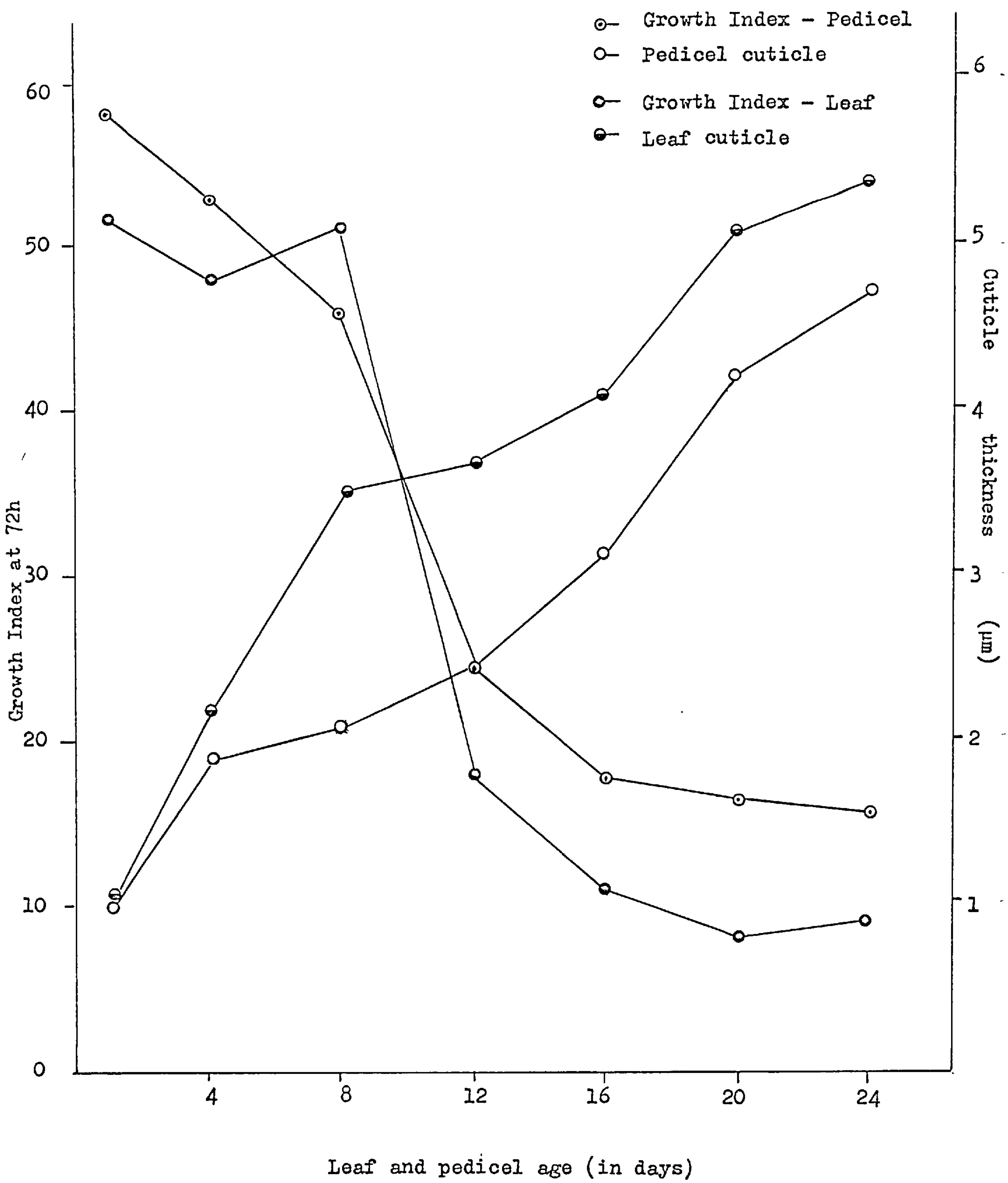
Before inoculation, cuticle thickness of the leaves and pedicels was measured. The GI of the fungus was estimated at 72h and sporulation was estimated after 7 days.

The results are very similar to those of Expt (8) in that susceptibility decreased rapidly with increase in age of tissue and pedicels remained susceptible for a longer period (Fig. 16). But there was no sporulation of any significance on the pedicels. However, there was a marked relationship between GI and the cuticle thickness for both leaf and pedicel (see Fig. 16) even though cuticle thickness was marginally less on the pedicel.

Expt (13) Pedicel inoculations

In Expt (12) some success was achieved in producing sporulation of the mildew on attached pedicels under laboratory conditions. But yet in the field under natural conditions flower pedicels appear to be particularly susceptible and are covered with closely packed masses of conidiophores. In an attempt to reproduce this effect 3 - 5 day old attached pedicels were

Fig. 16 Effect of age of attached leaves and pedicels on mildew growth



individually or mass inoculated, covered with polyethylene bags for 24h and maintained under the following conditions:

(i) in the $20 \pm 1^{\circ}$ cabinet with a 16h photoperiod

(ii) in the 18° cabinet with a 16h photoperiod

(iii) in the greenhouse at approx. $15 - 18^{\circ}$

(iv) the open rooftop of the Botany Dept. at South Kensington (September 1971)

Of these situations successful inoculations which resulted in good sporulation were obtained in 6 – 8 days with plants maintained in the 18° cabinet, but flowers of some greenhouse plants were also covered with sporulating mildew after 9 – 12 days. Limited sporulation in the 20° cabinet is difficult to explain since this temperature is generally regarded as favourable for mildew development. One possible explanation is that the cool/warm air blown directly into the 20° cabinet by a fan had a desiccating effect on the fungus.

(B) <u>Observation of mildew development on flowers under field conditions</u>

The development of mildew on flower buds was studied on rose bushes of cv. Frensham at Silwood Park. The plants were already heavily infected before the study was undertaken. Two hundred young buds (c. 6 – 7 mm long, 2 – 3 mm wide), still covered by the surrounding bracts, were tagged on 28 July 1971. About 10 – 25 pedicels were sampled at 3 – day intervals for 12 days and thereafter at 4 – 7 day intervals for a total of 29 days and were examined as follows:

(a) rate of growth of the flower (increase in pedicel length)

(b) visual assessment of mildew coverage on both receptacle and the pedicel

(c) celloidin peels removed from the surface, stained in lactophenol cotton blue and examined for mildew growth

(d) examined for growth abnormalities

(e) pedicels sectioned with a freezing microtome and the sections (15 μm) examined under the microscope.

These studies revealed that the flower buds were infected as soon as they emerged from the bracts, the calyx and the upper part of the receptacle being the first regions to get infected. Stained celloidin peels of these region showed that the initial inoculum consisted of germinating conidia and there was no evidence of mycelial growth. With the extension of the pedicel above the ring of bracts, the hyphal extensions of the germinating conidia spread downwards but it also received additional conidia from outside.

During the early stages of the development of the flower buds, the pedicels were covered with numerous reddish glands borne on stalks and they persisted on them for c. 7 - 9 days. The fungus spread in between and over the surface of these structures. Possibly these glands acted as spore traps although experimental evidence is lacking.

The growth of the flower was initially slow (Table 44), then proceded rapidly from 9 - 18 days. Maximum growth was attained after 18 - 20 days. The flowers started to open around the 17th day and by 20 days, over 90% of the flowers were fully open.

The observation indicated that the infection of the flower buds was usually completed within 7 - 9 days during which period the glandular structures persisted. This was followed by a period of maximum rate of extension of the pedicel but in a majority of the pedicels looked at, the fungus did not appear to spread or develop in the newly extended tissue. Instead, it was restricted to the upper (near the receptacle) portion of the pedicel.

About 80 - 90% of the infected buds grew normally and blossomed in about 3 weeks time without showing any adverse effects of the prescence of the parasite. However, a few severely infected buds were stunted and never opened. They remained attached for 2 - 3 weeks or longer and eventually decayed and dropped. In severely infected pedicels abnormal

Table 44. The development of rose powdery mildew on flowers of cv. Frensham in the field

Date	Total examined	No. infected buds	No. infected buds: No. with conidia present	No. infected buds: No. with sporulating colonies present	No. with swellings	Growth of flowers: Pedicel length(cm)*	
28.7.71	25	0	0	0	0	0.6	Pedicel covered with bracts
31.7.71	25	4	4	0	0	0.9	Pedicel exposed-covered with glandular hairs
3.8.71	25	19	19	15	0	1.0	Glands still present
6.8.71	25	25	25	25	2	1.3	Few glandular hairs still present
9.8.71	25	25	25	25	1	2.5	Rapid extension of pedicel
15.8.71	15	15	15	15	1	3.3	Flowers beginning to open
19.8.71	10	10	10	10	1	4.3	All open
26.8.71	10	10	10	9	0	4.0	Flowers withered

* Mean of 10 replicates

swellings were observed (Fig. 17). They occurred in a frequency of about 4 - 10% of infected buds anywhere along the length of the pedicel and could be detected as early as 8 - 10 days. Sometimes when only one side of the pedicel was infected, the swelling could be seen on that side.

Sections through infected pedicels revealed general browning of the epidermal cells (Fig. 18) but sometimes browning could be seen in 1 or 2 layers beneath the epidermis: The histochemistry of the infected tissue is discussed on p.160. Sections through swollen pedicels showed that occasionally there was a considerable amount of meristematic activity in the sclerenchymatous cell layer underneath the epidermis (Fig.19). This was possibly a defence mechanism to cut off the nutrient supply to the fungus although the latter was restricted to the epidermis. Often the swelling were due to hypertrophy of cortical cells. It is possible that auxins are involved with this phenomenon but experiments designed to determine auxin activity in diseased tissue failed to demonstrate its existence (Appendix 10 p.242).

Fig. 17. Infected pedicels of various ages of cv. Frensham showing relative increase in size. Uninfected pedicels of comparable age are on the left.

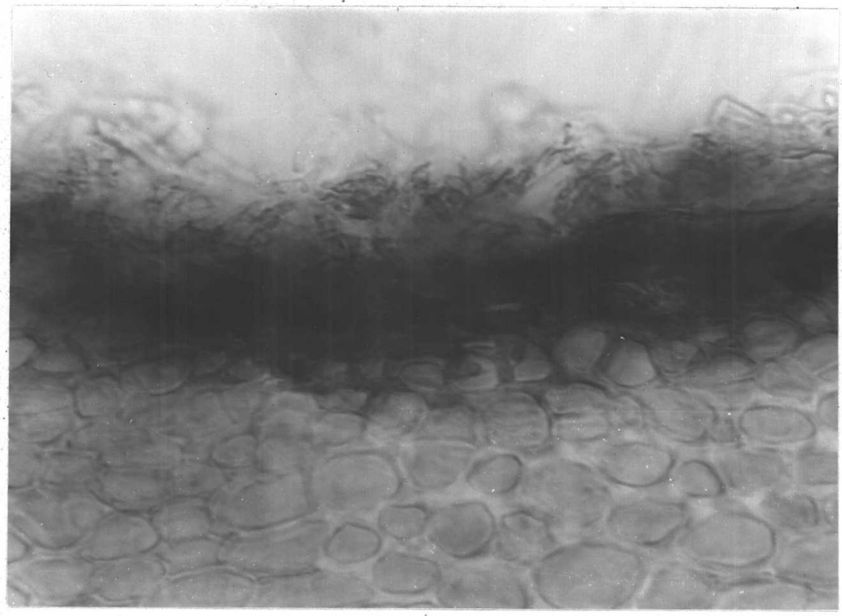


Fig. 18. Section of infected pedicel of cv. Frensham showing browning of epidermal and outer cortical cells X 3150 approx.

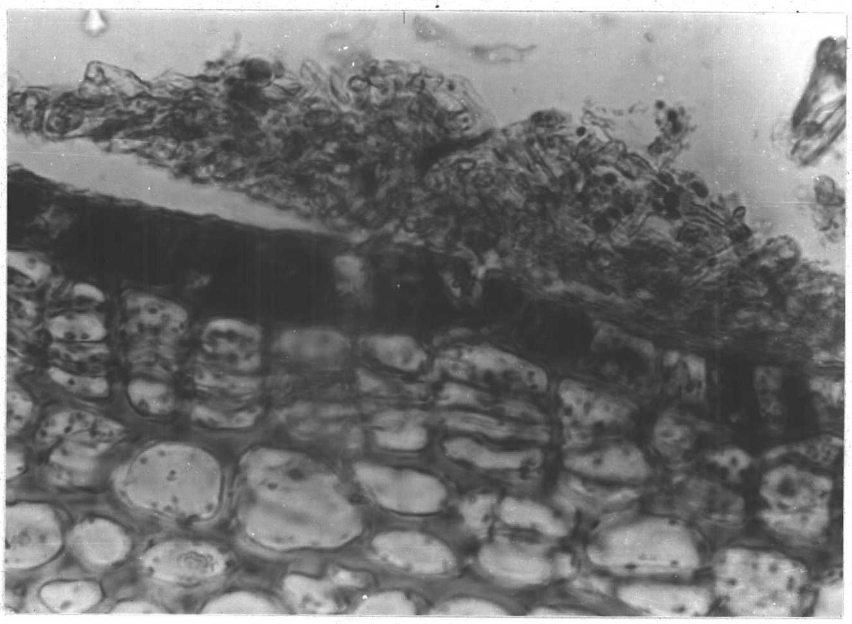


Fig. 19. Section of infected pedicel of cv. Frensham showing newly divided cells in the cortex X 3150 approx.

DISSCUSSION

Effect of leaf wetness

The experimental results indicate that periods of leaf wetness retarded the development of rose powdery mildew on detached leaves. The effect was most marked and persistent when leaves were wetted immediately after inoculation with conidia and the degree to which growth was retarded was directly related to the length of the wet period (Figs. 7 and 11, Tables 35 and 36).

Leaf wetness did not appear to inhibit the production of the first germ tube by the conidium. On the contrary, germination on mist-sprayed leaves was only marginally less than on dry leaves during the first 6h (Table 36). It did, however, reduce the number of germinated conidia that grew to a sporulating colony and increased the number that remained at the 1-germ-tube stage. Evidently the most susceptible phase occurs some 6 - 8h after inoculation when processes leading to penetration of the host tissue are initiated. Possibly surface wetness at this stage prevents the fungus from establishing contact with the host. Certainly in experiments where leaves were sprayed 6 or 12h after inoculation with conidia, the inhibitory effect on growth was less marked. This suggests that fungal development following initial penetration of the host is much less sensitive to surface wetness though here results must be interpreted with some caution because above values of c. 35 small differences in growth indices correspond to substantial differences in mycelial extension. Periods of leaf wetness following the initial penetration of the leaf could well have a considerable effect in delaying sporulation of the colony.

Even water sprays applied immediately after inoculation with conidia did not completely inhibit mildew development. In this instance, it is likely that colonies developed mainly from conidia which were between water droplets. In the field a similar situation could arise during periods of rain for the very young rose leaves which are most susceptible to mildew are those most difficult to wet and on these water stays in discrete droplets. Clearly, there are many aspects of the leaf wetness/mildew development interaction which require further study.

Mildew growth on pedicels

Severe infection of the pedicels and receptacles is a feature of powdery mildew development on the cv. Frensham in the field (Price, 1969) and casual observations on other cultivated roses suggest that these organs are more susceptible than other tissues. Price (1969) thought that the cylindrical form of the pedicel and its position might make it a particularly efficient spore trap but he could find no significant differences in numbers of powdery mildew conidia deposited per unit area of leaves and pedicels. One aim of the present investigation was to examine mildew growth rates on leaves and pedicels of comparable age in the expectation of quantifying the apparent differences in mildew development at these two sites. In the event, in experiments with both detached and attached leaves and flowers, few differences in growth, as assessed by an index were observed after the first 24h. The indications that development during the 6h or so was more rapid on pedicels may, however, be of some significance for it could mean that the minimum time required for infection i.e. the initiation of the first haustorium, is less. It could be that in the field, periods of favourable environment which are insufficient to allow conidial germination and mildew establishment on leaves suffice for these processes on pedicels. Nevertheless, in general, the results of these particular experiments on mildew growth were disappointing and the failure to reproduce on pedicels the extensive sporulation which is a feature of field infection suggests that some essential feature was not reproduced in the experimental system.

On the other hand, the rate of mildew growth *per se* need not necessarily be greater on pedicels than on leaves. Other factors could result eventually in more extensive colonization of the pedicels. One such factor is their longer period of susceptibility, by some 9 to 10 days, which these experiments clearly demonstrate. This appears to be linked with a slower rate of cuticle formation, there being for both pedicel and leaf, a marked inverse relationship between mildew growth and cuticle thickness. Possibly, too, in the field environmental differences between the two sites favour mildow development on the pedicel. For example, the upright position of the pedicel and its protection by the expanded receptacle may prevent water droplets from adhering during rain and so mildew develops unchecked during such periods whereas on leaves, as the leaf - wetness experiments indicated, mildew growth is delayed.

The abnormal growth of some infected pedicels is another feature of interest. This is mainly due to hypertrophy of the cortical cells, sometimes associated with meristemmatic activity, beneath the infected region. The significance of this reaction is not clear and requires further study. Although auxins are likely to be involved in this process, experiments failed to substantiate this supposition.

III. STRUCTURE OF THE INFECTED ROSE AND THE PATHOGEN

INTRODUCTION

Observations in the field of most varieties of cultivated rose attacked by the powdery mildew fungus Sphaerotheca pannosa suggest that the flower stalks are often more severely infected than other organs. Also the results of studies in Section II indicate that the increased susceptibility of the flower stalk is possibly associated with rapid initial development of the fungus on pedicels and with the slow rate of formation of the pedicel cuticle.

The primary aim of this investigation was to make a comparative study of the process of infection of leaves and pedicels in an attempt to understand the difference in susceptibility at the fine structural level. Available time allowed only the pedicels to be examined thoroughly and special attention was paid to the haustoria in the infected epidermal cells. The haustoria of Erysiphales, including the rose powdery mildew fungus were investigated as early as 1900 when Smith published an elaborate account on this subject. He concluded that the haustoria were isolated from the host protoplast by an extra - haustorial zone and identified a sac - like covering around haustoria. He considered that this sac - like sheath consisted of host cell wall debris. Since then numerous papers have been published on haustoria of fungal parasites and the use of electron microscope during the last decade has led to major advances in understanding the structure of host - parasite relationship. Recently, the literature connected with the fine structure of the host - parasite relationship has been reviewed by Ehrlich and Ehrlich (1971).

Almost all studies on the ultrastructure of powdery mildew haustoria have been made on species of Erysiphe (Ehrlich and Ehrlich,1963; McKeen et. al.,1966; Bracker, 1968; Stavely et. al., 1969; Kunoh and Akai, 1969; Edwards and Allen, 1970) and there are relatively few references to Sphaerotheca fuliginea (Hirata and Kojima, 1962; Dekhuijzen and van der Scheer, 1966). This investigation deals with the ultrastructure of the rose powdery mildew fungus S. pannosa.

MATERIALS AND METHODS

(1) Host plant material

Plants of cv. Frensham were raised from 25 - 35 cm long stem cuttings taken from bushes at Silwood Park. They were potted singly in 24 cm diam pots of John Innes no.1 compost, left in the open garden for 2 weeks and transferred to a greenhouse at the Chelsea Physic Garden on 1.11.71. Two to three weeks later new leaves developed and flower buds were produced after 6 - 8 weeks. The flower buds were tagged as described on p.104.

(2) Routine method for infecting pedicels

Plants were brought to the laboratory and 3 to 6 days after the pedicels were first visible they were individually inoculated with 24h conidia of S. pannosa as described on p.131. The inoculated plants were incubated at 18 $\pm$ 1^{o} in an illuminated growth cabinet with a 16h photoperiod. During the first 24h the plants were covered with polyethylene bags to provide a saturated atmosphere necessary for conidial germination. The diseased tissues were examined 5 - 12 days after inoculation. Appropriate controls were also examined.

(3) Histochemistry

Fresh material was sectioned (25 μm) with a freezing microtome and examined with the light microscope after staining with one of the following:-

(a) Lactophenol cotton blue for fungal material

Sections were stained in 0.1% cotton blue in lactophenol for 15 - 20 min , rinsed in water to remove excess stain and mounted in either water or 50% glycerine. Fungal material stain blue.

(b) Methylene blue for phenol cells (Beckman and Mueller, 1970).

Sections were mounted in water soluble methylene blue at concentrations of 0.05%, 0.1%, 0.5% and 1% (W/V). Phenol cells stain dark blue.

(c) Nitroso reaction for phenols and tannins (Jensen, 1962; Beckman and Mueller, 1970).

To sections immersed in a few drops of distilled water in a watch glass equal volumes of 10% $NaNO_3$, 20% Urea and 10% Acetic Acid were added in succession. After 3 - 4 min 2 volumes of 2N NaOH added. A cherry pink colour with the nitroso reaction indicates phenolic substances in cells

(d) Ferric Sulphate reaction for tannins (Jensen, 1962).

Sections were mounted in fresh solutions of one of the following:-

(i) 0.5% anhydrous ferric chloride in 0.1N HCl

(ii) 1% ferrous sulphate in 0.1N HCl

A blue precipitate (ferric sulphate) indicates tannins.

(e) Glutaraldehyde / Osmium tetroxide

Material fixed in glutaraldehyde/osmium tetroxide (p.156) as used in electron microscopy was also sectioned with a freezing microtome and examined under the light microscope. They were mounted in water without further staining.

(4) <u>Preparation for Transmission Electron Microscopy</u>

(a) <u>Fixation</u>

The following fixatives were used:-

(i) 1 - 2% aqueous $KMnO_4$ for 40 min to 2h at room temperature (Bracker, 1968).

(ii) A mixture containing glutaraldehyde (3%) and OsO_4 (5%) in 0.1M sodium cacodylate buffer (p^H 7) for 2h at 4° followed by 1% OsO_4 in 0.1M buffer for 2 - 4h at 4° (Littlefield's modification of Franke's glutaraldehyde / osmium tetroxide fixation procedure, personal comm.).

(iii) 2% glutaraldehyde in 0.1M sodium cacodylate buffer (p^H 7) for 10 - 24h at room temperature followed by washing in buffer, and 1% OsO_4 in 0.1M buffer for 3 - 4h.

Considerable difficulty was encountered in finding a suitable fixation method. Repeated tests for varying fixation times with methods (i) and (ii) gave poor results. The use of glutaraldehyde followed by OsO_4 gave better structural preservation than other methods. However, this method was found to be inadequate when tissues were fixed in glutaraldehyde for less than 18h.

Schedule for routine fixation with glutaraldehyde followed by osmium tetroxide (Glut / Os).

Pedicels were cut into c. 1mm pieces under 2% glutaraldehyde in 0.1M sodium cacodylate buffer, p^H 7. The specimens were then transferred to covered embryo cups containing the fixative and left at room temperature (18 - 20°) for 24h. The glutaraldehyde was obtained as a 25% aqueous stock solution (TAAB Laboratories).

Washing : The glutaraldehyde was drawn off with a Pasteur pipette and replaced by 0.1M sodium cacodylate buffer and left for 5 - 10 min. This was repeated twice. Care was taken to keep the specimens covered with buffer.

Post fixation : The buffer solution was replaced by 1% osmium tetroxide in 0.1M buffer and left for 3½h at 4°. The osmium tetroxide was prepared from a 4% (W/V) aqueous stock solution and mixed with distilled water and 0.2M buffer in proportions 1 : 1 : 2 respectively. The mixture was always prepared immediately before use although it could be kept for several days at 4° before it became discoloured.

(b) Dehydration

The osmium tetroxide fixative was withdrawn and the fixed material dehydrated at room temperature with a series of ethanol solutions graded in 10% steps. The solutions were changed at 10 min intervals. This was followed by 2 – 3 changes of absolute ethanol at 15 – 30 min intervals. The material was left overnight in a further change of absolute ethanol in a covered embryo cup at 5°.

(c) Embedding

The Epon resin mixture (3A + 1B) was used throughout. The absolute alcohol was renewed and left for 30 min before commencing the following treatments to introduce the resin. Epoxy propane was used as a link reagent to introduce resin to the tissues. The concentration was increased gradually, first by adding drops at 10 min intervals and then by replacing half of the mixture with epoxy propane at 3 x 10 min intervals. This was replaced by pure epoxy propane, which was renewed three times at 10 min intervals.

Epon resin mixture with the catalyst was introduced at this stage. One half of the epoxy propane was replaced by the resin mixture at 2 x 10 min intervals. Two thirds of the volume of epoxy propane / resin mixture was then replaced by the resin mixture and the embryo cups were left covered with lids for 1 – 2h at room temperature to allow the mixture to diffuse through the material. The lids of the embryo cups were then left half open overnight at room temperature to allow the epoxy propane to evaporate.

The following day 4 – 6 transfers to fresh resin at room temperature were made within the 24h. Afterwards, they were transferred to resin in flat polythene embedding cups (40 mm diam x 7 mm deep, of the type used as finger grips in the sliding doors of glass cabinets). Labels bearing the specimen numbers were also included. These were left overnight at room temperature and finally cured at 60° for 48h.

The hardened blocks were removed from the embedding cups and the embedded specimens were examined under the stereo – binocular microscope for selection of specimens and subsequent trimming in preparation for sectioning in the required planes.

(d) Sectioning

Rectangular blocks about 3 mm^3 containing the specimens were cut from the flat resin disks with a fine saw. These were mounted with Durofix on small lengths of perspex rod 8 mm in diameter in the orientation required and allowed to harden at 60° for 30 min. Each block was trimmed with a new razor blade under a binocular microscope to leave the specimen exposed at the apex in a small trapezoidal pyramid.

Sections were cut with a diamond knife on a LKB ultramicrotome. Those in the silver - grey range, and about 60 nm thick (Peachy, 1958) were picked up on 100 mesh copper grids with formvar film (see Appendix 11 p. 248) and dried.

(e) Section staining

Grids were immersed singly in drops of uranyl acetate (UA) placed in grooves cut in a polythene sheet 3 mm thick in a covered Petri dish and stained for 30 min at 60°. They were then held at the extreme edge with forceps and washed in a stream of glass distilled water from a Pasteur pipette and released into a large Petri dish of water where they were left for 5 - 10 min supported in grooves of a piece of reeded glass.

The grids were picked up with forceps and whilst still gripped at the extreme edge, the excess water between the prongs of the forceps was carefully withdrawn with the point of a triangle of filter paper.

They were immersed singly for 10 min at room temperature in drops of lead stain (Reynolds, 1963) contained in grooves in a polythene sheet which was enclosed in a Petri dish. Before adding the lead stain, a few moist pellets of NaOH were placed inside the Petri dish to absorb CO_2 and prevent the precipitation of lead carbonate. Access to the grids was gained by slightly lifting the Petri dish lid on one side.

Afterwards the grids were washed in a stream of water and then immersed in water in a Petri dish as described above. They were then picked up with forceps, washed in a further stream of water and dried as described above. Each grid was placed in a gelatin capsule. For convenience in maintaining the sequence of sections and grids about 10 - 12 capsules were previously glued between two strips of card and placed in a 35 mm film negative envelope.

Material was viewed and electron micrographs were taken with either an AEI EM 6 at 75 and 100 keV, an EM 6B at 60 keV or AEI EM 7 at 900 keV.

RESULTS

(1) Histochemistry of rose cells

(a) Laotophenol cotton blue

Haustoria in epidermal cells were stained dark blue. The cuticle and the host cell walls were also stained blue. The haustorium appeared as a globular structure enclosed by a granular sheath (see Fig. 5). These granules were probably the finger - like protrusions of the haustorial body seen in the EM. The other contents of the rose cells were either unstained or stained light blue.

(b) Methylene blue

The cytoplasm in all cells was stained blue or greenish blue with the highest concentration of the stain in the cell walls. The epidermal and collenchyma cells of the pedicel cortex (2 - 3 layers beneath the epidermis) were stained dark blue. Parenchymatous cells of the cortex, pith and vascular tissues were stained greenish blue. The staining characteristics of infected and uninfected tissues were similar.

(c) Nitroso reaction

Infected tissues gave a slight reaction to this test. Immediately after the addition of NaOH, the epidermal cells of the infected pedicels and the outer cortical cells turned light pink but the colour disappeared in 1 - 2 min. Heavily infected tissue also reacted similarly but the pink colour was seen to move out of the cells into the solution making the latter reddish - brown. No such reaction was observed with uninfected tissue.

(d) Ferric sulphate reaction

No blue precipitate indicating tannins was observed. All cells remained unstained.

(e) Glutaraldehyde / osmium tetroxide

Cell walls stained brown to black. Their contents remained relatively unstained except for those of the epidermal cells in the infected tissues where almost all the cells stained a shade of dark brown. The uninfected tissues reacted similarly but their epidermal cells stained less intensely. The cortical cells whose contents were opaque or very dense when viewed in the electron microscope were not seen to be stained.

(2) Fine structure of the host and the fungus haustorium

(a) Uninfected epidermis

The epidermal cells were approximately 10 x 11 µm in size bound by a wall 0.3 µm in thickness. The surface wall had a well marked cuticle (Fig. 20). The cells were highly vacuolated often with one or two large vacuoles and sometimes several small ones. The vacuolar contents when shown were usually of the fibrillar - granular type but sometimes the vacuoles contained small amounts of darkly staining electron opaque material.

The groundplasm mainly occupied thin regions between vacuoles or a vacuole and the wall. Numerous spherical to ovoid mitochondria with villi were present. Large lipid droplets that were not membrane bound were abundant. Chloroplasts were found in some cells which were presumed to be guard cells but often only starch grains were seen. Rough and smooth endoplasmic reticulum (ER) were abundant. One to three golgi bodies were also found in a single cell profile. The cells contained a single nucleus, c. 5 µm in diam bound by a double membraned nuclear envelope. A darkly staining nucleolus was often found in the centre of the nucleus.

(b) Infected epidermis

Invaded cells and adjacent uninfected epidermal cells were conspicuously different in that the vacuoles contained large amounts of electron opaque granular material (Fig. 21). The vacuolar contents were often shrunken by fixation (Fig.21). The haustorium occupied a large part

(c. one third its volume) of the invaded cell. The vacuole size was often correspondingly diminished to accomodate the haustorium. No other obvious changes were observed in the invaded cells. Mitochondria and chloroplasts were structurally normal and the golgi bodies were not unusually abundant. The host nucleus was frequently seen in the proximity of the haustorium and was often distorted to follow the contour of the invading organ. The contents of the nucleus were not different from that of the invaded cells.

(c) Cells of the pedicel cortex

Cells of the uninfected cortex were highly vacuolated often with a single large vacuole with a thin lining of cytoplasm adjacent to the cell wall. Most vacuoles were either devoid of contents or more likely their contents were not differentially stained. In some, a few scattered granules were seen. The nucleus was small (2–3 μm diam) and was often found abutting the cell wall with a thin covering of cytoplasm around it. Numerous spherical mitochondria and small vesicles containing lipidic substances were also found in the cytoplasm.E R and golgi bodies were frequently found.

In heavily infected tissue the large vacuoles of some cortical cells, and in particular those of the three layers below the epidermis were filled with extremely opaque granular material similar to that found in vacuoles of the infected epidermal cells (Fig. 22). The texture of this material varied from small discrete granules to large aggregates of coalescent amorphous globules. These dark cells were seldom distinguishable and then only with difficulty under the light microscope whereas in the E M they were conspicuous even without staining with UA/Pb. The cytoplasm of these cells was found adjacent to the cell wall. Cytoplasmic organelles were not easily seen presumably due to poor fixation which was a common problem with most infected tissue. Fig. 22 shows apparent cell division in a dark cell but the dividing wall is incomplete. It possibly increases its surface area but its significance is not known.

The frequency of dark cells in infected and uninfected pedicels was estimated by examination of sections in the electron microscope. The cell numbers were estimated from three sections from pedicels of each infection age group. Where sections were from one pedicel, sampling was limited to every 5th section. All cells in an area 10 epidermal cells wide by 6 cells deep were counted at x 1000 magnification. The frequency is expressed as a mean percentage of the total.

The results (Table45) show that there was a marked correlation between infection age and the number of dark cells. Dark cells also occurred in uninfected tissue, but these were relatively few.

(d) Haustorium

(i) Occurrence

Haustoria are restricted to the epidermal cells but not all cells are infected 12 - 14 days after inoculation of the pedicel. Usually there is one haustorium in a cell but occasionally two have been observed in severely infected tissue (Fig.23). A mature haustorium with its ensheathing layers occupies about one third of the volume of the host cell and is always enclosed with the host cytoplasm and the host nucleus lies beside it (Fig.24).

(ii) Morphology

The section of a haustorium (Fig. 24) shows a tabular neck approximately 3.5 µm in length and 0.6 - 1 µm diam connecting the superfi - cial cell (appressorium) with the ellipsoidal haustorial body. The appressorium connection has been broken in all sections observed. The body is surrounded by profiles of haustorial lobes one of which shows connection with it (Fig. 25). The haustorial body and the lobes are enclosed in a common matrix which is in turn surrounded by the host plasmalemma. This will be discussed in detail below. There is a septum at the junction of the neck and the body. The septum thins towards its centre and has a central pore (Fig. 26).

Table 45. Frequency of dark cells in epidermal and cortical cells in pedicels after various periods of infection

Age of infection (days)	Mean percentage dark cells			
	Infected pedicel		Uninfected pedicel	
	Cortex	Cortex & epidermis	Cortex	Cortex & epidermis
1	4	16.1	-	-
5	36.0	39.4	1	9.4
8	22.6	24.3	2	4.4
12	33.0	34.9	1.3	4.9

The lobes frequently show connections with the body in epidermal cells exposed in tangential and longitudinal sections of the pedicel (Figs. 27, 28). Transverse sections seldom show connections between these. This suggests a bipolar structure of the haustorium similar to that of Erysiphe graminis (Bracker, 1968). The lobes often show curving and reflex through 90° so that they lie close to the surface of the body. They are sometimes branched (Fig. 28). There is evidence that most lobes arise at the proximal (nearer the neck) region of the poles. The morphology of the lobes and their connections is particularly well shown in thick (½–1 µm) sections (Fig. 29). These were examined in an AEI EM 7 electron microscope at 900 keV or AEI EM 6 at 75 and 100 keV. The different accelerating potentials did not give significantly different images.

(iii) Haustorial cytoplasm

The cytoplams of the haustorial body is continuous with that of the lobes (Figs. 25, 27,28). A single nucleus c. 2.5 µm in diameter occupies the centre of the body (Fig. 23). The nucleus is bounded by a double membraned nuclear envelope. Only one haustorial nucleus showed a nucleolus. This was darkly stained. Mitochondria are abundant in the haustorial body and in the lobes. Their profiles suggest that they are elongated or cylindrical (Fig. 25). The cristae are plate - like as in other Ascomycetes (Bracker, 1968) but unlike those of rose mitochondria. ER profiles, many of which have ribosomes attached are frequent (Fig. 25,30). Free ribosomes are also found in the cytoplasm (Fig. 29). Large vacuoles or vesicles are sometimes seen in the body (Figs. 23, 30) but small, membrane - bound vesicles are observed more frequently (Fig. 25). Some of the vesicles show no contents but most include small vesicular profiles and granules with interconnections (Fig. 23, 25). Fig 30 shows a crystal lattice inside a vacuole. Osmophilic droplets without bounding membrane and which are probably lipids are also found in the cytoplasm of the body (Fig. 25). These are irregular in size and are sometimes coalescent. Darkly stained granules (c. 20 nm) in clusters about 0.1 µm diam are frequently found in the haustorial body and lobes in sections stained with uranyl acetate for the normal period (30 min) followed by lead for 20 min (Figs. 29, 30). The granules are visible with difficulty in sections stained with lead for shorter periods and are probably glycogen similar to the glycogen deposits in other fungi (Coffey et.al. 1972).

The cytoplasm of the neck differ in several respects from that of the body. It contains extensive vacuoles (Figs. 31, 32). Their membranes are not shown clearly but their contents are of the fibrillar - granular type. The cytoplasm of the neck also contains a few profiles of ER, many free ribosomes, a large multilayered membrane complex and osmophilic droplets which are probably lipidic in nature (Figs. 33, 34). The lipid droplets appear over vacuoles and groundplasm suggesting movement during sectioning (Fig. 31). Nuclei or mitochondria have not been observed in the neck. The septum between neck and body has a pore which is frequently occluded on each side by clusters of rounded and coalescent electron dense bodies (Fig. 26). These may be lipid droplets or Woronin bodies which are frequently found in this position but crystalline structure suggesting the latter (Bracker, 1967; Carroll, 1966) was not observed.

(iv) Wall structures, limiting membranes and sheathing substances.

The haustorial neck is slightly constricted at the point of penetration of the host cell wall (Figs. 24, 31). The penetration pore is a smooth - surfaced aperture about 0.3 µm in diam. No gap is shown between the surfaces of the fungal and the plant cell walls. The host cell wall shows fibrills which appear unaltered at the point of penetration except for a slight general darkening (Fig. 31). The structure of the cuticle also appear unaffected around the site of penetration. The fungal wall in the neck region is electron dense (Figs. 33, 34) but in the distal half it gradually changes in opacity and continues around the body and the lobes as an electron lucent layer (Fig. 25). In sections stained with UA / Pb for longer periods the fungal wall around the body and lobes appears fibrillar and darkly stained (Figs. 29, 30).

Surrounding the neck is a collar of material which extends over a wide area on the inner surface of the host cell wall and tapers in a cone towards the haustorial body (Figs. 31, 32, 33, 34). It usually ends near the septum between the haustorial neck and body (Fig. 34) but sometimes extends only just over half way to the septum (Fig. 32).

The ultrastructure of the collar is different from that of the host wall and the transition between the two surfaces in abrupt (Fig. 34). It is composed of fibrillar and irregularly and loosely - arranged vesicular material (Figs. 33, 34). The peripheral layer of the collar is stained more darkly and does not contain membranous material. Fibres are most conspicuous in this region (see Figs. 33,34). The outer margin of the collar may be smooth (Fig. 32) or irregular with small outgrowths arising from it (Fig 34). It is bounded by the plasmalemma of the host (Figs.33,34). The internal diameter of the collar is slightly larger than the neck leaving a cylindrical channel between it and the neck and continuous with the cytoplasm of the host (Figs. 33, 34, 35). The channel extends to or almost to the host wall (Figs. 34, 35). The inner margin of the collar has irregular and deep clefts (Fig. 35).

The collar causes invagination of the host plasmalemma (Fig.33) but in places it is difficult to follow this membrane around the rim of the collar and into the channel because of the high electron density of other material in this region. Parts of the fungal plasmalemma nearby are also obscure in some micrographs. However, by examination of a selection of micrographs the host plasmalemma can be distinguished in all regions with the possible exception of the inner surface of the rim of the collar.

There are discrete zones of electron opaque material on the inner and outer surface of the neck wall at a position approximately half way along the neck (Figs. 31, 32, 33, 34). The outer zones are more extensive than the inner and the outer and the inner lie respectively opposite each other (Figs. 32, 34). This observation and the appearance in tangential sections (Fig. 36) suggest that they are parts of two rings. The outline of the material is lobed (Fig. 32) and the similarity in texture of material in the cytoplasm of the neck suggests that the rings are lipidic in nature. The rings obscure the fungal and host membranes in these regions (Figs. 33, 34). The change in the staining characteristics of the wall mentioned above do not coincide with the ring structures but occur nearer the haustorial body (Figs. 33, 34). The fungus plasmalemma is densely stained for most of its length (Figs. 27, 28, 30,33) and is slightly undulated.

The wall and the ensheathing substances around the body and the lobes are different from those of the neck region. The transparent layer about 0.12 µm thick adjoining the fungus plasmalemma is probably the fungal wall (Figs. 25, 28). Outside the wall and forming a common matrix between the body and the lobes is the sheath matrix which is electron lucent except that it contains a reticulum of fine granules similar to the contents of plant vacuoles (Fig. 25). The sheath matrix is limited by a sheath membrane which is an interface between it and the host cytoplasm (Figs. 23,24). The sheath membrane is continuous around the haustorium and no pores are visible in it. Small vesicular profiles are seen near the periphery of the sheath membrane (Fig. 33). These may be invaginations of the membrane.Large clusters of these vesicular profiles are especially common below the junction of the neck and the body (Fig. 35).

The lengths of plasmalemmas of fungus and host in the body region of Fig. 21 were measured with a thread and found to be 58 cm and 28 cm respectively. This indicates that the surface areas differ by a factor of two (Freere and Weibel, 1967).

Occasionally the entire haustorium is completely enclosed by material indistinguishable from that of the collar (Fig. 37). The body is small and there are numerous small lobes. No septum or rings are seen. This poor development associated with the enclosure suggests that under some circumstances the papilla and collar material may provide a mechanism resisting infection.

Explanation of figures

The following abbreviations are used.

b	haustorial body	hw	host wall
c	collar	l	lipid droplet
ch	channel	lo	haustorial lobes
cl	chloroplast	m	mitochondria
cr	crystal lattice	mc	membrane complex
cu	cuticle	n	neck
er	endoplasmic reticulum	ne	nuclear envelope
fn	fungal nucleus	r	rings
fp	fungal plasmalemma	rb	ribosomes
fw	fungal wall	s	septum
g	golgi apparatus	si	sheath invagination
gl	glycogen granules	sm	sheath membrane
hn	host nucleus	sma	sheath matrix
hp	host plasmalemma	v	vacuole or vesicle

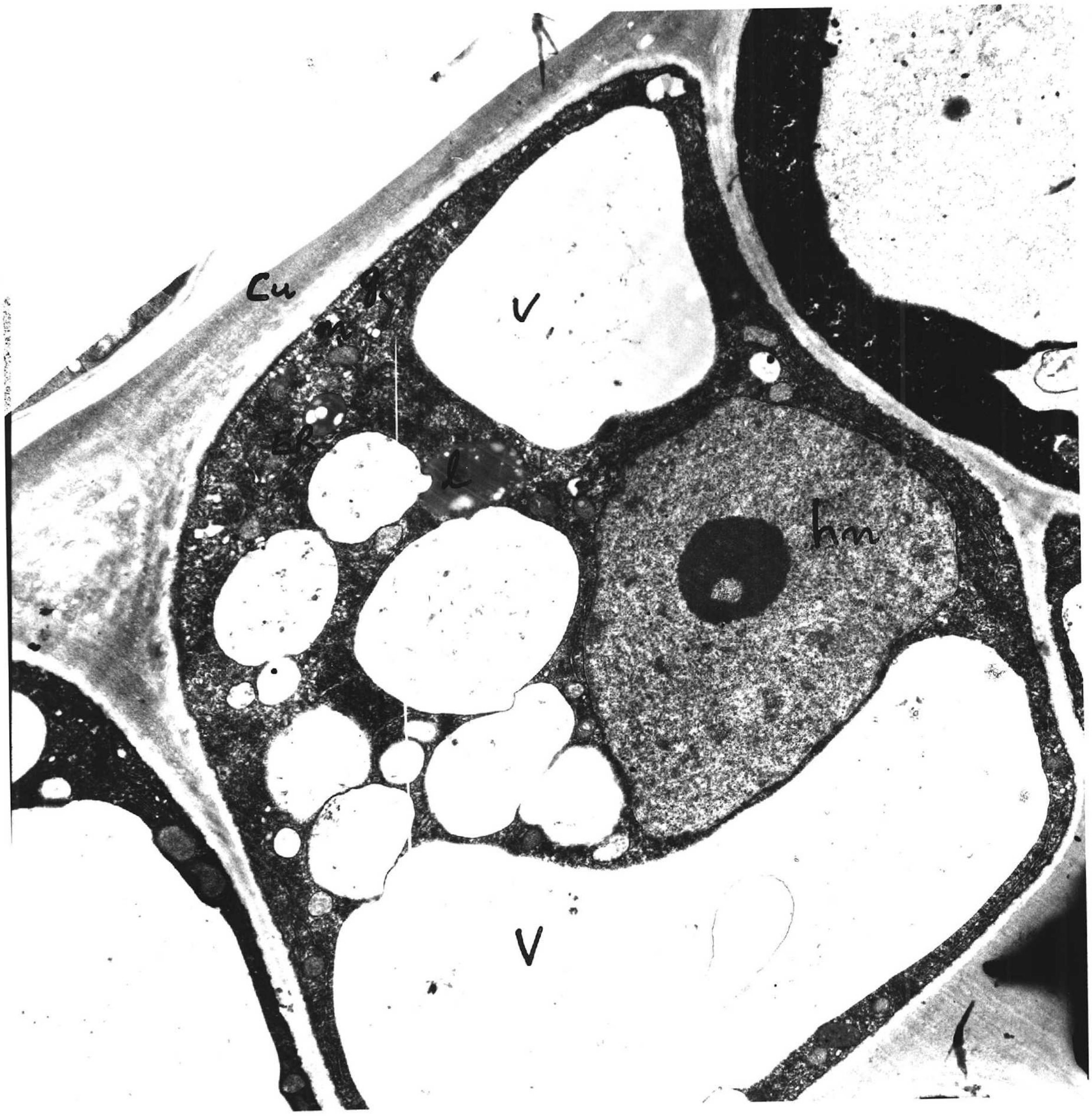


Fig. 20. Uninfected epidermal cell bound by a cell wall with a well marked cuticle (cu) on the surface wall. The cell shows a large nucleus (hn) with a nucleolus, extensive vacuoles (v), numerous mitochondria (m), lipid droplets (l) ,golgi apparatus (g) and ER. x 11,200.

Fig. 21. Epidermis with one infected cell showing a haustorium with the host nucleus (hn) beside it. The vacuoles (v) in all the cells contain electron dense material. x 10,900.

Fig. 22. Cells of the infected pedicel cortex showing the large vacuoles (v) containing electron opaque material of varying texture. Note dark cell with incomplete dividing wall. x 4,500.

Fig. 23. Infected epidermal cell showing two haustoria. The vacuoles (v) are smaller than in the uninfected cells and contain electron opaque material. The host cytoplasm shows ER and mitochondria (m). Note haustorial body (b) with nucleus (fn) bounded by a nuclear envelope (ne), mitochondria (m) and vesicles (v) containing granules with interconnections. Profiles of haustorial lobes (lo) also containing vesicles and mitochondria surround the body embedded in a sheath matrix (sma). Some lobes show connection with the body. The sheath membrane (sm) limits


b
b
v
sm
v
ne
fm
m
sma
ER
m

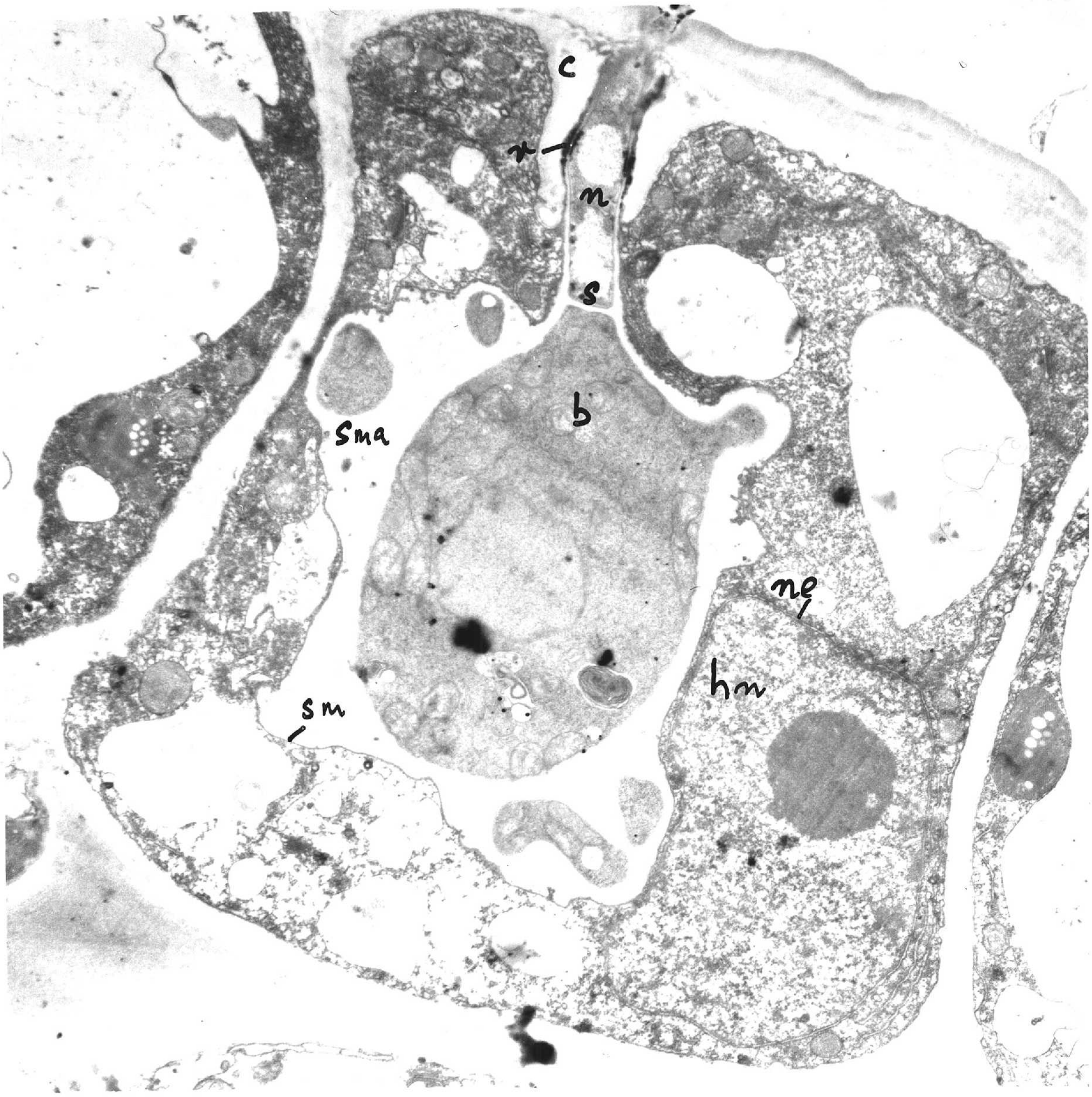


Fig. 24. Median section of a haustorium showing the tubular neck (n) separated from the ellipsoidal body (b) by a septum (s). The sheath membrane (sm) limits the electron lucent sheath matrix (sma) from the host cytoplasm. Host nucleus (hn) limited by nuclear envelope (ne) containing a nucleolus lies beside the haustorium. The rings (r) in the neck are shown. The collar (c) surrounds the base of the neck. x 11,300.

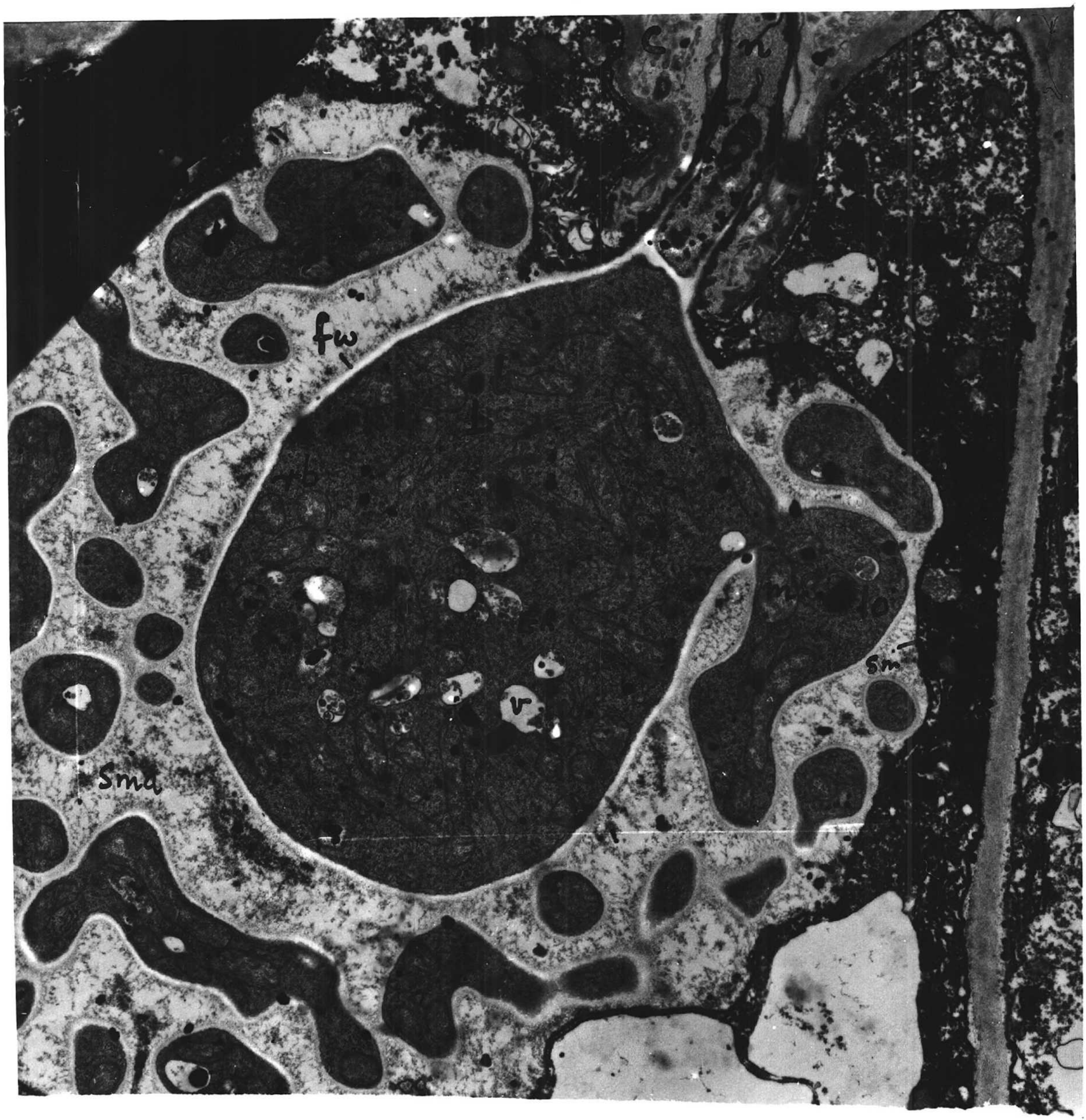


Fig. 25. Section of a mature haustorium showing a part of the neck (n) surrounded by the collar (c) and the body (b) and lobes (lo) one of which shows cytoplasmic connection. Note vesicles (v), mitochondria (m) lipid droplets (l), ribosomes (rb), ER in the body and lobes. The body and lobes are bound by an electron lucent fungal wall (fw). Also note the reticulum of fine granules in the sheath matrix (sma) bounded by the sheath membrane (sm). x 14,200.

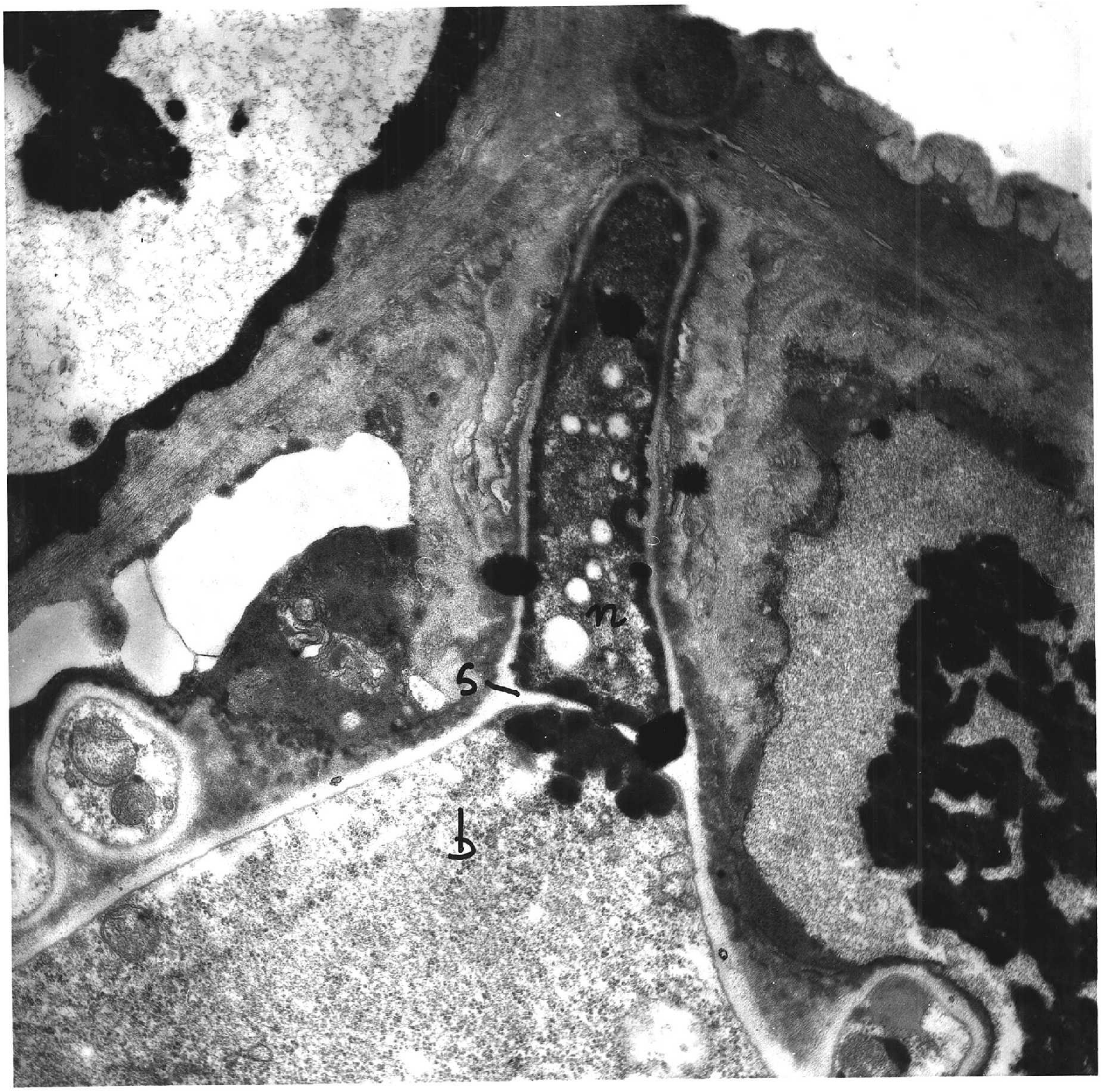


Fig. 26. Section of haustorial neck showing a septum (s) between the neck (n) and the body (b). The septal pore is occluded by a cluster of electron dense bodies. x 20,400.

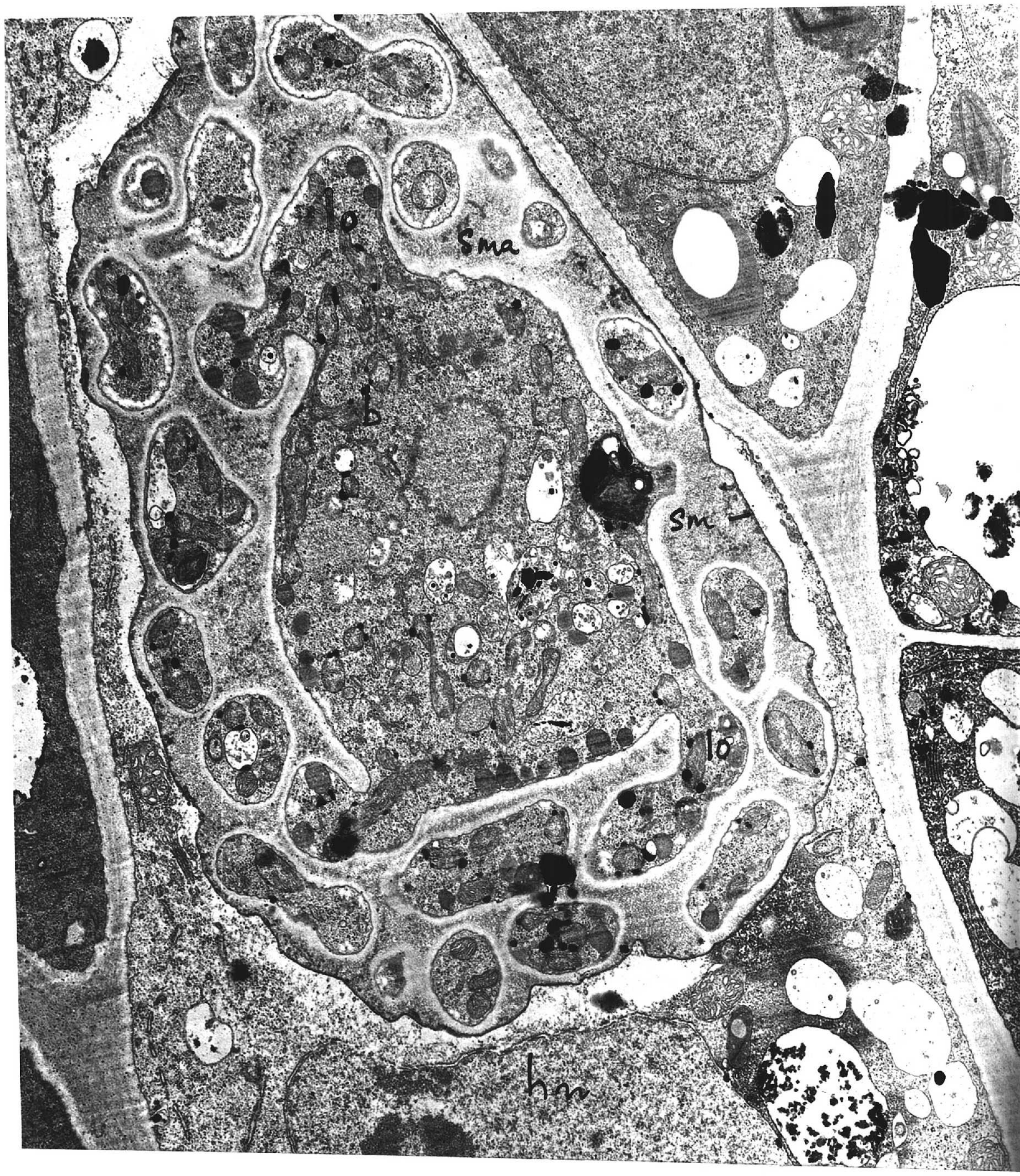


Fig. 27. Longitudinal section of the pedicel showing haustorial lobes (lo) connected to the body (b) at the two poles. Numerous profiles of branched lobes embedded in a membrane bound (sm) sheath matrix (sma) are seen around the body. The host nucleus (hn) lies beside the haustorium. x 13,900.

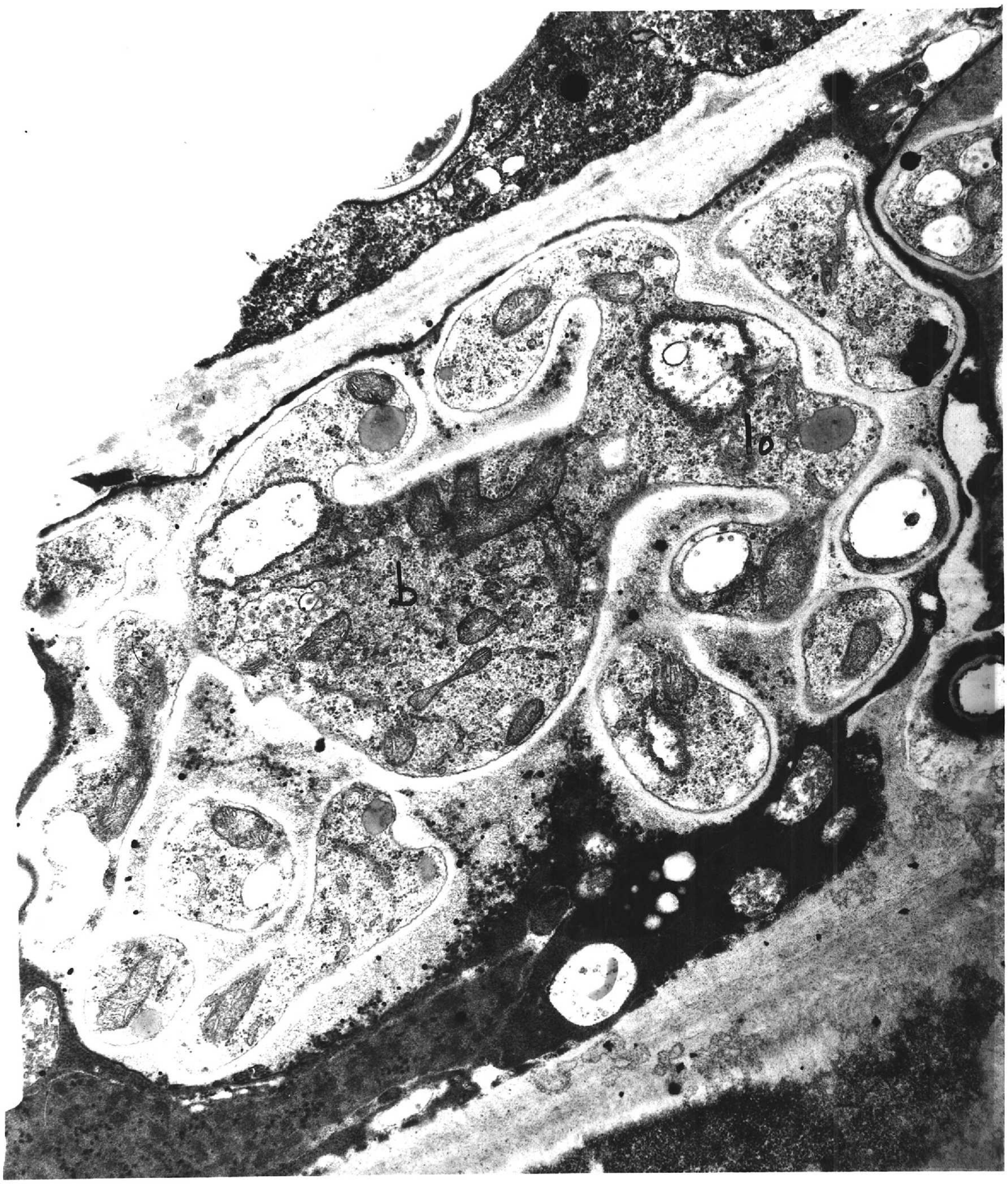


Fig. 28. L.S. pedicel showing polar connections of branched haustorial lobes (lo) with the body (b). x 20,200.

Fig. 29. A part of haustorial body with lobes (lo) in a section stained with UA for 30 min and Pb for 20 min showing glycogen granules (gl) in clusters. Free ribosomes (rb) in the cytoplasm are shown. x 39,000.

Fig 30. A part of haustorial body showing a crystal lattice (cr) in a vacuole (v). The fungal plasmalemma (fp) is undulated and darkly stained and the fungal wall (fw) shows fibrills. Glycogen granules (gl) are present in the fungus. x 30,400.

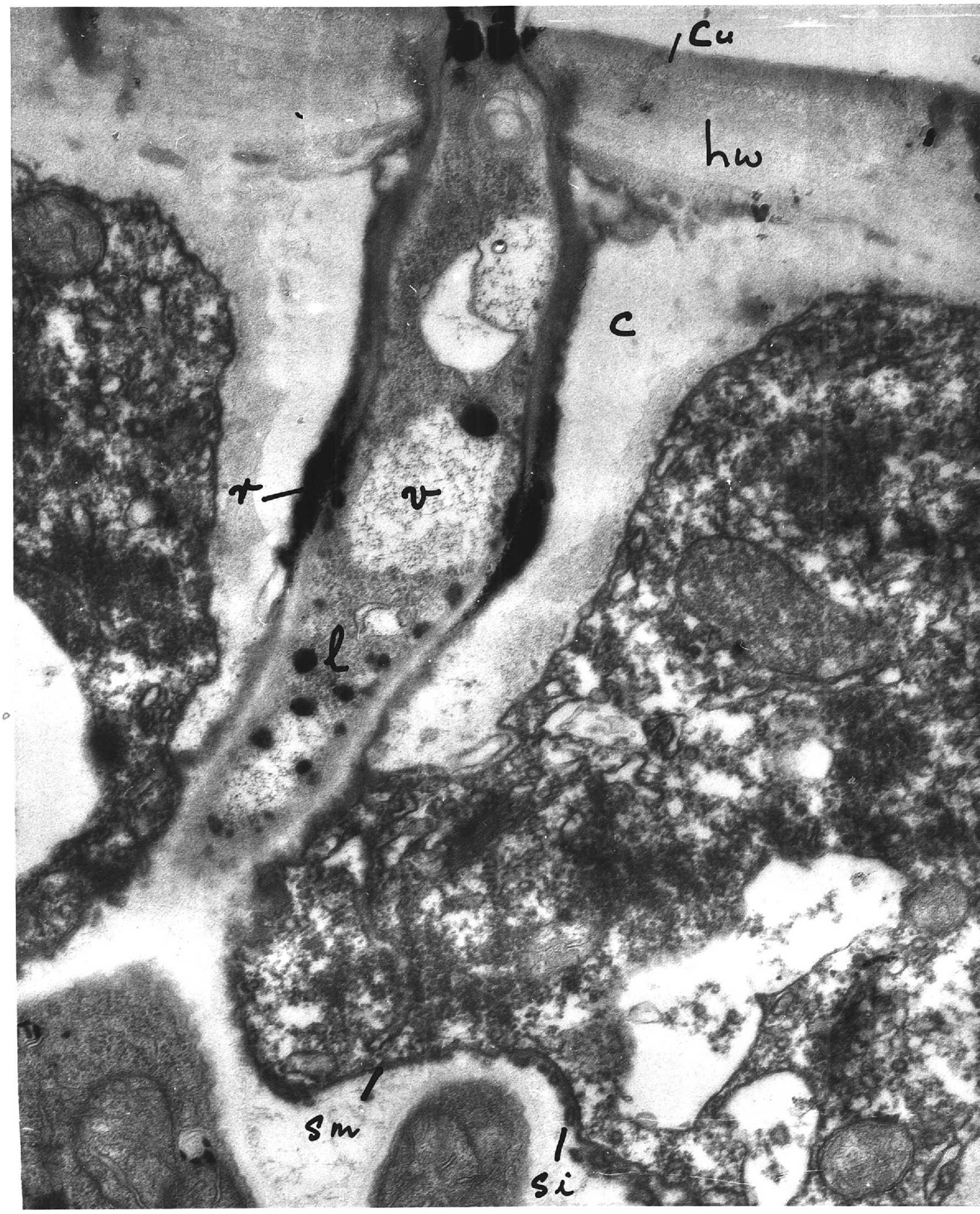


Fig. 31. Section through haustorial neck. Note the unaltered cuticle (cu), host wall (hw) and the constricted haustorial neck at the point of penetration. Also seen are the vacuoles (v), lipid bodies (l), the rings (r), collar (c) and invaginations (si) of the sheath membrane (sm) x 37,000.

Fig. 32. Haustorial neck showing the outer and inner rings (r) and droplets of electron dense material. Note collar (c), septum (s), vacuoles (v) containing granular material and a paucity of their contents. x 47,400.

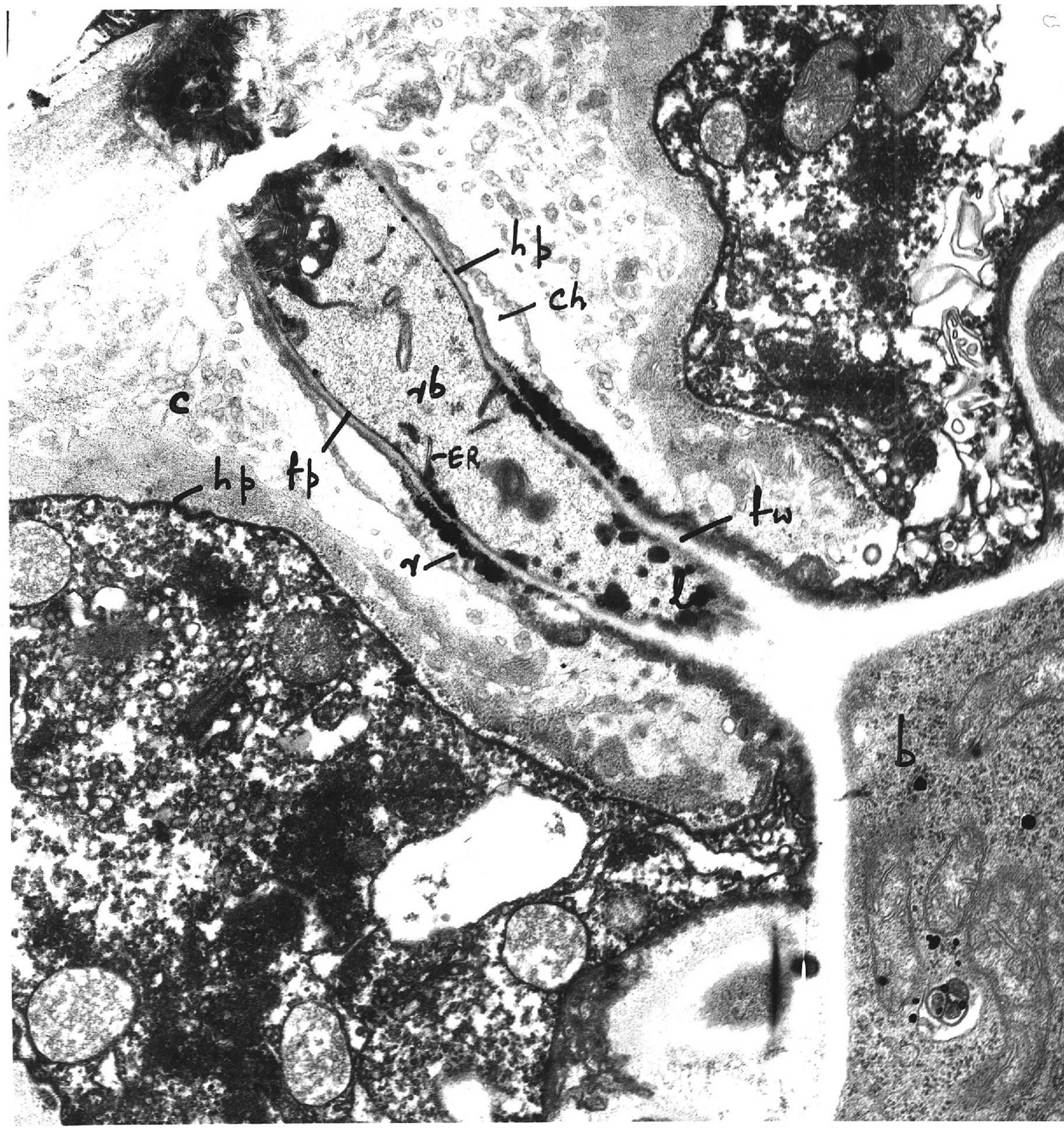


Fig. 33. Section through neck showing the channel (ch) surrounded by the collar (c). Parts of the host plasmalemma (hp) are seen lining the channel. The host plasmalemma covers the collar and continues over the rim of the channel. The fungus plasmalemma (fp) is darkly stained. Both plasmalemmas are obscure in the region of the rings (r). The fungal wall (fw) changes in its staining characteristics towards the body (b). Note lipid droplets (l), free ribosomes (rb) and ER in the neck cytoplasm. x 30,000.

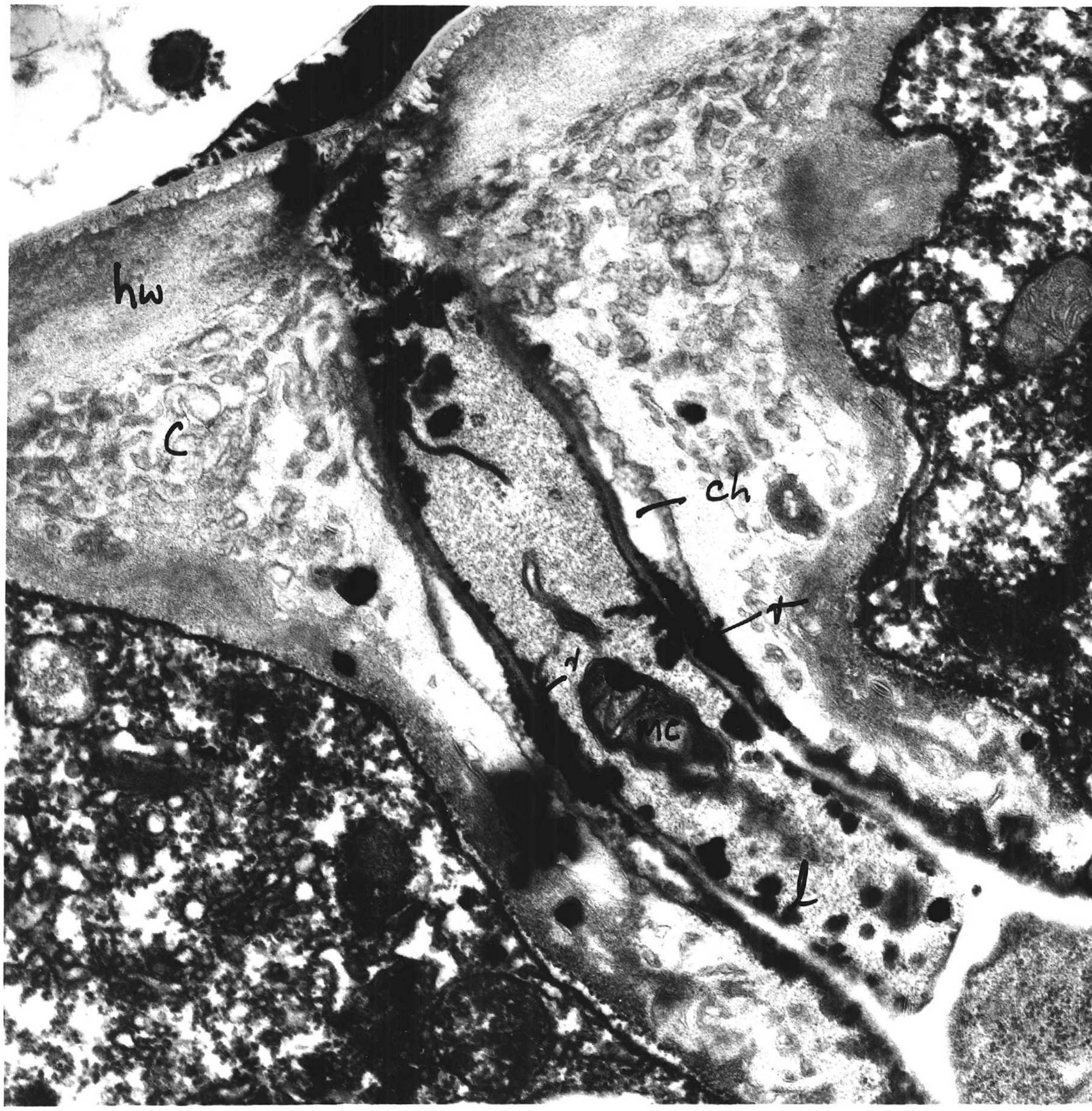


Fig. 34. Haustorial neck showing the channel (ch) formed by the collar (c) extending from the host wall (hw). Note the inner and the outer rings (r) on the fungus and host plasmalemmas, the change in wall appearence near the rings, the large number of lipid droplets (l) in the neck and the multilayered membrane complex (mc) in the neck cytoplasm. x 38,300.

Fig. 35. Section of neck showing channel almost extending to the host cell wall. The inner surface of the collar (c) is pitted. The sheath membrane (sm) with its invaginations (si) can be followed into the channel. x 19,800.

5

Fig. 36. Tangential section of the neck region of a haustorium showing deposits of dark material of the outer ring (r) . x 37,700.

Fig. 37. Enclosure of haustorium by material similar to and continuous with the collar (c). Note absence of septum and poor development of haustorial lobes. x 11,000

DISCUSSION

Dark cells

The dark cells frequently seen in large numbers in invaded tissue were probably phenol cells although they failed to give typical reaotions with the appropriate tests. Phenol cells are known to occur in many plants including various genera and species of Rosaceae (Politis, 1957). Beckman and Mueller(1970)believe that many "tannin" cells reported in earlier literature (Easau, 1965; Swain, 1965) are possibly phenol cells in active life whioh through chemical treatment have become decompartmentalized and polymerized to form tannins.

There are several reports of increases in phenols in plant tissue following injury or infection (Cruickshank, 1963; Kuć, 1966) but their funotion is still poorly understood. It has been suggested that following infeotion or injury they may play an important role in inhibiting enzymatic hydrolysis, directly inhibiting mycrobial development and in the process of repair (Beckman and Mueller, 1970).

Unlike in the cells of banana roots (Beckman and Mueller, 1970) or Eucalyptus tissue (Wardrop and Cornshaw, 1962) where phenols occur in specialized cells in rose cv. Frensham there appears to be a rapid activation of phenol synthesis following infection with S. pannosa. Further study in this connection would be most interesting and emphasis should be given to the development of satisfactory staining methods for phenols and phenol cells to be positively identified under the light microscope.

Haustorium

The haustorium of *Sphaerotheca pannosa* is similar in many respects to that of *Erysiphe graminis* (Ehrlich and Ehrlich, 1963; Bracker, 1968; Kunoh and Akai, 1969). Both have a central body from which arises branched lobes or protrusions which are embedded in a common sheath matrix and bounded by a sheath membrane that is continuous with the host plasmalemma. It conforms to the general conoept of struotures of obligate parasites in that it does not contact the host cytoplasm directly and penetration of the host plasmalemma does not occur. In E. graminis the finger - like protrusions are straight and project away from the central body and the sheath follows the contours of these structures (Bracker, 1968). In S. pannosa the protrusions are curved so that they lie close to the body separated from the host by a globular sheath. Hirata (1937), Hirata and Kojima (1962) and Dekhuijzen and van der Scheer(1967) showed similar curving protrusions to emerge from the central body of the haustorium of S. fuliginea from cuoumber but Dekhuijzen and van der Scheer (1969) reported that these protrusions do not always occur.

Penetration of the host cell wall seems to be by a precise process of entry similar to that described for Uromyces appendiculatus with Frenoh bean (Hardwick et.al. 1971) in that it is a smooth surfaced aperture with no swellings or signs of erosion. There is no clear evidence to suggest whether entry of the fungus is by chemical or mechanical means. A slight darkening of the host cell wall around the penetration site suggests chemical modification of the cell wall in this region and similar effects of epidermal wall around the penetration peg have been reported for other powdery mildews (McKeen et. al., 1969; Stavely et. al. 1969).

The collar which consists of an electron lucent material with embedded electron dense material surrounding the haustorial neck and with a space (the channel) in between is similar to that described for E. graminis (Bracker, 1968). It probably contains callose which is formed as a response of higher plants to injury (Currier, 1957) and structures in a similar position are associated with the haustoria of members of Peronosporales (Fraymouth , 1966), Phytophthora infestans (Blackwell, 1953), rusts (Ehrlich et. al., 1966; 1968; Heath and Heath, 1971; Heath, 1971;1972;

Coffey et. al.1972) and powdery mildews (Smith, 1900; McKeen et. al.,1966; Braoker, 1968). The structure of the collar suggests that it contains material other than callose which is stated to be electron lucent (Heslop - Harrison, 1967). The inner margin of the collar is deeply pitted in the channel region where there is considerable proliferation of the host plasmalemma. The vesicles in the inner zone of the collar possibly originated from these invaginations by outgrowth and subsequently trapping by the collar material. The outer margin of the collar is rough or smooth similar to that of E. graminis (Bracker, 1968).

The origin of the collar is not completely understood. In powdery mildews this is thought to develop as extensions of parts of reaction products which begin to form before penetration (Stanbridge et. al. 1971). The channel is similar to that of E. graminis (Bracker, 1968) but unlike E. graminis where the channel is filled with electron dense material the channel in S. pannosa shows distinguishable bounding membranes. The channel whioh is not present in young infections has been frequently observed in later stages (Stanbridge et.al. 1971). This suggests that the formation of the channel or atleast part of it must be either by dissolution of collar material or in S. pannosa by a different type of development not yet recorded.

The haustorial body is sheathed with an amorphous layer that is bounded by a sheath membrane which forms the interfaoe with the host ptotoplast and which is continuous with the channel membrane. Thus the sheath membrane almost certainly is a continuation of the host plasmalemma. The haustorium therefore lies in an extra - cellular zone. This is in acoord with Bracker (1968) who considers the sheath membrane as an extension of the host plasmalemma and the sheath as a zone between the haustorial wall and the host ectoplast.

The sheath is different from the collar.It is two layered. The inner layer is most likely of fungal origin and the outer layer possibly of host origin. However, Dekhuijzen and van der Scheer(1967) observed intact sheath and sheath membranes in isolated haustoria of S. fuliginea and concluded that both layers are of fungal origin.

As the sheath is extra - cellular it must play an important role in the transfer of metabolites from the host to the fungus. The contents of the sheath matrix are similar to those of plant vacuoles suggesting that it is fluid. The large number of finger - like lobes of the fungal wall increases its surface area considerably giving improved access to nutrients in the sheath matrix. The sheath membrane is small in area suggesting that its permeability is higher in that it limits the rate of exchange.

The dark zones or the rings that are found on or over each side of the plasmalemma of the host and the fungus in the neck region have not been recorded previously for any Erysiphales. These structures should be interpreted with some caution as it is possible that lipid material that is commonly found in the neck cytoplasm may have been extracted and deposited in special zones during fixation. However their definite structure and their consistency suggest that they are characteristic parts of the haustorium. Opaque ring structures have been described by Rice (1927) for rust fungi but it was Hardwick et. al.(1971) who first showed in Uromyces appendiculatus that it is a part of the fungal wall. It has also been seen in the necks of several other rust fungus haustoria (Heath and Heath, 1971; Zimmer, 1970; Coffey et. al. 1972).

Although rings observed in S. pannosa are somewhat similar to the rings described in other haustoria they differ in several important respects. These are:(1) Presence of deposits of material in two rings of characteristic size on the fungus and host membranes. The wall between the two rings is relatively electron lucent. The rings in other haustoria are seen as darkly staining zones of the whole width of the fungal wall between the plasmalemmas of host and the fungus. (2) Clearly defined outer margins of the rings which suggest that the rings are more definite structures than the diffuse rings described in other haustoria. For example the ring appears relatively diffuse in U. appendiculatus haustorium as shown in Fig. 10 b of Hardwick et. al. (1971).

The results for S. pannosa do not justify the distinction of a special part of the wall of the neck as a neckband (of which the ring is a part) in U. appendiculatus as described by Hardwick et. al 1971.

The function of these rings is unknown. Hardwick et. al. (1971) interpreted the neckband of U. appendiculatus as a part of the fungal wall marking the begining of a phase in which the parasite expands in the host cytoplasm. The more clearly defined dark rings of S. pannosa may perform a physiological function related to that of the Casparian strip in endodermal cells in plant roots forming a barrier to the extracellular passage of ions. The struoture of the sheath matrix indicates that it is fluid and hence the turgor pressure of the epidermal cell would be exerted upon it. Because it is extracellular, it and its metabolites in process of transfer to the fungal cytoplasm would tend to leak to the exterior along a path between fungal and host plasmalemmas in the neck region. To prevent leakage completely, the wall between these membranes should be impermeable and the membranes closely adherant to it. It is possible that the two rings are special structures with this adherant function. No deductions can be made on the permeability of the wall layers between the membranes but it may be significant that the gradual change in wall appearance occurs between the rings and the body.

IV. PRODUCTION OF ASEPTIC CONIDIA AND THEIR GERMINATION ON ARTIFICIAL SUBSTRATES

INTRODUCTION

There are several references in the literature to the growth of powdery mildews on plant tissue cultures (callus). All suggest that powdery mildews grow poorly in this situation (Morel, 1948; Heim and Gries, 1953; Sohnathorst, 1059; Croyier and Hildebrandt, 1962). Powdery mildew fungi have not been grown axenically but in recent years there have been several such reports for rust fungi which were previously regarded as obligate parasites(Williams *et al.*, 1966; Bushnell, 1968; Coffey, *et al.*, 1972 Bushnell and Stewart, 1971; Rajendran, 1972) and it has been suggested that the problem of culturing rust fungi *in vitro* is essentially nutritional (Williams *et al.*, 1966).

Germination of powdery mildew conidia on agar and on membranes has been briefly reviewed on pages 20 & 24. Longree (1939) found that germination of *S. pannosa* conidia on potato dextrose agar (PDA) and on plain agar was low. Yarwood *et al.* (1954) on the other hand found germination of *E. graminis* conidia was higher on nutrient agar than on plain agar. Weltzien - Stenzel (1959) recorded 53.6% germination of *U. necator* on plain agar compared to 81.4% on collodion membranes on water. Dickinson (1949) germinated *E. graminis* conidia on 2% gelatin and described the upright growth of germ tubes under certain conditions. Poor germination on agar substrates obtained by some workers is believed to be due to the presence of a film of moisture (Schnathorst, 1965).

Dickinson (1949) reported germination with appressorial formation of *E. graminis* conidia on double membranes — collodion membranes containing paraffin wax; the wax layer was sometimes penetrated and at the tip of the penetration tube a haustorium - like swelling was observed. De Waard (1971) germinated conidia of *E. graminis* f.sp. *hordei* and

S. fuliginea on cellulose membranes laid on different agar media and found that the growth of appressoria and their side branches was stronger on modified Czapek Dox agar and on PDA than on plain agar and this was thought to be due to an uptake of nutrients. De Waard (1971) also reported higher (88%) germination for E. graminis conidia on 25 μm thick cellulose membranes compared to low germination (25%) on 45 μm thick membranes and concluded that water permeability of the membranes is of importance in causing a humidity gradient favourable for germination of powdery mildew conidia. Penetration of membranes or haustorial formations were not observed. One of the prime requirements for culturing fungi axenioally is the production of aseptic propagules. With the rust fungi which grow within the host tissue, uncontaminated propagules (usually uredospores) can be obtained with ease by surface sterilizing the host material before pustules erupt. The sori which open 4 – 5 days later then produce abundant aseptic uredospores which can be used in growth experiments. Because powdery mildew fungi grow mainly on the surface of their hosts, the production of uncontaminated propagules is a more difficult problem. But it is possible that the mildew would regenerate from haustoria in surface sterilized leaf disks although no referenoe to such work is found in the literature.

This study was undertaken using E. cichoracearum from burdock with the ultimate aim of producing a batch of aseptic conidia to be used in an attempt to culture powdery mildews axenically. Germination of two other powdery mildews viz. S. mors-uvae from black currant and S. pannosa from rose on cellulose membranes laid on agar media was also studied.

MATERIALS AND METHODS

The host plants were grown in the greenhouse and the mildews were maintained on detatched leaves as described on p. 27.

(1) Surface sterilization of leaf disks

Unless otherwise stated, in all experiments 1.6 cm diam disks were cut from mildewed leaves with a no. 6 cork borer and surface sterilized in a 10% solution of Chloros (1.1% available chlorine) containing 0.001% Tween 80 for 10 min by shaking in a mechanical shaker at half speed. About 3 - 4 ml of Chloros solution was allowed for each leaf disk. The leaf disks were then washed in 4-5 changes of sterile distilled water, gently dried between sheets of sterile tissue (sterilized in the oven at 150° for 2h) and placed on 1% agar in Petri plates (5 - 8 disks/plate). In some experiments disks were floated on sterile distilled water in 6.5 x 3.5 x 2cm polystyrene boxes that were previously sterilized by exposing them with lids separated, to propylene oxide vapour for 24h in a desiccator (c. 1 ml propylene oxide/ 1 of air). All operations were done under aseptic conditions. The Petri plates and the boxes were then closed with lids and incubated at 17° in a cabinet with a 16h photoperiod. After 12 - 17 days they were examined under the binocular microscope without opening the lids for signs of regeneration of the mildew. These were then marked and later used for experiments or tested for the effectiveness of surface sterilization referred to as "tests for sterility".

(2) Test for "sterility"

Leaf disks were placed individually in 7 - 10 ml Nutrient Broth in McCartney bottles and incubated for 2 - 3 days at 25°. Turbidity of the medium was regarded as evidence of contamination of the leaf disks.

(3) Agar media

(a) Potato dextrose agar (Oxoid)

(b) V8 Juice agar

V8 Juice (Campbell's Soups Ltd.)	200 ml
Distilled water	800 ml
Oxoid agar	20 g

Preparation : The mixture was steamed until the agar had dissolved, then dispensed into medical flats and autoclaved for 20 min at 1 kg/cm^2.

(c) Modified Czapek Dox agar (Williams et al. 1966)

This contained the following:

Czapek minerals	
Sucrose	30 g
Difco yeast extract	1 g
Difco Bacto-agar	10 g
Distilled water	1

Medium autoclaved at 1 kg/cm^2 for 15 min.

(d) Modified Czapek Dox peptone agar (Williams et al. 1967)

To the above medium 0.1% Evan's peptone added.

(e) Modified Czapek Dox (Oxoid) agar

The following organic constituents were added to this medium:-

(1) 0.1% Difco yeast extract

(2) 0.1% Evan's peptone

(3) 0.1% Difco yeast extract and 0.1% Evan's peptone.

(4) Germination on membranes

The membranes used were PT 400 (uncoated cellulose) obtained from British Sidac Ltd., Lancashire. The membranes were laid on agar in Petri dishes (5 cm diam) and dusted with 24h conidia in a settling tower (c. 10 conidia per low power field, x 100). A piece of filter paper with glycerine was placed in the Petri dish cover to prevent any water condensation on the membranes. The plates were inverted, placed in a plastic box covered with the lid and incubated at 20° in a cabinet. Three to four replicates were prepared for each experiment and the results are given as an average of the total. Percentage germination of the conidia was assessed and the germ tube length was measured as described on p.31 - 32.

EXPERIMENTAL

(1) Production of aseptic conidia

(A) Surface sterilization of leaf disks

A series of experiments were carried out in an attempt to produce aseptic conidia from surface sterilized mildewed leaf disks. These are briefly outlined in Appendix 12 pages 249-265 and are discussed below with appropriate illustrations under several headings.

(i) Methods of surface sterilization and incubation of leaf disks

Best results were obtained when leaf disks were surface sterilized with 10% Chloros for 10 min followed by 3 - 5 changes in sterile distilled water (Expts. 1, 2). Disks surface sterilized this way remained green upto 18 - 21 days and they often appeared puckered and raised in the centre. Regrowth of the mildew was visible to the naked eye 14 - 17 days after treatment. The number of disks with the regenerated mildew was however very small (Expts. 1, 4, 6, 7, 10). Shaking mildewed disks in sterile water alone was insuffioient in removing contaminating organisms (Expt. 2). Although treatment with 10% Chloros alone without washing in water was sufficient in producing disks free from contaminants (Expt. 3A), almost all mildewed disks treated this way turned brown 5 - 7 days later (Expt. 3B). Chloros possibly had a toxic effect on the host tissue when allowed to remain on it. Substrate had no influence on regeneration of the mildew (Expt. 5) but agar plates enabled easier handling of the disks. Some leaf disks floated on water sank to the bottom. A temperature of 17° was found to be more favourable and browning of the surface sterilized disks was rapid at 22° (Expt. 4).

(ii) Age of mildew colony

The results of experiment 6 suggest that regrowth of the mildew was better (13.3%) from older cultures (14 day) than from young (9 day) cultures.

(iii) Method of testing for aseptic conidia

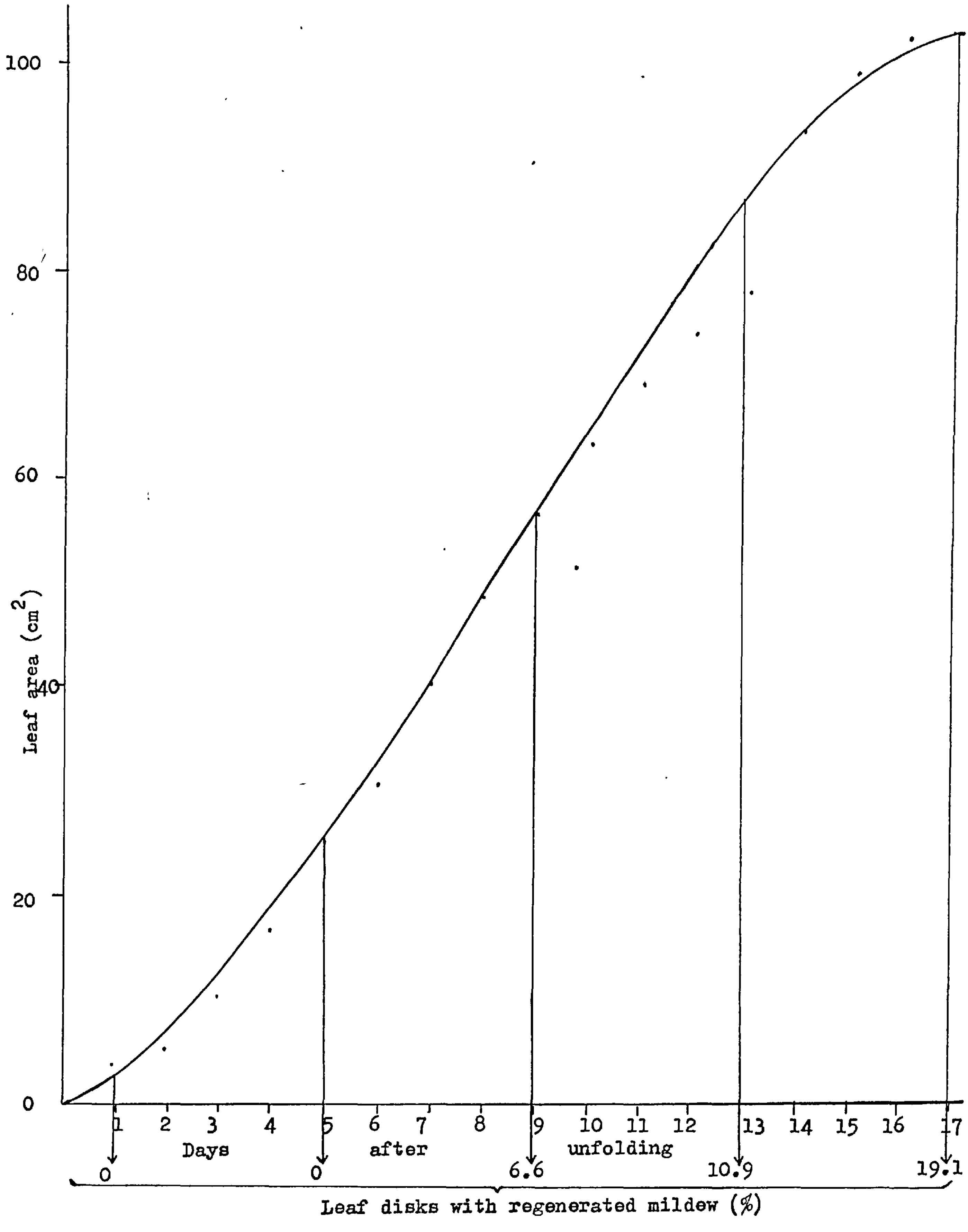
Experimental results indicated that nearly all the surface sterilized disks that developed the mildew were contaminated (Expts. 7, 10) although this did not necessarily mean that conidia themselves were contaminated. In none of the experiments was it possible to completely eliminate contaminating organism — mostly bacteria. Increased Chloros concentration and treatment time had harmful effects on the host tissue (Expts. 8 and 9) and did not eliminate the contaminating organisms (Expt. 9).

(iv) Longevity of treated material

In general, surface sterilized disks remained green and alive on agar/distilled water for c. 20 to 22 days but after that they soon turned yellow and eventually died. In an attempt to increase the longevity of leaf disks a preliminary experiment was conducted on the effect of kinetin on the development of the mildew and its effect on the host (Expt. 11). The results (see Appendix Table XI p. 254) indicate that 10 ppm kinetin increased the longevity of leaf disks for upto 5 - 6 weeks. But kinetin had little effect on the regeneration of mildew on the surface sterilized disks (Expt. 12) which remained alive for c. 3 weeks.

In an attempt to standardize the leaf material and to relate its age to regrowth of the mildew, several experiments were carried out. A quick method for estimating leaf area is described in Expt. 13. Making use of this method the growth pattern of the first four leaves in Burdock was studied (Expt. 14). The results indicated that the 4th leaf which unfolded c. 5 weeks after planting the seedlings was most suitable for inoculation. The relation of leaf age to mildew development and its subsequent regeneration after surface sterilization is studied in Expt. 15. The results (Fig. 38) indicate that with the increase in age of leaf the amount

Fig. 38 Relationship of leaf age to regeneration of mildew on surface sterilized disks.



of regrowth of the mildew increased; a maximum of 19.1% was obtained with the 17 day old leaves which had stopped expanding (see Fig.38). No growth was seen on 1 and 5 day old leaves although these supported abundant mildew. In a further experiment (Expt. 16) the susceptibility of the leaves was studied after maximum expansion. But the results showed that regrowth of mildew did not occur on these due to rapid senescence of the surface sterilized disks.

Experimental results did not indicate a definite relationship of leaf age or age of culture to regeneration of the mildew. Generally, regeneration was low and variable but there was some indication that this was more pronounced on older leaves and sometimes in older cultures (Expts. 6 and 15). Although it was assumed that regeneration occurred from haustoria, evidence (Expt. 17) suggest that this originated from hyphal portions from among numerous crevices on the leaf surface (Fig.39). Indeed complete removal of surface growth was not achieved even after physically removing the mildew with wet cotton wool (Expt. 18). Increased regeneration of the mildew from older leaves and older cultures was possibly due to the presence of a large number of grooves or crevices on them in which air bubbles may have formed during surface sterilization; hyphal portions trapped in these probably resumed their growth when the leaf surface was dried.

(B) A technique for growing *E. cichoracearum* aseptically on Burdock plants in the laboratory

Fifty aseptic burdock seedlings were produced by surface sterilizing seeds with 0.1% Mercuric chloride for 10 min (see Appendix 13 p.266). These (12 day old) were transferred singly under aseptic conditions to 20.5 cm high, 4 cm diam boiling tubes containing 50 ml Hoagland's nutrient medium solidified with 0.6% agar (Appendix 14 p. 267). The tubes were plugged with muslin - covered cotton wool bungs, stood in wooden racks

Fig. 39. Regeneration of fungus (E. cichoracearum) from a hyphal fragment on surface sterilized leaf disk after 8 days.

and placed in a 23.5 ± 1^{o} cabinet with a 16h photoperiod.

After 6 - 8 days the radicles grew into the medium and this was immediately followed by the divergence of the cotyledons and the upward growth of the shoots. The third leaf of each plant that developed 3 - 4 weeks later was inoculated with aseptic conidia (from surface sterilized leaf disk cultures) picked up at the tip of a sterile eyelash. Mildew developed on a total of 43 plants after 10 - 12 days and sporulated profusely for 2 - 3 weeks by which time the 4th and the 5th leaves had also developed and reached the cotton bungs. There was limited spread of the colonies on the inoculated leaves but no mildew developed on the other leaves even after 4 weeks. The plants however remained green and appeared to grow vigorously during this period. Only 3 plants showed visible contamination in the nutrient medium. A 'sterility' check of the mildewed - leaf portions in Nutrient Broth revealed that of the 43 infected plants 31 were free from contaminating organisms.

(2) Germination and growth on agar media

The primary aim of this investigation was to find a suitable medium for growing E. cichoracearum from Burdock in axenic culture. Two preliminary tests were done in connexion with this. viz. (a) Method of placing conidia on agar media and (b) conditions suitable for germination and growth on agar (Appendix 12 Expts. 19 and 20 respectively, p.263-4). The results showed that over 95% of the conidia transferred from surface sterilized disks were aseptic although in previous experiments it was shown that a majority of the disks supporting the aseptic colonies were contaminated. The germination of these aseptic conidia was low. This was possibly due to (1) the presence of old, non - viable conidia in the inoculum and (2) existence of a film of water on the agar surface. Drying the agar surface however did not improve germination (Appendix 12 Expt 21 p.265).

Germination and growth of E. cichoracearum on different agar media

The influence of different agar media on germination and growth of E. cichoracearum was studied using aseptic conidia. Single conidia or conidial chains picked up on a sterile eyelash were placed on the media in pre - marked positions on the underside of plates. The plates were incubated in the dark in a water saturated atmosphere (in covered germination trays containing water) at 17° in a cabinet and examined periodically for 5 - 6 weeks.

Results:

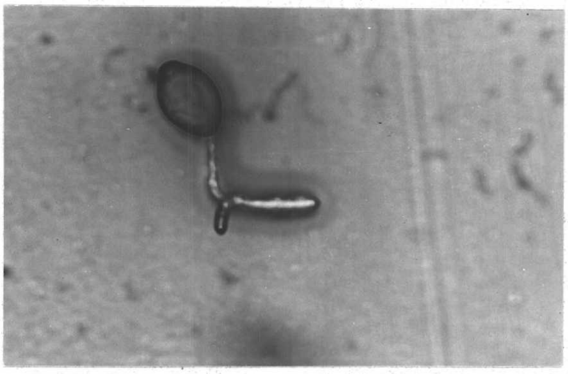
Germination on all media was low (Table 46). Often, germ tubes grew upright, away from the agar surface. Growth continued in these without branching until the germ tubes were about 5 - 10 times the long axis of the conidium. Septa were not observed. Those that grew along the agar surface were shorter and sometimes branched and formed club shaped structures (Fig. 40 a,b). Some produced forked germ tubes. Their growth usually ceased after 3 - 4 days. Branching was not seen on water agar where the germ tube growth ceased after 2 days. The different agar media did not appear to influence the growth of the germinated conidia.

(3) Germination on membranes

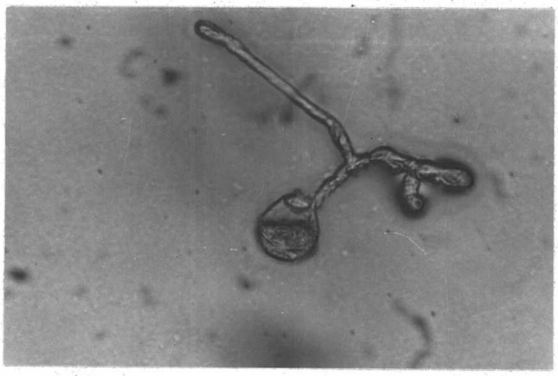
Poor germination of conidia and the limited growth of the germ tubes on agar media could possibly be due to the presence of a film of water on the agar surface. Therefore, germination of powdery mildew conidia on membranes laid on different agar media was investigated (see Materials and Methods p. 197). Twenty four - hour conidia of E. cichoracearum from burdock, S. mors - uvae from black currant and S. pannosa from rose were used in this investigation.

Table 46. Percentage germination of aseptic conidia of *E. cichoracearum* on different agar media

	Medium	% germination
1)	1% Water agar	36.0
2)	2% Water agar	16.0
3)	Potato dextrose agar	19.6
4)	V8 Juice agar	28.0
5)	Modified Czapek-Dox (Oxoid) yeast extract agar (pH 6.9)	29.0
6)	Modified Czapek-Dox (Oxoid) peptone agar (pH 5.7)	33.2
7)	Modified Czapek-Dox (Oxoid) yeast extract-peptone agar (pH 6.7)	26.4
8)	Modified Czapek-Dox agar, pH 6.4 (Williams *et al.* 1966)	39.0
9)	Modified Czapek-Dox peptone agar (Williams *et al.* 1967)	
	(a) pH 5.7	25.8
	(b) pH 6.4	29.4
	(c) pH 7.1	19.1



(a)



(b)

Fig. 40. Growth of Erysiphe cichoracearum on Czapek Dox agar after 3 days (a) and after two weeks (b). X 480 approx.

Results:

Different agar media did not influence conidial germination although this was somewhat higher for S. pannosa on modified Czapek Dox agar (Table 47). On germination, conidia of E. cichoracearum formed club shaped appressoria which often produced curved branches. Curved branches also frequently developed from the appressoria of S. pannosa but in S. mors - uvae, very few germinated oonidia formed branches. Branching was less frequent on water agar and the growth of germ tubes of S. pannosa stopped after 1 day on this substrate while it continued on Czapek Dox agar. In fact in all 3 species germ tube growth was stronger on Czapek Dox agar and PDA than on water agar (see Table 47) suggesting a nutrient uptake by the germinating conidia. Penetration of the membranes was not observed. Germ tube development could not be observed beyond 2 - 3 days due to heavy contamination.

Table 47. Germination of powdery mildew conidia on cellulose membranes laid on different media

Medium/Fungal species		Mean % germination *	Length of germ tubes (μm)+	
			after 1 day	after 2 days
Water agar	(i) E. cichoracearum	45.4	145.4	201.7
	(ii) S. pannosa	40.9	42.0	46.1
	(iii) S. mors-uvae	24.6	43.1	57.6
Modified Czapek-Dox yeast extract-peptone agar				
	(i) E. cichoracearum	44.3	229.8	266.6
	(ii) S. pannosa	61.0	82.0	99.4
	(iii) S. mors-uvae	38.5	56.4	76.4
Potato Dextrose agar				
	(i) E. cichoracearum	43.1	197.9	258.4
	(ii) S. pannosa	-	-	-
	(iii) S. mors-uvae	34.3	66.6	81.0

* Average of 3 replicates + Mean of 60 germ tubes

DISCUSSION

One of the major obstacles in studies pertaining to cultivation of parasitic fungi in tissue cultures or on artificial media is the production of viable, aseptic propagules. With the powdery mildew fungi although it is possible to produce aseptic conidia that can be used in such studies, the methods available for the production of aseptic mildew are not altogether reliable particularly when aseptic conidia are required in large numbers for experimentation. The results of this investigation suggest, though not conclusively, that the regenerated mildew originated from hyphal portions that were trapped in air bubbles during surface sterilization. Hence the production of aseptic conidia by this method appears to be a matter of chance. There was no indication of the regeneration of the fungus from haustoria.

Failure to grow powdery mildews axenically may be as much due to inability to reproduce the right physical conditions as it is to select the correct nutritional requirements. Indeed in the present experiments the ever present film of moisture on the agar surface may have interferred with the normal germination of conidia or prevented the germinated conidia from establishing contact with the agar surface. Since haustoria are the only structures of the fungus that can tolerate free moisture, it would be desirable to maintain the agar surface absolutely dry for successful germination of powdery mildew conidia and for their subsequent growth on artificial media. But yet, it is necessary to maintain the proper atmospheric moisture around the germinating conidia to achieve a greater degree of success. Success of axenic cultures of powdery mildews would therefore depend largely on the environmental factors which should be held as near optimal as possible. Due consideration should be given to the moisture stress of the specie under study, temperature requirements and also light conditions as this may affect haustorial formation/appressorial maturation by affecting the nucleus (Mount and Ellingboe, 1968).

With regard to the choice of substrate perhaps greater consideration should be given to adjusting PH and osmotic values approaching that of the host. In the present study better germination and growth was obtained on cellulose membranes laid on agar media. There was evidence to indicate enhanced growth on membranes laid on Czapek Dox agar and PDA than on water agar suggesting a nutrient uptake by the germinating conidia. These observations agree with the findings of De Waard (1971). If membranes are used to culture powdery mildews axenically it would be necessary to use those that are thin enough to ensure rapid penetration of the infection pegs and at the same time sufficiently permeable to water in causing a humidity gradient favourable for germination.

REFERENCES

BAILEY , N.T.J. (1964) Statistical methods in biology, 200 pp. London : Universities Press.

BARTLETT , A.B. (1964) Relation of nitrogenous substances in rose leaves to powdery mildew resistance. Diss. Abstr. 24 , 3045 - 3046.

BECKMAN , C.H. and MUELLER, W.C. (1970) Distribution of phenols in specialized cells of banana roots. Phytopathology , 60, 79 - 82.

BENT , K.J. (1967) Vapour action of fungicides against powdery mildews. Ann. appl. Biol. 60. 251 - 263.

BENADA , J. (1970) The effect of different conditions of cultivation of powdery mildew (Erysiphe graminis D.C.) on the infectivity. Phytopath. Z. 6 , 273 - 276.

BENTLEY , J.A. (1962). Analysis of plant hormones. In Methods of biochemical analysis , ed.D. Glick , pp 75 - 125.

BONNER , J. and GALSTON , A.W. (1959) Principles of plant Physiology.

BLACKWELL , E.M. (1953) Haustoria of Phytophthora infestans and some other species. Trans. Br. Mycol. Soc. 36 , 138 - 158

BLUMER , S. (1933). Die Erysiphaceen Mitteleuropas mit besonderer Berucksichitigung der Schweiz. Beitr. KryptogFlora Schweiz 7, 1 - 483.

BLUMER , S. (1967) Echte Mehltaupilze (Erysiphaceae) , 436 pp. Jena : Gustav Fisher Verlag

BRODIE , H.J. (1945). Further observations of the mechanism of germination of the conidia of various species of powdery mildew at low humidity. Canad. J. Res. C., 23 , 198 - 211.

BRODIE , H.J. and NEUFELD , C.C. (1942). The development and structure of the conidia of Erysiphe polygoni DC. and their germination at low humidity. Canad. J. Res. C. 20 , 41 - 62.

BRACKER , C.E. (1967) Ultrastructure of fungi. Ann. Rev. Phytopathol. 5 , 343 - 374.

BRACKER , C.E. (1968). Ultrastructure of the haustorial apparatus of Erysiphe graminis and its relationship to the epidermal cell of Barley. Phytopathology , 58 , 12 - 30.

BURCHILL , R.T. (1958) Studies on apple mildew. PhD. Thesis. University of Bristol.

BUSHNELL , W.R. (1968). In Vitro Development of an Australian isolate of Puccinia graminis f . sp. tritici. Phytopathology, 58 ,526 - 527.

BUSHNELL , W.R. and STEWART , D.M. (1971). Development of American isolates of Puccinia graminis f. sp. tritici on an artificial medium. Phytopathology , 61 , 376 - 379.

BUTLER , E.J. and JONES , S.G. (1955). Plant Pathology , 979 pp. London McMillan.

CARROLL , G.C. (1966).A study of ascosporogenesis in Saccobolus kerverni and Ascodesmis sphaerospora (Doctoral thesis, University of Texas). Cited in Ann. Rev. Phytopathol. 5 , Bracker , C.E.,(1967)

CHEREWICK , W.J. (1944) Studies on the biology of Erysiphe graminis DC. Canal. J. Res. C., 22 , 52 - 86.

CLAYTON , C.N. (1942). The germination of fungus spores in relation to controlled humidity. Phytopathology , 32 , 921 - 943.

COFFEY , M.D., PALEVITZ , B.A. and ALLEN, P.J. (1972). The fine structure of two rust fungi, Puccinia helianthi and Melampsora lini. Canad. J. Bot. , 50 , 231 - 240

COLE , J.S. and FERNANDES , D.L. (1970). Effect of light , temperature and humidity on sporulation of E. cichoracearum on Tobacco. Trans. Br. Mycol. Soc. 55, 345 - 353.

CORNER , E.J.H. (1935). Observations on resistance to powdery mildews. New Phytol. , 34 , 180 - 200.

COYIER , D.L. (1961) Biology and control of rose powdery mildew. PhD Thesis. University of Wisconsin.

COYIER , D.L. and HILDEBRANDT, A.C. (1962). Characteristics of rose powdery mildew in Wisconsin. Phytopathology , 52 , 6. Abstract.

CRUICKSHANK , I.A.M. (1963). Phytoalexins. Ann. Rev. Phytopathol. 1 , 351 - 374.

CURRIER , H.B. (1957). Callose substance in plant cells. Amer. J. Bot. 44, 478 - 488.

DEKKER , J. (1962). Mode of action of procaine hydrochloride and some derivatives on powdery mildews. Meded. Landbhoogesch . Opzoekstnts Gent 26 , 1378 - 1384.

DEKHUIJZEN , H.M. and van der SCHEER , c. (1967) . Electron microscopic observations on haustoria isolated from cucumber leaves infected with powdery mildew. Neth. J. Plant. Pathol. 73 , 121 - 125.

DEKHUIJZEN , H.M. and van der SCHEER , c. (1969). The ultrastructure of powdery mildew , Sphaerotheca fuliginea isolated from cucumber leaves. Neth. J. Plant Pathol. 75 , 169 - 177

DELP , C.L. (1954). Effect of temperature and humidity on the grape powdery mildew fungus. Phytopathology , 44 , 615 - 626.

DE WAARD , M.A. (1971) Germination of powdery mildew conidia in vitro on cellulose membranes. Neth. J. Pl. Path. 77 , 6 - 13.

DICKINSON , S. (1949). Studies in the physiology of obligate parasitism . 1. The stimuli determining the direction of growth of germ tubes of rust and mildew spores. Ann. Bot.(London) , 13 , 89 - 104.

EDWARDS , H.H. and ALLEN , P.J. (1970). A fine structure study of the primary infection process during infection of Barley by Erysiphe graminis f. sp. hordei. Phytopathology , 60 , 1504 - 1509.

EHRLICH , H.G. and EHRLICH , M.A. (1963) Electron microscopy of the host - parasite relationships in stem rust of wheat. Amer. J. Bot. 50 , 123 - 130

EHRLICH , M.A., SCHAFER , J.F. and EHRLICH , H.G. (1966). Association of host endoplasmic reticulum with haustoria of *Puccinia graminis* f. sp. tritici. *Phytopathology* *56* , 876 - 877. (abstr.).

EHRLICH , M.A. and EHRLICH , H.G. (1971). Fine structure of the host - parasite interfaces in mycoparasitism. *Ann. Rev. Phytopathol.* *9* , 155 - 184.

ELLINGBOE , A.H. (1968). Inoculum production and infection by foliage pathogens. *Ann. Rev. Phytopathol.* *6* , 317 - 330.

ESAU , K. (1965). *Plant Anatomy* (2nd ed.) 767 pp John Wiley and Sons.

FISHER , R.A. and YATES , F. (1963) statistical tables for biological , agricultural and medical research (6th ed.) London , Oliver and Boyd.

FRAMPTON , V.L. and LONGREE , K. (1941). The vapour pressure gradient above a transpiring leaf. *Phytopathology* , *31* , 1040 - 1042.

FRAYMOUTH , J. (1956). Haustoria of Peronosporales. *Trans. Br. Mycol. Soc.* *39* , 79 - 107.

FREERE, R.H. and WEIBEL , E.R. (1967) Stereological techniques in microscopy. *J. Micros. Soc.* *87* , 25 - 34.

GRAF - MARIN , A. (1934). Studies on powdery mildew of cereals. *Mem.* *157* , *Cornell agric. Exp. Sta.* 48pp.

GRAINGER , J. The ecology of *Erysiphe graminis* DC. *Trans. Br. Mycol. Soc.*, *31* , 54 - 65.

GRIFFEE , P. (1967) Msc Thesis. Imperial College of Science and Technology, London.

HAMMARLUND , C. (1925). Zur Genetik , Biologie , und Physiologie einiger Erysiphaceen. *Hereditas* 6 , 1 - 126.

HARDWICK , N.V. , GREENWOOD , A.D. and Wood , R.K.S. (1971). The fine structure of the haustorium of *Uromyces appendiculatus* in *Phaseolus vulgaris*. *Canad. J. Bot.* *49* , 383 - 390.

HASHIOKA , Y , (1937). Relation of temperature and humidity to *Sphaerotheca fuliginea* (Schlecht.) Poll. , with special reference to germination , viability and infection. *Trans. Nat. Hist. Soc. Formosa* , *27* , 129 - 145.

HEIM , J.M. and GRIES , G.A. (1953). The culture of *Erysiphe cichoracearum* on sunflower tumour tissue. *Phytopathology* *43* , 343 - 344.

HEATH , M.C. (1971).Haustorial sheath formation in cowpea leaves immune to rust infection. *Phytopathology* , *61* , 383 - 388.

HEATH , M.C. and Heath , I.B. (1971). Ultrastructure of an immune and a susceptible reaction of cowpea leaves to rust infection. *Physiol. Plant Pathol.* *1* , 277 - 287.

HEATH , M.C. (1972). Ultrastructure of nonhost reactions to cowpea rust. *Phytopathology* *62* , 27 - 38.

HESLOP - HARRISON , J. (1966). Cytoplasmic continuities during spore formation in flowering plants. *Endeavour* , *25* , 65 - 72.

HICKMAN , M.J. (1958) . Measurement of humidity. In notes on Applied Science No. 4., National Physical Laboratory.

HIRATA , K. (1937). The haustoria of some powdery mildews. *Ann. Phytopath. Soc. Japan* *6* , 319 - 334.

HIRATA , K. and KOJIMA , M. (1962). On the structure and the sac of the haustorium of some powdery mildews, with considerations on the significance of the sac. *Mycol. Soc.*, *3* , 43 - 46.

HIRST , J.M. (1953). Changes in atmospheric spore content: Diurnal periodicity and effects of weather. *Trans. Br. Mycol. Soc.* *36* , 375 - 393.

HORSFALL , J.G. (1956). Principals of fungicidal action. 279 pp. Chronica Botanica Company , Waltham , Mass.

HOSSAIN , S.M.M. (1959). Studies on the germination of the conidia of *Erysiphe graminis* DC. MSc Thesis. University of Southampton.

JACKSON , G.V.H. (1968). Investigations on the conidial germination of Sphaerotheca mors - uvae Cooke on glass slides and on leaf material. MSc Thesis. Imperial College of Science and Technology, London.

JENSEN , W.A. (1962) . Botanical Histochemistry . W.H. Freeman and Co., San Francisco , Calif. 408 pp.

JHOOTY , J.S. and McKEEN , W.E. (1965 b). Influence of host leaves on germination of asexual spores of Sphaerotheca macularis (Cooke). Canad. J. Microbiol. 11 , 539.

JHOOTY , J.S. and McKEEN , W.E. (1965 a). Water relations of asexual spores of Sphaerotheca macularis (Wallr. ex. Fr.) Cooke and Erysiphe polygoni DC. Canad. J. Microbiol. 11 , 531 - 538.

KNIGHT , T.A. (1818). On the prevention of mildew in particular cases. Trans. Hort. Soc. 2 , 82 - 90 . Cited by Yarwood, C.E. (1957).

KOHL , H.C. (1955). Mildew control on mist. The Bloomin' News. 7 (4): 17. Cited by Rogers (1959).

KOOPMANS , M.J. (1959). An in vitro evaluation of the toxicity of chemicals for Erysiphaceae. Meded. Landb. Hogesch. Gent , 24 , 821 - 827. Cited by Zaracovitis (1964 b).

KUĆ , J. (1966).Resistance of plants to infectious agents. Ann. Rev. Microbiol. 20 , 337 - 370.

KUNOH , H. and AKAI , S. (1969). The electron microscopic observation of haustoria of Erysiphe graminis hordei. Mycopathol. Mycol. Appl. 38 , 337 - 344.

LANGHANS , R. (1955). Mist starts roses. N.Y. State Flower Grow. Bul. 115 : 1 , 4. Cited by Rogers , 1959.

LEOPOLD , A.C. (1964). Plant Growth and Development . McGraw - Hill , 466 pp.

LONGREE , K. (1939). The effect of temperature and relative humidity on the powdery mildew of roses. Mem. 223, Cornell. agric. Expt. st 43 pp.

MANNERS , J.G. and HOSSAIN , S.M.M. (1963). Effect of temperature and humidity on conidial germination in Erysiphe graminis. Trans. Br. Mycol. Soc. 46 , 225 - 234.

McCLELLAN , W.D. (1942). Control of powdery mildew of roses in the greenhouse. Cornell. Univ. Agric. Expt. Sta. Bul. 785 , 1 - 39.

McKEEN , W.E.,Smith , R. , and MITCHELL , N. (1966). The haustorium of Erysiphe cichoracearum and the host - parasite interface on Helianthus annus. Canad. J. Bot. 44 , 1299 - 1306.

McKEEN , W.E., SMITH , R. and BHATTACHARYA , P.K. (1969). Alterations of the host wall surrounding the infection peg of powdery mildew fungi. Canad. J. Bot. 47 , 701 - 706.

MITCHELL , N.L. and McKEEN , W.E. (1970). Light and electron microscope studies on the conidium and germ tube of Sphaerotheca macularis Canad. J. Microbiol. 16, 273 - 280.

MORRISON , R.M. (1964). Germination of conidia of Erysiphe cichoracearum. Mycologia. 56 , 232 - 236.

MOREL , G. (1948). Recherches sur la culture associee de parasites obligatoires et de tissus vegetaux. Ann. des. Epiph. 14 , 1 - 11 (Cited in Heim and Gries , 1953).

MOUNT , M.S. and ELLINGBOE , A.H. (1968). Effects of ultraviolet radiation on the establishment of Erysiphe graminis f. sp. tritici on wheat. Phytopathology , 58 , 1171 - 1175.

MOUNT , M.S. and SLESINSKI , R.S. (1971). Characterization of primary development of powdery mildew. In Preece , T.F and Dickinson , C.H. (ed) , Ecology of leaf surface micro - organisms.

NAIR , K.R.S. and ELLINGBOE , A.H. (1965). Germination of conidia of E. graminis f. sp. tritici . Phytopathology 55 , 365 - 368.

NOUR , M.A. (1958). Studies on Leveillula taurica (lev). Arn and other powdery mildews. Trans. Br. Mycol. Soc. 41 , 17 - 38.

PADY , S.M. , KRAMER , C.L. and CLARY , R. (1969). Sporulation of some species of Erysiphe. Phytopathology 59 , 844 - 848.

PEACHEY , L.D. (1958) J. biophys. biochem. Cytol. 4 , 233.

PERIES , O.S. (1962). Studies on strawberry mildew caused by Sphaerotheca macularis (Wallr. ex. Fries) Jac. Ann. appl. Biol. 50 , 211 - 23

POLITIS , J. (1957).Uber die Tannenoplasten oder Gerbstoffbildner der Rosaceen. Protoplasma 48 , 261 - 268.

PRABHU , A.S. , RAJENDRAN , V. and PRASADA , R. (1962). Moisture requirements for germination of conidia of E. graminis tritici E. Marchal. Indian. Phytopath. 15 , 280 - 286.

PREECE , T.F. (1965) . Early stages of American Gooseberry mildew. Plant Pathology 14,83 - 86.

PRICE , T.V. (1969) PhD. Thesis. Studies of overwintering , epidemiology and control of Sphaerotheca pannosa on rose. Imperial College of Science and Technology , London.

PRICE , T.V. (1970). Epidemiology and control of powdery mildew (Sphaerotheca pannosa) on rose. Ann. appl. Biol., 65 , 231 - 248

PURKAYASTHA , R.P. and DEVERALL , B.J. (1965). The detection of antifungal substances before and after infection of leaves (Vicia faba L.) by Botrytis spp. Ann. appl. Biol. 56 , 269.

RAJENDRAN , R.B. (1972). Cytology of Puccinia graminis f. sp. tritici on an artificial medium. Mycologia 64 , 591 - 598.

RAMSEY , J.A. , BUTLER , C.G. and SANG , J.H. (1938) . The humidity gradient at the surface of a transpiring leaf. J. Exp. Biol., 1' 255 - 265.

REYNOLDS , E.S. (1963). The use of lead citrate at a high p^H as an electron opaque stain in electron microscopy. J. Cell. Biol. 17 208.

RICE , M.A. (1927). The haustoria of certain rusts and the relation between host and pathogen. Bull. Torrey Bot. Club , 54 , 63-153.

ROGERS , M.N. (1959). Some effects of moisture and host plant susceptibility on the development of powdery mildew of roses, caused by Sphaerotheca pannosa var. rosae. Mem. 363 , Cornell agric. Expt. Sta. 38 pp.

SCHNATHORST , W.C. (1959). Spread and life cycle of the lettuce powdery mildew fungus. Phytopathology , 49 , 464 - 468.

SCHNATHORST , W.C. (1960). Effect of temperature and moisture stress on the lettuce powdery mildew fungus. Phytopathology , 50 , 304-308.

SCHNATHORST , W.C. (1965) Environmental relationships in the powdery mildew Ann. Rev. Phytopathol. 3 , 343 - 366.

SCHAFFNIT , E. and VOLK , A. (1930). Beitrage zur Kenntnis der Wechselbez - iehungen zwischen Kulturpflanzen , ihren Parasiten und der Umwelt. (11. Mitteilung). Uber den Einfluss der Enahrung auf die Empfanglichkeit der Pflanzen fur Parasiten (11 Teil).Phytopath. Z. 1. 535 - 574.

SCHAFFNIT , E. (1932).Aus unserer Versuchstatigkeit mit Wein , Obst - und Gartengewachsen. Mitt. dt LandwGes. 47 , 471 - 472 ; 487 - 489.

SHIPTON , W.A. and BROWN , J.E. (1962). A whole cleaning and staining technique to demonstrate host - pathogen relationships of Wheat stem rusts. Phytopathology. 52 , 1313.

SHAW , L. (1935). Intercellular humidity in relation to fireblight susceptibility in apple and pear. Cornell. agric. Exp. Sta. Mem. 181 , 40 pp. (cited by Rogers , 1959).

SMITH , H.C. and BLAIR , I.D. (1950). Wheat powdery mildew investigations. Ann. appl. Biol. 37 , 570 - 583.

SMITH , G. (1900). The haustoria of the Erysipheae. Bot. Gaz. 29 , 153 - 18

SOMMERS , E.and HORSFALL , J. (1966). The water content of powdery mildew conidia. Phytopathology , 56 , 1031 - 1035.

STAVELY , J.R., PILLAI , A., HANSON , E.W. (1969). Electron microscopy of the development of Erysiphe polygoni in resistant and susceptible Trifolium pratense. Phytopathology 59 , 1688 - 1693.

STANBRIDGE , B., GAY , J.L. and WOOD , R.K.S. (1971). Gross and fine structural changes in Erysiphe graminis and Barley before and during infection. In Ecology of leaf surface Micro - organisms, (edited by Preece , T.F. and Dickinson , C.H.), 367 - 379.

STOLL , K. (1941). Untersuchungen uher den Apfelmehltan Podosphaera leucotrica (Ell. and Ev.) Salm. Forschungdienst, 11 , 59 - 70.

SPRAGUE , R. (1949). Apple powdery mildew spray trials at the Tree Fruit Station in 1949. Proc. Washington State Hort. Ass., 45 , 47 - 50. Abstr. in Rev. appl. Mycol. 30 , 475.

SWAIN , T. (1965). The tannins, pp. 552 - 579. In J. Bonner and J.E. Varner (ed), Plant Biochemistry. Academic Press , New York , London.

TOMKINS , R.G. (1932). Measuring germination. Trans. Brit. Mycol. Soc. 17 , 147 - 149.

THUT , F.H. (1939). The relative humidity gradient of stomatal transpiratio Amer. J. Bot. 26 , 315 - 319.

WARDROP , A.B. and CORNSHAW , J. (1962). Formation of phenolic substances in the ray parenchyma of Angiosperms . Nature 193 , 92.

WEINHOLD , A.R. (1961). Temperature and moisture requirements for germination of conidia of Sphaerotheca pannosa from peach. Phytopathology 51 , 699 - 703.

WEINHOLD , A.R. and ENGLISH , H. (1964). Significance of morphological barriers and osmotic pressure in resistance of mature peach leaves to powdery mildew. Phytopathology, 54 , 1409 - 1414.

WELTZIEN - STENZEL , M. (1959). Studies on the germination biology of conidia of Uncinula necator (Schwein.) Burr. Hofchen - Briefe , 12 , 29 - 52.

WILLIAMS , P.G., SCOTT , K.J. and KUHL , J.L. (1966). Vegetative growth of Puccinia graminis f. sp. tritici in vitro. Phytopathology 56 , 1418 - 1419.

WILLIAMS , P.G., SCOTT , K.J., KUHL , J.L. and MACLEAN , D.J. (1967). Sporulation and pathogeneoity of Puccinia graminis f. sp. tritici grown on an artificial medium. Phytopathology 57 , 326 - 327.

WOODWARD , R.C. (1927). Studies on Podosphaera leucotricha (Ell. and Ev.) Salm. 1. The mode of penetration. Trans. Brit. Mycol. Soc. 12 , 173 - 204.

YARWOOD , C.E. (1936). The tolerance of Erysiphe polygoni and certain other powdery mildews to low humidity. Phytopathology 26 , 845 - 859.

YARWOOD , C.E. (1936 a). Diurnal peridiocity of Erysiphe polygoni . J. Agric. Res. 52 , 645 - 657.

YARWOOD , C.E. (1939) Control of powdery mildews with a water spray. Phytopathology 29 , 288 - 290.

YARWOOD , C.E. (1949). Effect of soil moisture and nutrient concentration on the development of bean powdery mildew. Phytopathology 39 , 780 - 788.

YARWOOD , C.E. (1950). Water content of fungus spores. Amer. J. Bot. 37 , 636 - 639.

YARWOOD , C.E. (1951). Fungicides for powdery mildews. Proc. llnd. Int. Congress of Crop Proteotion , 500 - 521.

YARWOOD , C.E. (1952). Some water relations of Erysiphe polygoni conidia. Mycologia. 44 , 506 - 522.

YARWOOD , C.E. (1957). Powdery mildews. Bot. Rev. 23 , 235 - 301.

YARWOOD , C.E. and HAZEN , W.E. (1944). The relative humidity at leaf surfaces. Amer. J. Bot. 31 , 129 - 135.

YARWOOD , C.E., SIDKY , S., COHEN , M. and SANTILI , V. (1954). Temperature relations of powdery mildews. Hilgardia , 22 , 603 - 622.

ZARACOVITIS , C. (1964 a). Factors in testing fungicides against powdery mildews. PhD Thesis. University of Bristol.

ZARACOVITIS , C. (1964 b). Factors in testing fungicides against powdery mildews. The germination of the conidia in vitro. Annls. Inst. Phytopath. Benaki. N.S. 6 , 73 - 106.

ZARACOVITIS , C. (1966). The germination in vitro of conidia of powdery mildew fungi. In M.F. Madelin (ed.), The Fungus Spore , Colston Papers No. 18, 273 - 287.

ZIMMER , D.E. (1970). Fine structure of Puccinia carthami and the ultrastructural nature of exclusionary seedling resistance of safflower. Phytopathology 60 , 1157 - 1163.

APPENDIX

Appendix 1

Table(i) showing relative humidity of air over various saturated salt solutions (after Hickman, 1958).

Saturated salt solution	Temperature (°C)						
	5	10	15	20	25	30	35
	Relative humidity						
K_2SO_4	98	98	97	97	97	96	96
KNO_3	96	95	94	93	92	91	89
KCl	88	88	87	86	85	85	84
$(NH_4)_2 SO_4$	82	82	81	81	80	80	80
NaCl	76	76	76	76	75	75	75
$NaNO_3$	-	-	-	66	65	63	62
$NH_4 NO_3$	-	72	69	65	62	59	55
$Na_2Cr_2O_7 \cdot 2H_2O$	59	58	56	55	54	52	51
$Mg (NO_3)_2 \cdot 6H_2O$	58	57	56	55	53	52	50
$Mg Cl_2 \cdot 6H_2O$	34	34	34	33	33	33	32
KOH^2	14	13	10	9	8	7	6

Lactophenol cotton blue

Phenol (as crystals)	20 g
Lactic acid	20 g
Glycerine	40 g
Cotton blue	0.1 - 0.5 g
Distilled water	20 ml

Appendix 2

Table II Time to assess germination of *S. mors-uvae* conidia

Replicate number	24 h incubation				48 h incubation				72 h incubation			
	No. conidia	No. germ-inated	% germin-ation	Angle trans-formed	No. conidia	No. germ-inated	% germin-ation	Angle trans-formed	No. conidia	No. germ-inated	% germin-ation	Angle trans-formed
1	214	52	24.2	29.5	232	83	35.7	36.6	102	18	17.6	24.8
2	212	47	22.1	28.1	320	65	20.3	26.7	124	37	29.8	33.1
3	215	26	12.0	20.3	257	56	21.8	27.8	100	11	11.0	19.3
4	202	41	20.2	26.8	182	68	37.3	37.6	103	17	16.5	23.9
5	218	26	11.9	20.2	267	78	29.2	32.7	106	27	25.5	30.3
6	204	59	28.9	32.5	343	82	23.9	29.2	108	21	19.4	26.1
7	201	65	32.3	34.6	278	51	18.3	25.6	104	36	34.6	36.0
8	205	84	40.9	39.6	-	-	-	-	110	20	18.2	25.2
9	202	56	27.7	31.7	-	-	-	-	123	30	24.4	29.6
10	212	69	32.5	34.7	-	-	-	-	-	-	-	-

24 h incubation:
$\bar{x}$ % germination = 25.1
$\bar{x}$ transformed = 29.8
S D = 6.2

48 h incubation:
$\bar{x}$ % germination=25.7
$\bar{x}$ transformed =30.9
S D = 4.8

72 h incubation:
$\bar{x}$ % germination = 22.1
$\bar{x}$ transformed = 27.6
S D = 5.1

$\bar{x}$ = Mean

Appendix 3

Table III **Effect of increase in age of lesions in the field on germination of *S. mors-uvae* conidia**

Lesion age (weeks)	Germination site: Glass		Leaf	
	No. counted	No. germinated	No. counted	No. germinated
1	200	164	200	172
	200	160	200	136
	200	163	200	140
	200	149	200	148
	$\bar{x}$ = 79%		$\bar{x}$ = 74.5%	
	S D = 3.4		S D = 8	
		t = 1.14		
2	200	132	200	150
	200	148	200	162
	200	152	200	124
	200	120	200	142
	$\bar{x}$ = 69%		$\bar{x}$ = 72.2%	
	S D= 7.3		S D = 7.9	
3	200	184	200	156
	200	176	200	150
	200	180	200	172
	200	178	200	162
	$\bar{x}$ = 89.7%		$\bar{x}$ = 80%	
	S D = 1.7		S D= [illegible].6	

Table IIIcontd....

	200	140	200	168
	200	136	200	158
4	200	136	200	144
	200	148	200	146
	$\bar{x}$ = 70%		$\bar{x}$ = 77%	
	S D = 2.8		S D = 5.5	
	100	62	100	81
	100	68	100	64
5	100	66	100	70
	100	68	100	83
	100	79	100	80
	$\bar{x}$ = 68.6%		$\bar{x}$ = 75.6 %	
	S D = 6.3		S D = 8.2	
	99	81	52	40
	114	75	60	38
6	57	36	96	62
	102	57	44	36
	72	36	60	42
	$\bar{x}$ = 64.1%		$\bar{x}$ = 69.9%	
	S D = 12.0		S D = 8.1	

t = 1.11

$\bar{x}$ = mean percentage germination

Appendix 4.

Griffee's slide cleaning method

1. Wash hands with Chemico - rinse thoroughly

2. Rub paste of Chemico over coverslips with fingers.

3. Rinse thoroughly under running tap water ($\frac{1}{2}$ h).

4. Wash in distilled water (3 changes, 3 min each change).

5. Dry coverslips between new filter paper.

6. Store under alcohol

7. Remove from alcohol and dry as above.

8. Before use wash in distilled water. Blot dry.

Appendix 5

Table IV Percentage germination of conidia of different powdery mildew fungi at 20° in a saturated atmosphere

Fungus	Source of conidia	Date of sampling	Germination site	Mean % germination
1) *Erysiphe graminis*	Barley	1.7.70	Glass	89.4
			Leaf	85.1
		2.7.70	Glass	82.5
			Leaf	88.3
		3.7.70	Glass	82.5
			Leaf	82.3
2) *E. cichoracearum*	Burdock	26.6.70	Glass	95.4
			Leaf	92.1
		30.6.70	Glass	90.7
			Leaf	91.3
		2.7.70	Glass	93.3
			Leaf	93.1
		21.7.70*	Glass	87.7
			Leaf	91.0
		25.8.70*	Glass	96.2
			Leaf	96.7
3) *E. cichoracearum*	Michelmas Daisy	29.7.70	Glass	36.7
			Leaf	39.2
		5.8.70*	Glass	26.2
			Leaf	28.2

Table IV contd...

4) *E. horridula*	Forget-me-not	29.7.70	Glass	96.0
			Leaf	96.0
		5.8.70*	Glass	90.0
			Leaf	93.7
5) *E. fisheri*	Groundsel	11.8.70	Glass	86.4
			Leaf	89.8
		12.8.70	Glass	83.8
			Leaf	85.8
		13.8.70	Glass	81.5
			Leaf	89.7
6) *E. martii*	Red clover	11.8.70	Glass	81.2
			Leaf	89.6
		12.8.70	Glass	92.8
			Leaf	91.6
		13.8.70	Glass	82.5
			Leaf	89.2
7) *E. polygoni*	Knotgrass	11.8.70	Glass	85.8
			Leaf	97.2
		12.8.70	Glass	88.8
			Leaf	95.0
		13.8.70	Glass	82.5
			Leaf	88.7
8) *E. polygoni*	*Delphinium* sp	12.7.70	Glass	86.2
			Leaf	85.0
		13.7.70	Glass	90.0
			Leaf	89.5
		14.7.70	Glass	92.0
			Leaf	87.7
9) *E. polygoni*	Pea	2.9.70	Glass	82.5
			Leaf	85.5
		5.9.70*	Glass	83.7
			Leaf	83.7

Table IV contd...

10) <u>E. lamprocarpa</u>	Planyain	2.9.70	Glass	64.0
			Leaf	64.5
		7.9.70*	Glass	65.0
			Leaf	62.0
11) <u>E. galeopsidis</u>	Hedge woundwort	2.9.70	Glass	86.0
			Leaf	85.0
		5.9.70*	Glass	81.7
			Leaf	83.2
12) <u>E. galeopsidis</u>	Deadnettle	9.9.70	Glass	78.6
			Leaf	81.4
13) <u>E. artemesia</u>	<u>Artemesia</u> sp	9.9.70	Glass	20.2
			Leaf	25.2
14) <u>E. communis</u>	Turnip	2.9.70	Glass	94.7
			Leaf	95.5
		5.9.70	Glass	93.5
			Leaf	94.7
15) <u>E. communis</u>	Swede	2.9.70	Glass	95.0
			Leaf	96.7
		5.9.70	Glass	92.0
			Leaf	94.5
16) <u>E. heraclei</u>	Parsnip	9.9.70	Glass	97.0
			Leaf	97.8
		10.9.70	Glass	95.5
			Leaf	97.0
17) <u>Oidium sp</u>	Marrow	5.9.70	Glass	95.0
			Leaf	92.6
		9.9.70*	Glass	90.2
			Leaf	91.7

Table IV contd...

18)	Oidium begoniae	Begonia	2.9.70	Glass	92.7
				Leaf	95.5
			5.9.70	Glass	92.5
				Leaf	95.0
19)	Microsphaera sp.	Oak	19.7.70	Glass	97.0
				Leaf	98.5
			21.7.70	Glass	95.7
				Leaf	93.7
			29.7.70*	Glass	94.5
				Leaf	95.2
			5.8.70	Glass	94.2
				Leaf	94.5
			11.8.70*	Glass	87.0
				Leaf	97.0
			25.8.70*	Glass	92.2
				Leaf	92.0
20)	Podosphaera leucotricha	Apple	29.7.70	Glass	23.5
				Leaf	30.0
21)	Sphaerotheca mors-uvae	Black currant	19.7.70	Glass	88.2
				Leaf	83.7
22)	S. mors-uvae	Gooseberry	12.7.70	Glass	83.2
				Leaf	69.7
			13.7.70	Glass	69.8
				Leaf	73.2
			14.7.70	Glass	87.5
				Leaf	84.0
			19.7.70*	Glass	86.5
				Leaf	86.7
			26.7.70	Glass	87.0
				Leaf	87.5

Table IV contd...

23) *S. fuligenea*	Cucumber	23.7.70	Glass	80.8
			Leaf	79.7
		24.7.70	Glass	82.5
			Leaf	86.0
		25.7.70*	Glass	87.2
			Leaf	90.5
		2.8.70	Glass	77.7
			Leaf	79.0
24) *S. pannosa*	Rose	12.7.70	Glass	72.7
			Leaf	69.0
		14.7.70	Glass	77.0
			Leaf	74.1
		19.7.70*	Glass	77.0
			Leaf	79.0
		5.8.70*	Glass	82.7
			Leaf	83.3
		11.8.70*	Glass	80.7
			Leaf	83.2
		25.8.70*	Glass	66.6
			Leaf	78.5
		9.9.70*	Glass	87.2
			Leaf	89.6
25) *Uncinula necator*	Vine	11.8.70	Glass	87.4
			Leaf	93.6
		12.8.70	Glass	91.4
			Leaf	92.6
		13.8.70	Glass	89.5
			Leaf	91.0
		25.8.70*	Glass	94.7
			Leaf	97.0
26) *E. bicornia*	Sycamore	5.9.70	Glass	66.6
			Leaf	67.0

* Fresh samples

Appendix 6.

Shipton and Brown staining technique

1. Whole leaves or leaf pieces immersed in 10 - 15 ml alcoholic lactophenol cotton blue (1 part lactophenol cotton blue* to 2 parts 95% alcohol)
2. Solution containing the leaf sections brought to boil and simmered for 1 min.
3. After the leaves sank, solution brought to boil again for ½ min.
4. The leaf sections were then left in the stain for c. 48h at room temperature.
5. Sections were removed, rinsed in water and placed in chloral hydrate (2 g chloral hydrate to 2 ml water) for 30 - 40 min.
6. Mounted on a slide in 50% glycerine

*	Phenol	10 g
	Glycerine	10 ml
	Lactic acid	10 ml
	Aqueous aniline blue	0.02 g
	Distilled water	10 ml

Periodic - Schiff Method (Preece, 1965)

1. Tissue decolourized in methanol (24h)
2. Immersed in 1% Periodic acid* for 1 min.
3. Washed in distilled water 5 min.
4. Immersed in decolourized basic fuchsin** (5 min)
5. Washed in sulphurous acid+ solution for 5 min.
6. Immersed in distilled water for 5 min.
7. Mounted in distilled water for examination.

   (Host tissue — colourless, fungus — bright red).

Periodic acid*

Freshly prepared 1% solution in distilled water.

Decolourized basic fuchsin**

Basic fuchsin (1 g) dissolved in 200 ml distilled water by boiling. The solution was cooled , 20 ml 1N HCl added and mixed well. Crystals of potassium metabisulphite (1 g) was added and shaken until dissolved. The flask was plugged with a rubber bung and left overnight in the dark. The straw coloured solution was stored in the dark.

Sulphurous acid+

Potassium metabisulphite solution (10%)	10 ml
1N HCl	10 ml
Distilled water	200 ml

(solution prepared just before used)

Formalin fixation

Portions of leaves mounted on stubs were placed on a wire mesh support and placed in a 17.8 x 10.2 x 3.8 polystyrene box containing 100 ml formalin (30% formaldehyde) at the bottom. The box was closed with the lid and the leaf tissues fixed in formalin vapour for 40 min at room temperature.

Appendix 7

Table V Effect of 8 h leaf wetness on the Growth Index of *S. pannosa*

Incubation time (h)	Treatment and replicate	Grade & no. spores 0	1	2	3	4	Sum of grades	Growth Index
	A 1	72	28				28	
	2	70	30				30	7.6
6	3	66	34				34	
	B 1	76	24				24	
	2	72	28				28	6.5
	3	73	27				27	
	C 1	74	26				26	
	2	72	28				28	7.1
	3	68	32				32	
	A 1	30	54	16			86	
	2	24	60	16			92	23.8
	3	18	56	26			108	
24	B 1	24	74	2			78	
	2	22	75	3			81	19.0
	3	30	70				70	
	C 1	16	62	22			106	
	2	18	62	20			102	25.5
	3	28	50	18	4		98	

Table V contd...

	A	1	24	20	8	22	26	206	
48		2	21	30	9	17	23	191	47.5
		3	19	40	8	14	19	174	
	B	1	30	44	0	14	12	134	
		2	26	52	2	10	10	126	31.5
		3	23	56	6	10	5	118	
	C	1	20	50	8	6	16	148	
48		2	20	52	4	8	16	148	37.3
		3	27	39	5	13	16	152	
	A	1	25	28	5	13	29	193	
		2	26	24	0	4	46	220	50.7
		3	24	34	0	6	36	196	
	B	1	28	59	1	5	7	104	
72		2	29	55	0	2	14	117	29.4
		3	24	54	4	2	16	132	
	C	1	24	26	8	12	30	198	
		2	24	30	5	10	31	194	47.8
		3	26	32	4	10	28	182	

A = Control
B = Wetted immediately after inoculation
C = Wetted 6 h after incubation

Appendix 8

Table VI Some data on the effect of 8 h leaf wetness immediately after inoculation on the Growth Index of *Sphaerotheca pannosa* 72 h after incubation at 20°

Sample number	Treatment and replicate	Grades and no. of spores			Sum of grades	Growth Index
		0	1	>2		
Leaf 1	A 1	1	0	5		
	2	1	3	4	32	64
	3	0	1	5		
	B 1	1	3	0		
	2	3	3	1	9	30
	3	3	1	1		
Leaf 2	A 1	1	2	4		
	2	0	1	3	27	79.4
	3	0	2	4		
	B 1	3	0	0		
	2	2	2	1	6	21.4
	3	4	2	0		
Leaf 3	A 1	0	1	5		78.9
	2	0	1	4	30	
	3	1	4	3		
	B 1	2	3	0		
	2	1	4	0	13	34.2
	3	4	4	1		

Table VI contd....

Leaf 4	A 1	1	1	4	33	86.8
	2	0	1	5		
	3	0	1	6		
	B 1	2	3	1	14	41.1
	2	1	4	0		
	3	1	5	0		
Leaf 5	A 1	0	0	3	19	67.8
	2	1	4	2		
	3	1	1	2		
	B 1	2	2	0	11	45.8
	2	0	4	0		
	3	0	3	1		

A = control, B = 8 h wetness

Mean Growth Index A = 75.4, B = 34.5

Appendix 9

Table VII Some data on the growth of attached leaves and pedicels of rose

Plant Growth		Day 1	Day 2	Day 4	Day 7	Day 10	Day 14	Day 17	Day 19	Day 23
Flower bud	Pedicel length (cm)	0.2	0.22	0.5	1.1	1.7	2.5	2.7	2.8	2.8
	Cuticle thickness (μm)	1.1 (0.2)	2.1 (0.2)	2.4 (0.4)	3.1 (0.2)	3.7 (0.6)	4.7 (0.4)	4.8 (0.3)	4.9 (0.3)	- -
Leaf	Leaf area (cm^2)	1.3	3.2	4.0	6.5	6.9	7.9	7.9	-	-
	Cuticle thickness (μm)	1.2 (0.3)	2.5 (0.4)	2.8 (0.2)	4.0 (0.5)	6.5 (0.4)	6.5 (0.4)	6.4 (0.4)	6.6 (0.3)	- -

S D in brackets

Appendix 10.

Tests for auxin activity in diseased and healthy rose tissue

The *Avena* straight growth tests of Bentley (1962) were used for the bioassays of auxins. Hulled oat seeds soaked overnight in distilled water were sown on damp blotting papers in germination trays and incubated at 25^{o} in a constant temperature room (CTR) in the dark. After 3 days, they were removed to a room illuminated by red light and the apical 3mm of the young coleoptiles that had developed were cut off. The 10mm portions below this cut were used for the bioassays. The coleoptile segments were allowed to soak in distilled water for 8h before tests were begun.

Expt (i) Crude extracts:

Diseased and healthy tissues of leaves and pedicels of cv Frensham of comparable age were collected from plants at Silwood Park on 28.8.71, brought to the laboratory and frozen at -18^{o}. Equal fresh weights (3.5 g) of tissue were cut up and ground to a pulp in 3 ml distilled water with a mortat and pestle. The extract was squeezed through several layers of muslin to remove the pulp, the liquid fraction made up to 12 ml with distilled water and centrifuged at 3300 G for 15 min to remove the remaining suspended material. The supernatants were transferred to glass tubes and held in an ice bath until the assays were performed. All operations were done in the dark.

Fifteen coleoptile sections were placed in 10 ml of the extract in staining blocks, covered with glass lids and incubated at 25^{o} in the CTR in the dark. Coleoptiles in distilled water were used as controls. After 48h, they were removed and measured to determine the average change in length for each treatment.

Expt (ii) Purified extracts

Healthy and diseased pedicels of greenhouse grown cv. Perle d' Alcanada were harvested and immediately frozen at -18°. Equal fresh weights (2.5 g) of tissue (flower stalks only) were macerated in 90 ml methanol in the Omnimixer for 2 x 20 seconds at half speed and filtered twice through a Buchner funnel. The filtrate was evaporated to dryness in a rotary evaporator under vacuum at 40°. The residue was dissolved in 10 ml methanol, transferred to a smaller flask and evaporated to dryness as above.

After standing overnight at -18° the filtrate was dissolved in 5 ml distilled water and 10 ml of ether was added to this. The p^H of the mixture was adjusted to 2 with Orthophosphoric acid. The aqueous phase was separated three times with ether and the ether fractions were collected. The ether extract was partitioned four times with 5% sodium bicarbonate(p^H 9) and the organic layer was discarded. The alkaline solution was acidified to p^H 2 and extracted twice with ether and the ether fraction evaporated to dryness in a rotary evaporator.

The residue was dissolved in 0.2 ml ethanol and strip loaded (2 inches) on a cleaned Whatman's No.4 chromatography paper. The chromatograms were run in isopropanol : ammonia : water (10:1:1 v/v) in the dark at room temperature (c. 22°) until the solvent front had travelled 17.7 cm. After drying under the fan, each chromatogram strip from the origin to the front was divided into 10 equal parts, each section (1.7 c wide) corresponding roughly to 0.1 Rf unit. A portion of the chromatogr below the front was used as a control. Each section was cut into smaller pieces, placed in 5 ml distilled water in a 5 cm diam. Petri dish and swirled around to ensure thorough mixing. All operations were done either in the dark or in red light.

In each Petri dish 8 coleoptile segments were placed, covered with lid and the dishes were incubated in the dark at 25° in the CTR. After 48h the average change in length of the coleoptiles was determined.

Results

Although the results are not analysed satistically there is no evidence to indicate cell elongation of any significance in the crude and purified extracts of host tissue (Tables VIII - X).

Appendix 10

Table VIII Avena coleoptile bioassay of crude extracts of healthy and diseased rose tissue

Replicate number	Length of coleoptiles (cm)				
	Crude extracts from				Control
	Diseased Pedicel	Healthy Pedicel	Diseased Leaf	Healthy Leaf	(distilled water)
1	1.1	1.3	1.3	1.0	1.4
2	1.0	1.3	1.1	1.0	1.4
3	1.1	1.3	1.3	1.1	1.4
4	1.2	1.1	1.2	1.1	1.5
5	1.0	1.1	1.4	1.1	1.4
6	1.1	1.2	1.2	1.0	1.4
7	1.2	1.4	1.1	1.1	1.4
8	1.0	1.2	1.1	1.2	1.5
9	1.0	1.3	1.3	1.1	1.3
10	1.0	1.3	1.2	1.1	1.4
11	1.0	1.2	1.3	1.0	1.5
12	1.1	1.4	1.2	1.1	1.4
13	1.1	1.2	1.2	1.0	1.4
14	1.0	1.2	1.2	1.1	1.3
15	1.1	1.2	1.4	1.2	1.4
Total	16.0	18.7	18.5	16.2	21.1
Mean	1.0666	1.246	1.233	1.08	1.406

Appendix 10

Table IX Avena coleoptile bioassay of purified extracts of diseased pedicels of cv. Perle d' Alcanada

Replicate No.	Coleoptile length (cm)										
	Rf										Control (distilled water)
	0.1	0.2	0.3	0.4	0.5	0.6	0.7	0.8	0.9	1.0	
1	1.6	1.5	1.5	1.7	2.0	1.6	1.3	1.5	1.6	1.7	1.7
2	1.5	1.6	1.5	1.7	1.6	2.0	1.5	1.5	1.4	1.6	1.7
3	1.6	1.5	1.5	1.9	1.8	1.5	1.4	1.5	1.4	1.6	1.6
4	1.7	1.7	1.9	1.6	1.5	1.5	1.4	1.5	1.6	1.7	1.6
5	1.7	1.6	1.6	1.6	1.6	1.6	1.5	1.6	1.5	1.6	1.5
6	1.5	1.4	1.6	1.5	1.8	2.0	1.4	1.6	1.4	1.6	1.4
7	1.5	1.6	1.5	1.9	1.5	1.6	1.6	1.5	1.7	2.0	1.7
8	1.8	1.7	1.5	1.6	1.7	1.5	1.4	1.4	1.4	1.5	1.4
Total	12.9	12.6	12.6	13.5	13.5	13.3	11.5	12.1	12.0	13.5	12.6
Mean	1.6	1.5	1.5	1.6	1.6	1.6	1.4	1.5	1.5	1.6	1.5

Appendix 10

Table X <u>Avena coleoptile bioassays of purified extracts of healthy pedicels of cv. Perle d'Alcanada</u>

Replicate No.	Coleoptile length (cm)										
	Rf										Control (distilled water)
	0.1	0.2	0.3	0.4	0.5	0.6	0.7	0.8	0.9	1.0	
1	1.6	1.7	1.4	1.6	2.1	1.9	1.6	1.5	1.5	1.5	1.7
2	1.8	1.9	1.4	1.6	1.9	2.0	2.0	1.3	1.5	1.6	1.7
3	1.8	1.8	1.7	1.7	1.7	1.9	1.8	1.5	1.6	1.7	1.5
4	1.9	1.5	1.8	1.5	1.9	1.7	1.9	1.6	1.5	1.7	1.5
5	1.6	1.4	1.6	1.5	1.9	1.9	1.6	1.3	1.8	1.4	2.0
6	2.0	1.6	1.6	1.8	1.7	1.7	2.3	1.4	1.5	1.4	2.0
7	1.9	1.4	1.7	1.7	1.8	1.6	2.1	1.6	1.5	1.6	1.5
8	1.6	1.7	1.6	1.6	1.8	2.3	1.7	1.6	1.5	1.9	1.5
Total	14.3	13.0	12.8	13.0	14.8	15.0	15.0	11.8	12.4	12.8	13.4
Mean	1.7	1.6	1.6	1.6	1.8	1.8	1.8	1.4	1.5	1.6	1.6

Appendix 11.

Filming grids

1) Microscope slides (3" x 1") were cleaned with 'Vim' by rubbing with fingers washed under the tap, rinsed and stored in distilled water. The slides were finally dried for use by blotting off excess water between sheets of filter paper and wiping with vellin tissue.

2) Cleaned slides were held in tweezers and dipped into formvar solution (0.6% formvar in chloroform), withdrawn slowly and evenly and allowed to dry.

3) An area about 5 x 1.5 cm was scored on one face of the slide using a needle.

4) The scored surface was moistened by breathing heavily onto it and the slide was gently lowered at an angle (about 45^{o}) into a bowl of water. The formvar film was then floated off from the slide. The surface of the water was swept with a clean glass rod to remove any surface film before lowering the formvar filmed slide.

5) Grids were placed onto the surface of the film, dull side down and the whole film with grids was picked up on nylon gauze.

6) Filmed grids were dried and stored in a desiccator.

Appendix 12.

Experiments on the production of aseptic conidia and their germination and growth on agar media.

Expt (1) Standard method/12 day mildew cultures

Method : 10 min in 10% Chloros + 5 washings, dried with sterile tissue and placed on 1% agar, incubated at 17°.

No. disks sterilized	Regeneration of mildew	
	No. disks	%
35	4	11.4

Expt (2) Method of eliminating surface sterilization with Chloros/non-infected disks. The effectiveness of the methods outlined below was checked by a 'sterility' test immediately after treatment (10 disks/treatment).

Method	% Disks contaminated
(1) Control (no treatment)	100
(2)a. 1 min sterile water* + 5 washings	100
b. 1 min 10% Chloros + 5 washings	60
(3)a. 2 min sterile water + 5 washings	100
b. 2 min 10% Chloros + 5 washings	30
(4)a. 5 min sterile water + 5 washings	100
b. 5 min 10% Chloros + 5 washings	30
(5)a. 10 min sterile water + 5 washings	100
b. 10 min 10% Chloros + 5 washings	20

* 0.001% Tween 80 solution

Expt (3) A. Eliminating the number of washings after Chloros treatment/non infected leaf disks.

The treated disks were immediately checked for sterility.

Method	% Disks contaminated
(1) 10 min 10% Chloros + 5 washings	0
(2) " " " + 2 "	0
(3) " " " + 1 "	0
(4) 10 min 10% Chloros only	0

B. Surface sterilization of 16 day mildew cultures with Methods 3 and 4 of Expt 3A

The treated disks were dried and plated on 1% agar.

Method	No. disks sterilized	Regeneration of mildew		Remarks
		No. disks	%	
10 min 10% Chloros +1 washing	42	1	2.3	Browning in 5-7 days
10 min 10% Chloros only	50	0	0	Browning in 5 days

Expt 4 Effect of temperature on regeneration of mildew/17 day culture

Method - as for Expt 1

Temp (^{o}C)	No. disks sterilized	Regeneration of mildew		Remarks
		No. disks	%	
17^{o}	25	1	4	Disks reasonably green afte 15 days
22^{o}	25	0	0	All disks turned bro wn after 1 days.

Expt (5) Effect of substrate on regeneration of mildew/12 day culture

Method - As for Expt 1.

Substrate	No.disks sterilized	Regeneration of mildew No.disks	%	Remarks
1% Agar(plates)	30	0	0	Easier handling
Distilled water (in polystyrene box)	30	0	0	Disks(some) tend to sink

Expt (6) Age of mildew on regeneration

Method - As for Expt 1

Mildew age (days)	No. disks sterilized	Regeneration (% disks)
9	40	5.0
12	30	3.3
14	30	13.3
21	50	2.0

Expt (7) Effectiveness of surface sterilization on the production of aseptic conidia/13 day cultures.

Method - As for Expt 1. Two hundred leaf disks were surface sterilized and examined after 17 days. The disks bearing the regenerated mildew were checked for sterility.

No.disks sterilized	Regeneration of mildew No.disks	%	% discs with aseptic conidia
200	41	20.5	4.8

Expt (8) Increasing Chloros concentration and treatment time on the production of aseptic conidia/14 day cultures.

Method - Similar to Expt 7 but the disks were treated as follows.

Treatment	No.disks sterilized	Regeneration of mildew no.disks	%	Remarks
(1) 15 min 10% Chloros + 5 washings	50	0	0	Disks normal
(2) 10 min 15% Chloros + 5 washings	50	4	8	Browning of disks after 10-12 days
(3) 15 min 15% Chloros + 5 washings	50	0	0	

Expt (9) A check on the effectiveness of method 2 Expt 8

Two hundred mildewed disks (12 day cultures) were surface sterilized accordingly. One hundred were checked for sterility immediately after treatment. The other hundred were dried and plated on Nutrient Agar.

	% Disks contaminated	Regeneration of mildew (%)
Nutrient Broth	51	-
Nutrient Agar	67*	2

* From direct observation of nutrient agar plates-mostly bacteria and <u>Penicillium</u> sp and <u>Rhizopus</u> sp. Over 80% disks turned brown after 15 days.

Expt (10) Repeat of standard method (= Expt 1)/14 day culture

Sterility check made on all disks with the regenerated mildew

No.disks sterilized	Regeneration of mildew % disks	% Disks with regenerate mildew that are sterile
100	9	11.1

Expt (11) Effect of Kinetin on the longevity of leaf disks and the development of mildew on them.

Disks (1.6 cm diam) cut from 10 day old leaves were inoculated with 24h conidia and floated on the following solutions in polystyrene boxes. These were maintained at 17°:-

0.5 ppm, 3 ppm, 5 ppm and 10 ppm Kinetin solutions; distilled water used as a control.

Results - see Table XI

Expt (12) Effect of Kinetin on regeneration of mildew on surface sterilized disks/12 day cultures.

Medium	No.disks sterilized	Regeneration of mildew No.disks	Regeneration of mildew %
Distilled water (control)	35	2	5.7
10 ppm Kinetin solution	40	1	2.5

Appendix 12

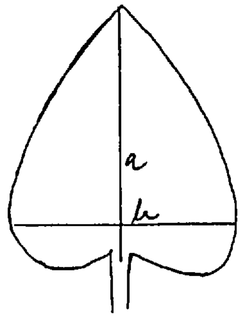
Table XI Effect of kinetin on longevity of leaf disks and mildew development

Medium	Appearance of the leaf disk cultures							
	Week 1	Week 2	Week 3	Week 4	Week 5	Week 6	Week 7	Week 8
Distilled water	-	Good	Good but yellowing started	Dying	Dead			
0.5 p.p.m. kinetin	-	Good	Disks yellowing Mildew good	Yellow Mildew dying	Dead			
3 p.p.m. kinetin	-	Good	Good	Good but small patches of yellow appearing	Dying	Dead		
5 p.p.m. kinetin	-	Good	Good	Good	Disks good Mildew dying	Disks yellow Mildew dead	Dead	
10 p.p.m. kinetin	-	Good	Good	Good	Good	Good Mildew dying on some	Disks yellowing & Mildew dying	Dead

Expt (13) A quick method of estimating the area of (attached) leaves.

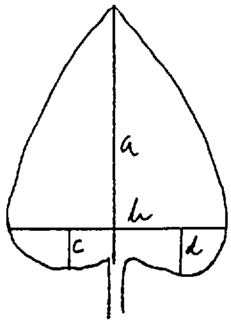
The actual area of 16 detached leaves of different sizes was determined by tracing them on cm graph paper. This was compared with 2 methods outlined below:

(a)



Area = Length (a) x breadth at the widest place (b)

(b)



$$\text{Area} = a \times \frac{b + (2 \times c) + (2 \times d)}{2}$$

TableX11 gives the detailed leaf area measurements. The estimated value obtained by method (b) closely agreed with the actual leaf area (Fig. i).

Expt (14) The pattern of growth of Burdock leaves 1 - 4.

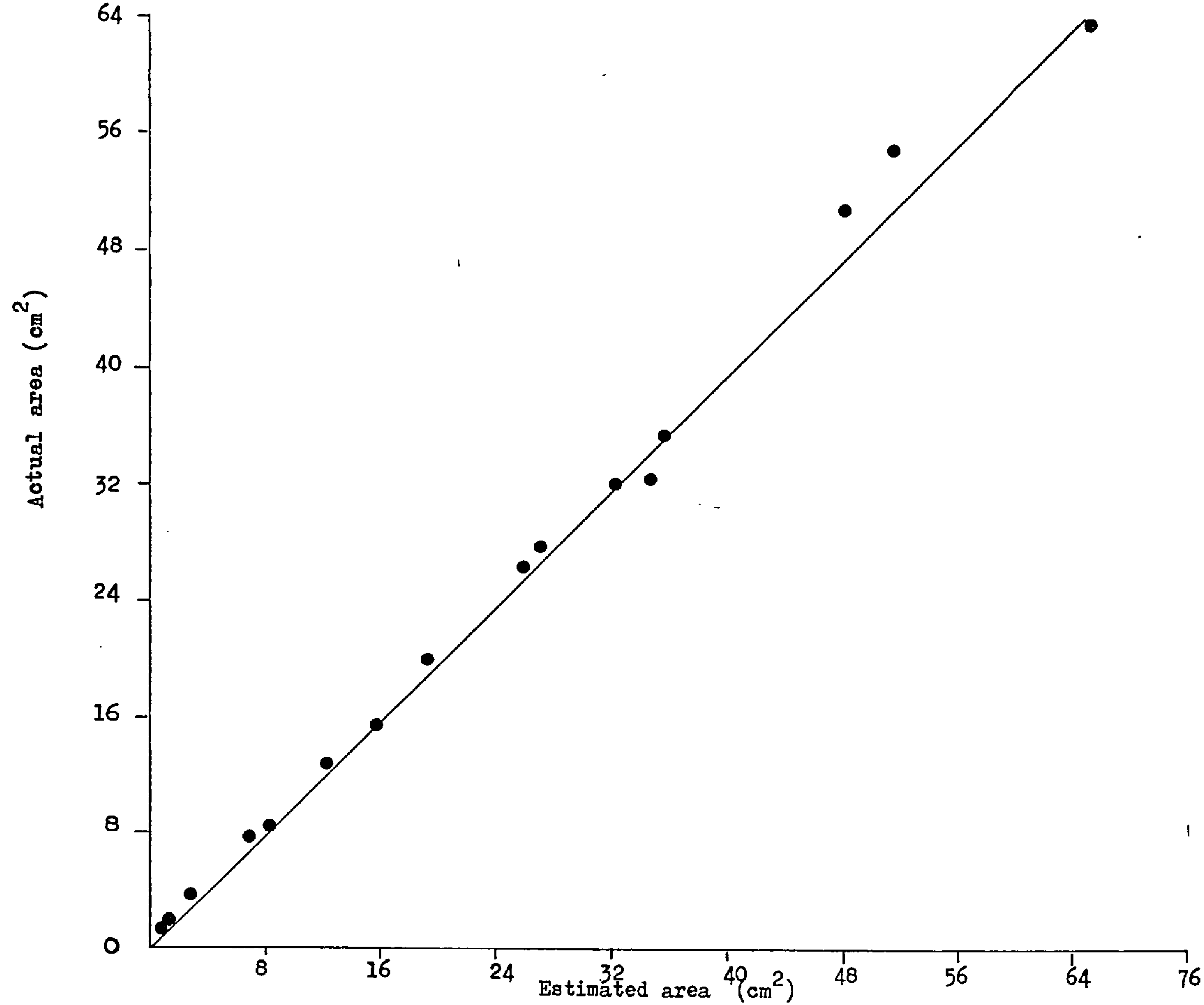
Burdock plants were raised from seed as described on p. and maintained in the greenhouse from November 1969 to January 1970. The first four leaves were tagged as they unfolded (day of unfolding was regarded to day 1) and their areas were estimated by method (b) of Expt 13 at 3 day intervals for 21 days.

The first leaf usually unfolded 12 - 14 days after

TableX11 Leaf area estimation

No.	Leaf	area	(cm^2)
leaves	actual area	Method (a)	Method (b)
1	1.7	1.5	1.6
2	1.7	1.9	1.7
3	3.6	3.8	3.2
4	7.8	7.9	7.3
5	8.5	9.6	8.5
6	12.9	14.1	12.5
7	15.6	15.5	15.7
8	20.0	22.4	19.1
9	26.3	28.1	25.4
10	27.7	39.1	26.6
11	31.9	39.1	32.2
12	32.2	37.2	34.6
13	34.1	38.4	35.1
14	50.9	64.0	48.0
15	54.5	59.3	51.2
16	63.6	73.8	64.8

Fig. i Relationship of actual leaf area to estimated leaf area in Burdock



planting and the second followed 10 - 12 days later. The third and the 4th successive leaves unfolded at shorter intervals of 5 - 7 days.

The leaf area determinations based on an average of 25 replicates for each leaf are given in Fig. ii . The growth pattern was similar in all four leaves. The maximum rate of expansion was between 3 and 9 days after unfolding. After 13 - 15 days growth ceased but the leaves remained funotional for a further 13 - 15 days. The first two leaves however turned yellow 3 - 4 days after they reached their maximum size. The fourth leaf was the largest of the four and remained green for a longer period.

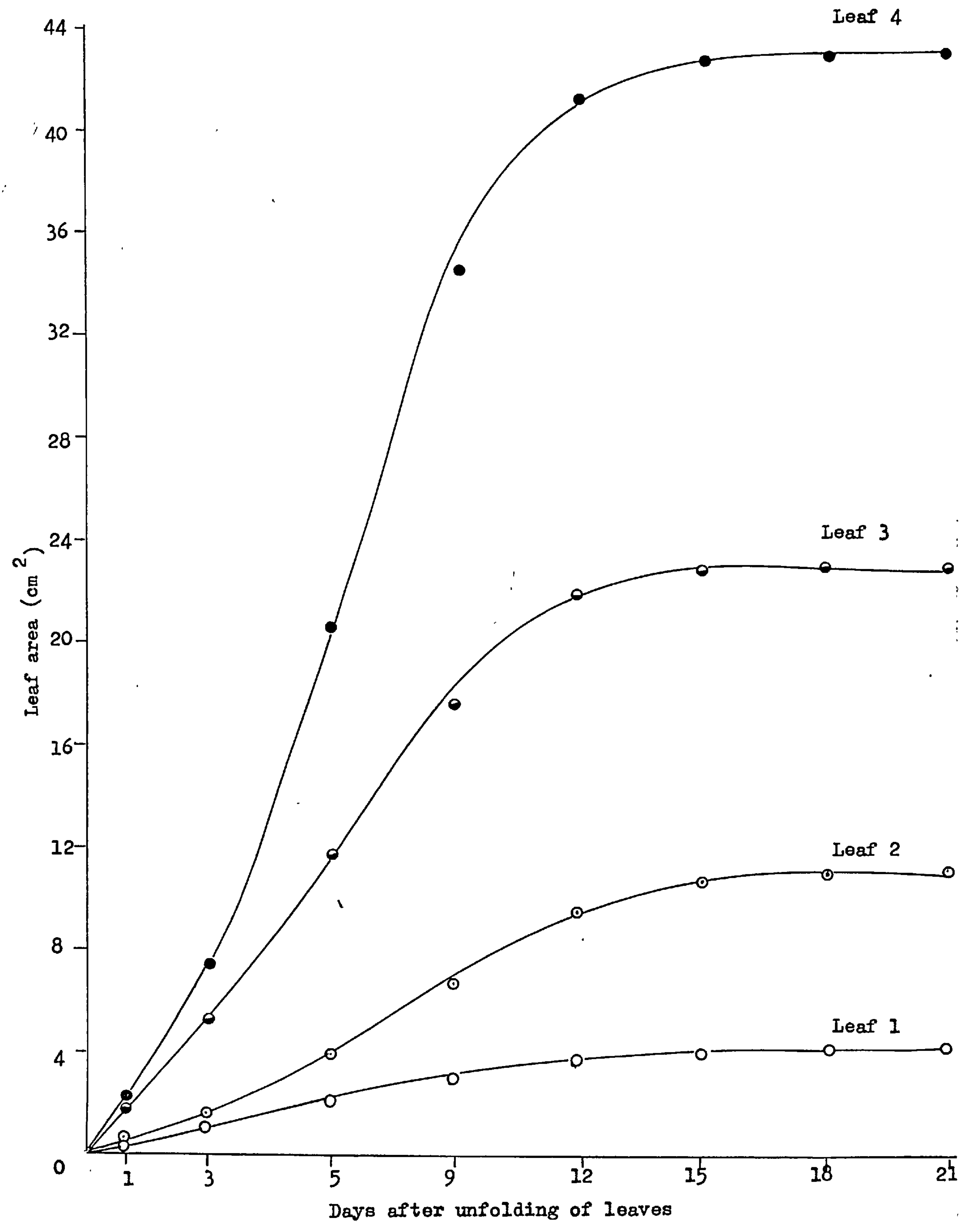
Expt (15) Effeot of leaf age on the development of mildew and its subsequent regeneration after surface sterilization.

Burdook plants were grown in pots and maintained in the greenhouse. The fourth leaf was tagged soon after it unfolded and its area was estimated daily. Four leaves at a time were detached 1, 5, 9, 13 and 17 days after unfolding, inoculated with 24h conidia and maintained in germination trays at 17°. The one - day old leaves with shorter petioles were floated on distilled water in a 17.8 x 10.2 x 3.8 polystyrene box, closed with lid and incubated as above.

The mildew developed 8 - 10 days later on all leaves including the older ones although on these mildew growth was sparse. After 12 days, 1.6 om diam disks were cut from well - mildewed areas, surface sterilized and examined 15 days later, after incubating at 17°. All disks with the regenerated mildew were tested for 'sterility' in Nutrient Broth.

Daily measurements of leaf area are given in Table X111.A summary of results of the relation of leaf age to regeneration of mildew on surface sterilized disks is given below.

Fig. ii Growth pattern of the first four leaves of Burdock.



Appendix 12

Table XIII Increase in leaf area in relation to age

		Leaf Area /cm^2 / Days after unfolding																
		1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17
	1	6.9																
D 1	2	6.7																
	3	3.7																
	4	4.3																
	5	2.9	6.1	12.5	24.0	32.5												
D 5	6	3.4	8.3	18.0	32.1	44.6												
	7	5.0	8.1	22.0	25.1	27.7												
	8	2.6	4.0	6.3	14.1	17.1												
	9	2.4	3.9	6.3	11.1	18.1	21.0	30.1	45.1	61.8								
D 9	10	2.1	2.6	5.0	7.2	15.1	16.9	29.1	42.1	53.9								
	11	2.2	3.1	6.0	10.2	18.1	20.0	36.1	52.1	68.9								
	12	2.3	3.0	6.4	11.6	.20.1	25.6	31.1	39.2	44.4								
	13	3.0	5.1	6.3	18.0	25.0	34.5	46.1	55.3	60.0	65.1	68.0	70.3	71.4				
D 13	14	3.2	5.1	8.7	15.1	28.1	37.5	48.1	52.1	62.2	66.3	68.1	70.0	74.1				
	15	3.1	4.0	7.5	15.1	21.5	29.1	39.1	46.5	52.1	54.1	58.1	60.1	63.1				
	16	3.3	4.6	9.6	13.9	26.0	32.5	39.1	42.1	46.7	50.1	55.1	57.0	61.5				
	17	3.4	4.9	7.6	14.5	27.0	35.0	40.5	48.7	53.2	63.2	73.7	82.4	88.0	91.4	95.4	98.0	98.1
	18	4.4	6.2	12.1	16.1	21.5	36.9	49.1	56.2	61.4	70.1	78.3	85.7	87.5	92.0	93.0	96.0	96.3
D 17	19	3.9	4.8	11.6	18.6	29.0	38.7	46.5	50.1	58.9	66.1	74.1	84.5	90.1	99.3	109.2	117.2	119.0
	20	3.9	4.9	12.2	15.6	24.0	31.2	41.1	48.3	57.2	69.0	73.0	80.3	86.3	59.4	96.7	96.9	97.4
Total		72.7	78.7	158.1	262.3	395.4	358.9	476.0	577.8	674.7	504.0	548.4	590.3	622.0	372.1	394.3	408.1	410.8
		20	16	16	16	16	12	12	12	12	8	8	8	8	4	4	4	4
Mean		3.6	4.9	9.8	16.3	24.7	29.9	39.6	48.1	56.2	63.0	68.5	73.7	77.7	93.0	98.5	102.0	102.7

Leaf age (days)	No.disks sterilized	% Disks with regenerated mildew	% Disks with aseptic conidia
1	20	0	0
5	30	0	0
9	45	6.6	0
13	55	10.9	0
17	68	19.1	0

The highest number of disks where mildew regrowth occurred was obtained with the 17 day old leaves which had reached their maximum size. No regrowth occurred on 1 and 5 day leaves which died one week after surface sterilization. All disks with the regenerated mildew were contaminated.

Expt (16) Susceptibility of leaves after maximum expansion

The 4th leaf of each of 10 plants was tagged and its area determined until it stopped expanding. Two and five days after maximum expansion five leaves were detached, inoculated and tested as in Expt (15).

The leaves reached their maximum size 16 days after unfolding. They were therefore sampled on the 18th and 21st day. The mildew developed on both sets of leaves around the 10th day after inoculation. A slight yellowing was seen on the 21 day leaves at the time of surface sterilization (12 days after inoculation) but the mildew growth on these seemed uneffected

Results:

Leaf age (days)	No.disks sterilized	Regeneration of mildew no.disks	%
18	50	0	0
21	50	0	0

No growth occurred on any of the treated leaf disks. Nearly 75% of the disks turned yellow and appeared dead 10 - 12 days after treatment.

Expt (17) Source of origin of the regenerated mildew.

Three hundred and fifty mildewed disks (12 day cultures) were surface sterilized and incubated on agar at 17°. At 2,4,6, 8,10,12 and 14 day intervals fifty disks were cleared in methanol for 24h, stained in lactophenol cotton blue for 30 min to 1h, washed in distilled water and examined under the binocular microscope for regrowth of the mildew.

days after surface sterilization	No. disks examined	No. disks with reg. mildew
2	50	0
4	50	0
6	50	0
8	50	5
10	50	2
12	50	5
14	50	8

The earliest stage at which mildew growth was detected was 8 days after treatment. Evidence suggested that mildew regrowth originated from hyphal portions in grooves on the leaf surface. Although dead conidia were stained, dead hyphae did not take up the stain. There was no evidence of regeneration of mildew from conidia.

Ten days after treatment, the mycelial growth was extensive on disks where mildew regrowth occurred. Hence their origins could not be traced. Sporulation was observed in 14 days but only 2 - 3 conidia per conidiophore were formed at this stage.

Expt (18) Effect of physical removal of the mildew from the leaf surface on its regeneration.

The surfaoe growth was removed from mildewed leaves (13 day oultures) by rubbing gently but firmly with a piece of wet cotton wool. The leaves were blot dried. Disks, 1.6 cm diam were cut from these and incubated at 17° on 1% agar for 2 - 3 weeks.

Examination of the leaf surface immediately after rubbing with cotton wool showed that not all surface growth was removed this way. Always a few strands of hyphae, conidia in clumps and conidiophore were seen especially in the leaf surface grooves.

Regrowth of the mildew was obtained on 36% of the disks after 12 days.

Expt (19) Methods of placing conidia on agar media

Conidia were transferred to modified Czapek Dox yeast extract agar plates using the following methods

(i) by blowing 24 h conidia from an infected leaf on to plates kept at the bottom of a settling tower.

(ii) conidia picked up singly/or in chains on a sterile eyelash mounted on a matchstick (sterilized in propylene oxide vapour for 24 h) and placed on agar in pre - marked positions (on the underside of plates). Source of inoculum : (a) aseptic conidia from surface sterilized disks (36 transfers in three plates) and (b) infected leaf — 24h conidia (100 transfers in 10 plates).

% Germination after 72 h		
Mass transferred conidia from infected leaf	Single conidial transfers	
	aseptic conidia	infected leaf
55 (Heavy contamination)	18.3 (2 transfers contaminated)	35 (all contaminated)

Expt (20) Effect of r.h. and light on the growth of E. cichoracearum in axenic culture

Conidia from aseptic cultures were transferred singly / chains on to Czapek - Dox yeast extract agar (pH 6.1) plates as in Expt 19 and incubated at 17° under the following conditions.

(i) Dark (Plates covered with black cloth)
- (a) in a saturated atmosphere (in germination tray)
- (b) in the open cabinet

(ii) Light
- (a) in a saturated atmosphere
- (b) in open cabinet

Results: Germination $<10\%$

Contamination <2-3%

Growth ceased after 2-3 days.

Expt (21) Effect of drying the agar surface on germination

Modified Czapek - Dox yeast extract agar plates were left half open and dried in a sterile oven (sprayed with thymol) at 37° for 2½h. Aseptic conidia were placed on the agar surface (after cooling) as in Expt (19) method ii, and incubated at 17° in (a) a saturated atmosphere and (b) the open cabinet

Results were similar to those of Expt 20.

Appendix 13.

Methods of surface sterilization of Burdock seeds

One hundred seeds were sterilized by the methods outlined below, placed on either Potato Dextrose Agar or V8 juice agar (10 seeds / plate) and incubated at 25^{o} for 12 - 14 days.

(i) Chloros (full strength) :- 10, 15 and 20 min followed by 4 - 6 changes in sterile distilled water.

(ii) Sodium hypochlorite :- 3, 5, 10, 20 and 30% solutions for 10 min followed by 4 - 6 changes in distilled water.

(iii) 0.1% Mercuric chloride :- ½, 1, 2, 3, 5, 8, 10, 12 and 15 min followed by 4 - 6 changes in sterile distilled water.

Of these methods, 10 min in 0.1% Mercuric chloride produced the best results. Although the amount of germination was reduced by nearly half (43%) relative to the control (96%) nearly 50% of the germinated seeds (22) were free from contamination. Germination was however delayed by 5 - 7 days in the treated seeds. Their subsequent growth in soil in the greenhouse was normal.

Appendix 14.

Hoagland's solution (Bonner and Galston, 1959)

Ca $(NO_3)2.4H_2O$	0.95 g
KNO_3	0.61 g
$MgSO_4.7H_2O$	0.49 g
Ferric tartrate	0.005 g
$NH_4\ H_2PO_4$	0.12 g
Distilled water	1 litre
Minor elements	1 ml

The medium solidified with 0.6% Difco Bacto - agar and autoclaved at 1 kg / cm^2 for 15 min.

Minor elements:-

H_3BO_3	0.6 g
$MnCl_2.4H_2O$	0.4 g
$ZnSO_4$	0.05 g
$CuSO_4.5H_2O$	0.05 g
$H_2MoO_4.7H_2O$	0.02 g
Distilled water	1 litre